9 ECOLOGICAL IMPACT

9.1 Introduction

9.1.1 This chapter presents the assessment of the ecological and nature conservation impacts that may arise from the construction and operation of the proposed Hung Shui Kiu New Development Area (HSK NDA) (herein referred to as the "Project"), as based on the revised recommended outline development plan (Revised RODP). A location plan, the Revised RODP, and description of the development are presented in **Figure 1.1**, **Figure 2.2** and **Chapter 2**, respectively. The aim of the ecological impact assessment is to provide information to identify, predict, and evaluate potential ecological impacts.

9.2 Environmental Legislation, Policies, Plans, Standards and Criteria

9.2.1 This assessment makes reference to the following HKSAR Government ordinances, regulations, standards, guidelines, and documents when identifying ecological importance of habitats and species, and evaluating and assessing potential impacts of the Project on the ecological resources:

- The Environmental Impact Assessment Ordinance (Cap. 499), and its subsidiary legislation, which provides guidelines on the environmental impact assessment process.
- Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) Annex 8 – recommends the criteria to be used for evaluating habitat and ecological impact.
- EIAO-TM Annex 16 – sets out the general approach and methodology for assessment of ecological impacts arising from a project or proposal, to allow a complete and objective identification, prediction and evaluation of the potential ecological impacts.
- EIAO Guidance Note No. 3/2010 – provides guiding principles on the approach to assess the recommended environmental mitigation measures in EIA reports.
- EIAO Guidance Note No. 6/2010 clarifies the requirements of ecological assessments under the EIAO.
- EIAO Guidance Note No. 7/2010 provides general guidelines for conducting ecological baseline surveys in order to fulfil requirements stipulated in the EIAO-TM.
- EIAO Guidance Note No. 10/2010 introduces some methodologies in conducting terrestrial and freshwater ecological baseline surveys in order to fulfil requirements stipulated in the EIAO-TM.
- EIAO Guidance Note No. 11/2010 introduces general methodologies for conducting marine ecological baseline surveys.
- The Country Parks Ordinance (Cap. 208) provides for the designation and management of country parks and special areas. Country Parks are designated for the purpose of nature conservation, countryside recreation and outdoor education. Special areas are created mainly for the purpose of nature conservation.
- The Forests and Countryside Ordinance (Cap. 96) prohibits felling, cutting, burning or destroying of trees and growing plants in forests and plantations on Government land. Related subsidiary regulations prohibit the selling or possession of listed, restricted and protected plant species.

- Under the Wild Animals Protection Ordinance (Cap. 170), designated wild animals are protected from being hunted, whilst their nests and eggs are protected from injury, destruction and removal. All birds and most mammals, including marine cetaceans, are protected under this Ordinance.

- The Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586) gives effect to the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) in Hong Kong. It restricts the import and export of species listed in CITES Appendices so as to protect wildlife from overexploitation or extinction. The Ordinance is primarily related to controlling trade in threatened and endangered species and restricting the local possession of them.

- The Town Planning Ordinance (Cap. 131) provides for the designation of coastal protection areas, Sites of Special Scientific Interest (SSSIs), Conservation Area, Country Park, Green Belt or other specified uses that promote conservation or protection of the environment.

- Chapter 10 of the Hong Kong Planning Standard and Guidelines (HKPSG) covers planning considerations relevant to conservation. This chapter details the principles of conservation, the conservation of natural landscape and habitats, historic buildings, archaeological sites and other antiquities. It also describes enforcement issues. The appendices list the legislation and administrative controls for conservation, other conservation related measures in Hong Kong and Government departments involved in conservation.

- The Water Pollution Control Ordinance (Cap. 358) which provides the main statutory framework for the declaration of water control zones (WCZ) to cover the whole of Hong Kong and the establishment of water quality objectives.

- ETWB TCW No. 5/2005 Protection of Natural Streams/Rivers from Adverse Impacts arising from Construction Works provides an administrative framework to better protect all natural streams/rivers from the impacts of construction works. The procedures promulgated under this Circular aim to clarify and strengthen existing measures for protection of natural streams/rivers from Government projects and private developments.

- DSD Practice Note No. 1/2015 Guidelines on Environmental and Ecological Considerations for River Channel Design presents the essential environmental and ecological considerations that should be taken into account in the design of river channels.

9.2.2 This section also makes reference to the following international conventions and national legislation

- The IUCN Red List of Threatened Species provides taxonomic, conservation status and distribution information on taxa that have been evaluated using the IUCN Red List Categories and Criteria. This system is designed to determine the relative risk of extinction, and the main purpose of the IUCN Red List is to catalogue and highlight those taxa that are facing a higher risk of global extinction. The IUCN Red List also includes information on taxa that are either close to meeting the threatened thresholds or that would be threatened were it not for an ongoing taxon-specific conservation programme.

- The People's Republic of China National Protection Lists of Important Wild Animals and Plants – lists detailed Category I and Category II key protected animals and plant species under Mainland Chinese Legislation. The list was last updated in November 2002.

- The Convention on Biological Diversity (CBD) opened for signature at the Rio Earth Summit in 1992 with three main objectives: to conserve biodiversity, to ensure sustainable use of the components of biodiversity, and to share the benefits arising from the use of genetic resources in a fair and equitable manner. There are currently over 190 Parties to the Convention, including China. In May 2011, the CBD was formally extended to Hong Kong. The Environment Bureau and the Agriculture, Fisheries and Conservation Department have embarked on an exercise to develop a city-level Biodiversity Strategy and Action Plan (BSAP) under the CBD.

9.3 Assessment Approach and Methodology

Assessment Area

9.3.1 In accordance with Clause 3.4.10.2 of the EIA Study Brief No. ESB-291/2015, the assessment area for the purpose of the terrestrial ecological impact assessment includes areas within 500 m distance from the Project area and any other areas likely to be impacted by the Project and associated works (refer to **Figure 9.1**).

9.3.2 No marine works would be undertaken under the Project. The assessment area for aquatic ecology is the same as the water quality impact assessment (as described in **Chapter 5**) and therefore includes Deep Bay WCZ and North Western WCZ (refer to **Figure 10.1** for locations).

Literature Review

9.3.3 Relevant findings from available studies/surveys regarding the ecological characters of the assessment area have been reviewed and collated in accordance with Sections 3.4.10.4 (i) and (ii) of the EIA Study Brief:

Reports and Studies from Private Sectors and Government

- Agreement No. CE 66/96 - Planning and Development Study on North West New Territories (NWNT), Additional Services: Feasibility Study on Development for Additional Population Growth in North West New Territories (PlanD, 2001); Environmental Assessment of the Planning and Development Study on North West New Territories (PlanD/TDD, 1999);
- North West New Territories Planning and Engineering Study (CEDD, 2003);
- New Nature Conservation Policy (AFCD, 2015);
- Territory-wide long-term biodiversity monitoring on selected taxa groups (2002 to 2016) (AFCD, 2016);
- Register of Sites of Special Scientific Interest (SSSIs) (PlanD, 2015);
- Memoirs of Hong Kong Natural History Society;
- Avifauna of Hong Kong (Carey *et al*., 2001);
- GeoInfo Map from Lands Department;
- Porcupine! – newsletter of Department of Ecology & Biodiversity of University of Hong Kong;
- Biodiversity – newsletter of Agriculture, Fisheries and Conservation Department (AFCD);

- Annual Report and other Publications of The Hong Kong Bird Watching Society; and

- Egretry Counts in Hong Kong with particular reference to the Mai Po Inner Deep Bay Ramsar Site. Report by Hong Kong Bird Watching Society to the Agriculture, Fisheries and Conservation Department (2000-2015) (Kwok *et al*., 2000; Wong and Kwok, 2001; Wong, 2002; Wong and Woo, 2003; Anon, 2004, 2005a, 2006a, 2007a, 2008a, 2009a, 2011a, 2012a, 2012b, 2013a, 2014a; 2015).

Relevant EIA studies

- Agreement No. CE 66/96 - Planning and Development Study on North West New Territories, Technical Paper No. 13 Environmental Impact Assessment (PlanD, 1999a);

- Agreement No. CE 66/96 - Planning and Development Study on North West New Territories, Technical Paper No. 13 Environmental Impact Assessment. (PlanD, 1999b);

- Feasibility Study for Additional Cross Border Links Stage 2, Environmental Assessment Technical Report (PlanD, 2000);

- Deep Bay Link (DBL) Environmental Impact Assessment (HyD, 2002a);

- Shenzhen Western Corridor (SWC) Environmental Impact Assessment (HyD, 2002b);

- Upgrading and Expansion of San Wai Sewage Treatment Works and Expansion of Ha Tsuen Pumping Station Environmental Impact Assessment (DSD, 2002); and

- Hang Hau Tsuen Channel at Lau Fau Shan Environmental Impact Assessment (CEDD, 2009).

9.3.4 Relevant data from the literature review has been drawn upon and included in the ecological assessment presented below.

Ecological Surveys

9.3.5 Based on the literature review, ecological surveys were carried out to fill information gaps, verify the information collated, and to fulfil the requirements of the EIA Study according to the EIA Study Brief No. ESB-291/2015.

9.3.6 A twelve-month programme of ecological field surveys (covering both dry and wet seasons) was undertaken from October 2011 to September 2012. **Table 9.1** summarises the survey programme. The methodologies adopted for the ecological surveys are presented below.

Table 9.1 Ecological Survey Programme

Proposed Survey	Dry Season						Wet Season						Wet Season		Dry Season			Wet Season	
	2011						2012						2013	2014	2015		2016	2016	
	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	May	Sep	Jan	Dec	Jan	May	Jun
Habitat and Vegetation	✓	✓								✓									
Avifauna Survey		✓		✓		✓		✓		✓		✓							
Avifauna Survey (Night Survey)						✓				✓									
Breeding Ardeid Flight Path Survey										✓									
Butterfly, Dragonfly and Damselfly Survey		✓					✓	✓		✓		✓							
Herpetofauna Survey						✓		✓		✓		✓							
Herpetofauna Survey (Night Survey)						✓				✓									
Terrestrial Mammal Survey				✓		✓		✓				✓							
Terrestrial Mammal Survey (Night Survey)						✓				✓									
Freshwater Communities Survey						✓				✓									
Intertidal Survey				✓						✓									
Additional Site Visits																			
Habitat and Status Update of Ecologically Sensitive Areas, Including: i. San Sang San Tsuen Egretry ii. Ngau Hom Shek Egretry iii. Marsh Areas at Fung Kong Tsuen iv. Fishponds and Mitigation Ponds v. Natural Watercourses vi. Agricultural Areas vii. Intertidal Area													✓	✓	✓	✓	✓		
Additional Ecological Surveys (at Newly Added Sites, i.e. Service Reservoir Sites)																			
Habitat and Vegetation														✓			✓		
Avifauna Survey														✓			✓		
Butterfly, Dragonfly and Damselfly Survey														✓			✓		
Herpetofauna Survey														✓			✓		
Terrestrial Mammal Survey														✓			✓		
Additional Ecological Surveys (at Newly Added Sites, i.e. Watercourse 5 and Watercourse 6; and San Sang San Tsuen Egretry)																			
Freshwater Communities Survey																		✓	
Breeding Ardeid Flight Path Survey at San Sang San Tsuen Egretry																		✓	✓

Note:
The ticks (✓) denote the timing of the surveys for the faunal or floral groups assessed under the Study.

9.3.7 Given the long study period and changes to the Revised RODP (see **Chapter 2**), additional surveys and site verification were conducted to fill in any residual information gaps, where relevant. Additional surveys verified and updated the ecological conditions and collected additional baseline information from the assessment area. This included:

- Since no egrets were recorded at Ngau Hom Shek egretry during the breeding season in 2012 during the site check, no flight line survey was conducted for this egretry. However, according to Anon (2012b), one nest was recorded in 2012. Therefore, an additional site visit was conducted at Ngau Hom Shek egretry during breeding season in 2013 to check the egretry status (i.e. active or abandoned);

- Additional surveys were conducted during September 2014, January 2015, December 2015 and January 2016 to update the habitat status, mapping and vegetation of the assessment area. Incidental records of fauna species of conservation importance, observed during the additional surveys and site verification, were also included. These additional surveys covered ecologically sensitive areas which are typically subject to relatively rapid change due to human influence and natural processes, including San Sang San Tsuen egretry, marsh areas (e.g. marsh at Fung Kong Tsuen), ponds, natural watercourses, intertidal area and agricultural areas;

- Habitat/vegetation and fauna surveys were undertaken at areas newly added to the assessment area (i.e. service reservoir sites at Ngau Hom Shek knoll and southeast of Yuen Long Highway) in September 2014 and January 2016;

- Additional freshwater communities survey was undertaken at watercourses newly added to the assessment area (i.e. Watercourse 5 at the west of Kau Lee Uk Tsuen and Watercourse 6 on the foot of Yuen Tau Shan) in May 2016; and

- Additional breeding ardeid flight path surveys were undertaken at San Sang San Tsuen egretry in May 2016 and June 2016.

9.3.8 Given that no marine works are required for the Project and the water quality impacts from the operation phase of the Project to marine waters of the Deep Bay WCZ and North Western WCZ would be minimal (**Chapter 5** refers), no marine ecological survey, including marine mammal survey, was deemed necessary.

Ecological Survey Methodology

Habitat Mapping and Vegetation Survey

9.3.9 A habitat map of suitable scale (1:5000) showing the types and locations of terrestrial habitats within the assessment area (refer to **Figure 9.2**) was prepared by making reference to existing aerial photography, GeoInfo Map from Lands Department[1] and then a ground-truthing exercise of the assessment area was undertaken. Habitat and vegetation survey was conducted in accessible areas of the different habitat types and along paths (including the walk transects presented in **Figure 9.1**) to characterise vegetation in the assessment area. Ecological characteristics of each habitat type, including size, vegetation type, species richness and abundance, community structure, seasonal patterns, inter-dependence of the habitats and species and presence of any features of ecological importance were defined and characterised. Representative photographs of the habitat types and/or any important ecological features identified were taken.

[1] Source: Base Map Data from Lands Department in 2016 (GeoInfo Map: http://www1.map.gov.hk/gih3/view/index.jsp).

9.3.10 Nomenclature and conservation status of plant species follow Xing *et al.* (2000), Wu and Lee (2000), Siu (2000), AFCD (2003) and AFCD (2004). Reference was also made to Corlett *et al.* (2000), Hu *et al.* (2003), Hong Kong Herbarium (2004), and Hong Kong Herbarium and South China Botanical Garden (2007; 2008; 2009; 2011).

Fauna Surveys

9.3.11 Ecological surveys on terrestrial faunal groups (i.e. mammal survey, avifauna survey, herpetofauna survey, odonate and butterfly survey) were carried out along the walk transects shown in **Figure 9.1**. Locations for point count of freshwater fish and aquatic invertebrate surveys are also presented in **Figure 9.1**. Vantage points for egretry are presented in **Figure 9.4A – 9.4C**.

Avifauna Survey

9.3.12 The avifauna species at various habitats were recorded visually and aurally during wet and dry season. The location of any avifauna species of conservation importance encountered was recorded, along with notable behaviour (e.g. breeding behaviour such as nesting and presence of recently fledged juveniles, roosting, and feeding activities).

9.3.13 A night-time avifauna survey was undertaken once during both wet season and dry season to survey nocturnal avifauna species. Avifauna within the assessment area were surveyed quantitatively using the transect survey method. The survey included detection of calling birds and direct observation of individuals, as appropriate. Ornithological nomenclature follows Carey *et al.* (2001), Viney *et al.* (2005) and the most recent updated list from The Hong Kong Bird Watching Society (HKBWS).The breeding ardeid flight path survey was undertaken at the San Sang San Tsuen egretry during the breeding season to investigate the status and nesting activities of the egretry. One survey in July 2012 was conducted for the new egretry discovered at San Sang San Tsuen during the course of a reconnaissance survey in July 2012. Additional ardeid flight path surveys were undertaken at San Sang San Tsuen in May 2016 and June 2016. The location of the egretry was mapped and the physical environment including identification of trees supporting nests, their size, and crown spread, and evidence and severity of human disturbance within or in close proximity to the egretry. The distribution and abundance of nests from all ardeid species was identified from ground level using binoculars. The nest status (vacant or occupied) and presence of adult/juveniles were also recorded.

9.3.14 Flight line surveys were conducted half an hour before dawn and lasted for about two hours from at least one vantage point (refer to **Figure 9.4A – 9.4C** for egretry location and vantage points). The flight line of ardeid individuals were followed by means of high power binoculars to identify their landing locations and potential foraging grounds, wherever possible. If the ardeids flew out of sight prior to landing, the location at which they were lost from sight was recorded. Flight paths of ardeids were plotted on a base map, and the approximate flight height was estimated and recorded.

Mammal Survey

9.3.15 Terrestrial mammal surveys were conducted in areas which may be potentially impacted by the Project and which could be potentially utilised by terrestrial mammals. The surveys focused on searching for field signs such as droppings, footprints, diggings or burrows left by larger terrestrial mammals in wet and dry seasons. Mammal identification was made as accurately as possible from the field signs encountered. In addition, any mammal directly observed was identified. Nomenclature of mammals follows Shek (2006). Flying mammals were also taken into account during the mammal survey through the use of bat detectors.

Herpetofauna Survey

9.3.16 Herpetofauna surveys were undertaken by transect count method quantitatively. Direct observation and active searching were conducted during day and night time. Diversity, relative abundance and habitat of the occurrence of herpetofauna were recorded. All reptiles and amphibians sighted were counted and recorded. Nomenclature of amphibian and reptile follow Chan *et al.* (2005) and Karsen *et al.* (1998), respectively.

9.3.17 Records of calling amphibians formed the bulk of the data collected, but this was also supplemented when possible by visual observation of eggs, tadpoles and adult frogs and toads. Amphibian survey focused on areas suitable for amphibians (e.g. forest, shrubland, watercourses).

9.3.18 During reptile surveys, careful searches of appropriate microhabitats and refugia (e.g. stones, crevices, ponds, leaf litter/debris, rotten log) were undertaken. All reptiles observed were identified. In addition to active searching, observation of exposed, basking or foraging reptiles was also recorded, as appropriate.

Odonate and Butterfly Survey

9.3.19 Butterflies and odonates within the assessment area were surveyed visually by transect count method during the dry and wet seasons. Diversity, relative abundance and habitat of the occurrence of butterflies and dragonflies were recorded. Nomenclature of butterfly follows Lo (2005) and nomenclature of odonate follows Tam *et al.* (2011).

Freshwater Communities Survey

9.3.20 Watercourses, storm drains and drainage lines which carry water only under heavy rainfall (e.g. contour lines which form part of the catchment but are not incised or do not hold water after a storm event) had been identified during the desktop study. However, freshwater surveys were conducted by active searching and direct observation only at permanently flowing streams within the assessment area during the dry and wet seasons. Representative locations in natural and modified watercourses which are accessible and would be directly impacted by the Project were chosen for surveys (see **Figure 9.1**), where a combination of kick sampling, turning over boulders and other means of active searching for stream organisms was undertaken, as appropriate to the conditions. Organisms encountered were recorded and identified to the lowest possible taxon level. Nomenclature of fish and assessment of geographic distribution status of the species in Hong Kong follow Lee *et al*. (2004).

Intertidal Survey

9.3.21 Surveys on intertidal communities were conducted twice (once in wet season and once in dry season of 2012) at representative survey locations within the assessment area (**Figure 9.1**). At each survey location, a qualitative walk-through survey was conducted to identify the intertidal flora and fauna (e.g. Horseshoe Crab and seagrass) present and their occurrence in the survey location. This walk-through survey helped assess whether the sampling exercise in the quantitative survey collected representative data (e.g. the number and type of species encountered) and whether the sampling effort was deemed adequate.

9.3.22 After the walk-through survey, quantitative survey was conducted using line transect method to establish the ecological profile on the intertidal habitats. One line transect was deployed at each survey location. Transects of a maximum length of 100 m were laid perpendicular to shoreline from high water mark down to low water mark during the low tide period (tide level below 1 m). Along each transect, standard ecological sampling quadrat (dimensions 0.5 m x 0.5 m) was laid at 1 m intervals (or other suitable quadrat dimension and interval distance depending on the field situation). Intertidal epifauna and

flora within each quadrat were identified and enumerated. In general, mobile fauna were counted in terms of abundance per unit area. Sessile organisms such as barnacles, oysters and algae were estimated in terms of percentage cover per fixed area. Intertidal fauna were identified to the lowest practicable taxon.

9.3.23 At survey locations in mudflat and sandy shore, a core sample using a 1,000 cm^3 hand core sampler was also taken from each water level (i.e. low tide, mid tide and high tide) and collected infauna were sieved, counted and identified to the lowest taxonomic level as far as possible.

Evaluation of Conservation Importance/Value and Impact Significance

Conservation Importance/Value

9.3.24 Impact significance depends on the magnitude and scale of an impact, and the conservation importance/value of the species or habitat(s) likely to be affected. Evaluation of ecological importance of the species and habitat(s) associated with the assessment area was based on the criteria outlined in Tables 2 and 3 of the EIAO-TM Annex 8.

9.3.25 Table 2 of Annex 8 (EIAO-TM) specifies criteria for evaluating the importance of a site/habitat. This includes: naturalness; size; diversity; rarity; re-creatability; fragmentation; ecological linkage; potential value; nursery/breeding ground; age; and abundance/richness of wildlife. These criteria have been considered when evaluating the importance of habitats recorded within the assessment area.

9.3.26 Table 3 of EIAO-TM Annex 8 specifies three criteria by which species conservation importance may be measured:

- Protection status (local – Hong Kong; Mainland Chinese; or international), making special reference to legally protected species and/or those under international conventions for conservation;
- Geographical distribution, with higher conservation value afforded to species with more restricted geographical ranges (locally or regionally). More weight shall be given to species which are endemic to Hong Kong or South China; and
- Rarity, with higher conservation value afforded to species which are internationally "rare" than to species which are only regionally or locally "rare".

9.3.27 For the purpose of this ecological assessment, species protection status is based on local and People's Republic of China legislation, standards and guidelines, outlined in **Section 9.2** above.

9.3.28 In terms of assessing geographical distribution, published references specialising in the distribution of specific faunal groups in Hong Kong have been utilised. For general distribution status of the species in Hong Kong, references are made to Fellowes *et al*. (2002) and the Hong Kong Biodiversity Database (AFCD, 2015). For the purpose of this ecological assessment, species which were identified as having a "rare" distribution were assessed as being of conservation importance.

9.3.29 The assessment of faunal restrictedness and rarity has been based on the assessment by Fellowes *et al*. (2002). This paper examines the local (Hong Kong), regional (South China) and global restrictedness of native fauna species occurring in a wild state in Hong Kong, combined with an assessment of the vulnerability of populations, using the most reliable and up to date information available at that time, and assigns a rating to each species accordingly. The following ratings are assessed: Local Concern (LC), Regional Concern (RC) and Global Concern (GC).

9.3.30 Some species are considered to be relatively secure in Hong Kong at present, but since the Hong Kong population is of regional or global importance, all Hong Kong localities are of potential regional and global importance, respectively. These species were therefore assigned as either Potential Regional Concern (PRC) or Potential Global Concern (PGC).

9.3.31 The Fellowes *et al*. (2002) assessment also considers restrictedness in nesting and/or roosting sites. For the purpose of this ecological impact assessment, species falling into this category have only been assessed as of having conservation importance, if they were recorded nesting/roosting within the assessment area during the surveys.

Impact Significance

9.3.32 The potential ecological impacts arising from the Project were assessed following the criteria outlined in Table 1 of the EIAO TM Annex 8. This included:

- Habitat Quality – particular emphasis was given to the evaluation of habitats;
- Species – particular emphasis was given to the evaluation of species importance;
- Size / abundance – impacts of greater significance generally result where a larger habitat area or greater species numbers are affected;
- Duration – short-term and long-term impacts;
- Reversibility – consideration of temporary reversible and permanent irreversible impacts; and
- Magnitude – consideration of the magnitude and scale of the Project element.

9.3.33 This also included consideration of direct, indirect, secondary, cumulative, adverse and beneficial impacts of the proposed development on the ecological resources.

Survey Limitations

9.3.34 Some parts of the assessment area (e.g. privately owned land, active construction sites, and fenced land) were inaccessible during the surveys. Access to the woodland/shrubland/grassland/uphill natural watercourse habitats in the west of the assessment area was also limited. In order to survey the inaccessible areas, flora and fauna species in these habitats were recorded with the use of binoculars where possible.

9.3.35 In addition, as the intertidal survey location P4a was not accessible during the wet season in January 2012, intertidal survey was conducted at a new survey location P4b. These limitations did not affect the survey objectives from being achieved.

9.4 Baseline Conditions

Literature Review

Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.4.1 "Coastal Protection Area" ("CPA") and "Conservation Area" ("CA") are recorded outside the Project area within the assessment area. Three egretries (Ngau Hom Shek egretry, Shenzhen Bay Bridge egretry and San Sang San Tsuen egretry) have also been recorded within the assessment area; however, only San Sang San Tsuen egretry is located within the Project area (refer to **Figures 9.1**). Sha Chau and Lung Kwu Chau Marine Park can be found within the aquatic ecology assessment area.

Coastal Protection Area

9.4.2 A "CPA", covering 20.52 ha and designated under the Ha Tsuen Outline Zoning Plan (OZP) (Statutory Plan No. S/YL-HT/10), is located outside the northern Project area along Deep Bay. Deep Bay is a shallow bay with a deeply intended coastline of around 17 km located east of the Pearl River Delta. It is fed by the Shenzhen River, Yuen Long River and Dasha River, and is heavily influenced by the discharge of the Pearl River estuary. The substrate of Outer Deep Bay is predominantly sand, while that of inner Deep Bay is almost exclusively mud (EPD, 2009). The coastline of the "CPA" comprises mainly mudflat and mangrove while the inland part of the "CPA" comprises mainly village/orchard and developed area, as well as small patches of ponds, agricultural area and plantation.

Conservation Area

9.4.3 A "CA", covering approximately 183 ha and designated under the Ha Tsuen OZP, Ping Shan OZP, and Lam Tei and Yick Yuen OZP (Statutory Plan Nos. S/YL-HT/10, S/YL-PS/16, S/TM-LTYY/8 and S/YL-TYST/10), is located west of the Project area on the eastern slope of Yuen Tau Shan and southeast of Yuen Long Highway. The majority of the "CA", except for a small area adjacent to the Kong Sham Western Highway (KSWH) falls outside the Project area. Outside the Project area, the "CA" comprises mainly shrubland and grassland, as well as some natural watercourse and small patches of village/orchard and plantation. While inside the Project area, the small area of "CA" comprises plantation and developed area (i.e. roads and reinforced concrete flood storage pond with limited plant species recorded).

Egretries

9.4.4 Three egretries occur within the assessment area (Ngau Hom Shek egretry, Shenzhen Bay Bridge egretry and San Sang San Tsuen egretry); however, only San Sang San Tsuen egretry is located within the Project area (**Figure 9.1**). Historical information of the egretries is also provided below, while the breeding ardeid flight path surveys were conducted on 8 June 2012, 6 July 2012 and 10 July 2012; and survey results are presented in **Appendix 9.5** and **Figure 9.4A**. Additional ardeid flight path surveys were undertaken at San Sang San Tsuen egretry in May 2016 and June 2016, and survey results are presented in **Appendix 9.5** and **Figure 9.4B – 9.4C**.

9.4.5 Ngau Hom Shek egretry is located in an open storage site at Ngau Hom Shek adjacent to Deep Bay Road and approximately 300 m outside the northern Project area (**Figure 9.3B** refers). It is built on bamboo clumps and has supported Chinese Pond Heron (*Ardeola bacchus*) and Little Egret (*Egretta garzetta*). The Ngau Hom Shek egretry was first recorded in 2000 and the total number of nests recorded at the site over the years is presented in **Table 9.2**. The site was abandoned between 2005 and 2007 and in 2011 and 2013. It was recorded to have re-established in 2014. Some of the nesting ardeids from the Ngau Hom Shek egretry might have relocated to Ngau Hom Sha Egretry, which is outside the Project area and located 1.5 km south-east to the Ngau Hom Shek egretry, during its years of abandonment.

Table 9.2 Number of Nests Recorded from Ngau Hom Shek Egretry during the Waterbird Monitoring Programme from 2000-2015

Year	Number of Nests of Little Egret	Number of Nests of Chinese Pond Heron	Total Number of Nests Recorded
2000[1]	11	4	15
2001[2]	1	6	7
2002[3]	5	8	13
2003[4]	5	8	13
2004[5]	1	1	2
2005[6]	0	0	0
2006[7]	0	0	0
2007[8]	0	0	0
2008[9]	4	5	9
2009[10]	5	15	20
2010[11]	8	10	18
2011[12]	1	2	3
2012[13]	0	1	1
2013[14]	0	0	0
2014[15]	2	2	4
2015[16]	2	6	8

Note:

[1]Kwok *et al.* (2000); [2]Wong and Kwok (2001); [3]Wong (2002); [4] Wong and Woo (2003); [5] Anon (2004a); [6] Anon (2005a); [7] Anon (2006a); [8] Anon (2007a); [9] Anon (2008a); [10] Anon (2009a); [11] Anon (2010a); [12] Anon (2012a); [13] Anon (2012b); [14] Anon (2013); [15]Anon (2014); [16]Anon (2015)

9.4.6 San Sang San Tsuen egretry is located at the edge of San Sang San Tsuen near KSWH (**Figure 9.3D**) within the Project area. It was built on two bamboo clumps standing less than 10 m from each other and has supported Chinese Pond Heron (*Ardeola bacchus*) and Little Egret (*Egretta garzetta*). It was first recorded in 2012 (Anon, 2012b) and the total number of nests recorded increased from seven to 15 between 2012 and 2013 (Anon, 2012b, 2013); however, the nest recorded decreased to four in 2015 (Anon, 2014) (**Table 9.3**).

Table 9.3 Number of Nests Recorded from San Sang San Tsuen Egretry during the Waterbird Monitoring Programme from 2012-2015

Year	Number of Nests of Little Egret	Number of Nests of Chinese Pond Heron	Total Number of Nests Recorded
2012[1]	4	3	7
2013[2]	8	7	15
2014[3]	4	2	6
2015[4]	3	1	4

Note:

[1] Anon (2012b); [2]Anon (2013); [3]Anon (2014); [4]Anon (2015)

9.4.7 Shenzhen Bay Bridge egretry is located within the assessment area adjacent to Shenzhen Bay Bridge and approximately 220 m outside the northern Project area (**Figure 9.3B** refers). This egretry was discovered in 2014 by HKBWS (Anon, 2014). Bamboo was utilised by two ardeid species, Chinese Pond Heron and Little Egret, as nesting substrate. This egretry remained size increased to 30 nests in 2015 (Anon, 2015) (**Table 9.4**).

Table 9.4 Number of Nests Recorded from Shenzhen Bay Bridge Egretry during the Waterbird Monitoring Programme from 2012-2014

Year	Number of Nests of Little Egret	Number of Nests of Chinese Pond Heron	Total Number of Nests Recorded
2014[1]	10	9	19
2015[2]	22	8	30

Note:
[1]Anon (2014); [2]Anon (2015)

Sha Chau and Lung Kwu Chau Marine Park

9.4.8 Sha Chau and Lung Kwu Chau Marine Park, covering a total sea area of about 1,200 ha, is located outside the Project area at the western edge of the North Western WCZ. The marine environment of the Marine Park is greatly influenced by the Pearl River freshwater run-off, which is signified by high organic and sediment loading. Marine organisms found in this region are highly adapted to this marine environment of low salinity and high turbidity (AFCD, 2015c). This Marine Park supports rich fisheries resources and serves as an important feeding ground for the Indo-Pacific Hump-backed Dolphin, locally known as Chinese White Dolphin (*Sousa chinensis*). The majority of dolphin sightings have been on the eastern coast, the northwest and southwestern tips of Lung Kwu Chau, as well as near the northeast and southwest coast of Sha Chau. In 2000, 5,580 m^3 of artificial reef was deployed within the Marine Park under the Hong Kong Artificial Reef Project to enhancement of marine resources and provide feeding stations for Chinese White Dolphin (HKARP, n.d.).

Habitats and Vegetation

9.4.9 A mosaic of habitats including woodland, shrubland, natural watercourse, plantation, wet and dry agricultural area, fishponds and mitigation ponds, modified watercourse, stormwater drain/recreational pond, grassland, village/orchard and developed area/wasteland, has been previously recorded within the Project area. Marsh, mudflat/coastal waterbody and mangrove have also been identified outside the Project area within the 500 m assessment area.

Developed Area/ Wasteland

9.4.10 Developed area within the assessment area comprised mainly residential dwellings, open storage areas, light industrial factories and transport infrastructure such as roads and railway/light rail. While, wasteland was mainly dominated by herbaceous and ruderal vegetation and was mainly recorded in heavily disturbed or previously developed areas within the assessment area (PlanD, 2001). Dominant species recorded in the developed area/wasteland included the exotic White Popinac (*Leucaena leucocephala*), Mile-a-minute Weed (*Mikania micrantha*) and Annual Bluegrass (*Ipomoea cairica*), as well as the native Elephant's Ear (*Macaranga tanarius*) and Common Lophantherum (*Lophatherum gracile*). No plant species of conservation importance have been recorded in this habitat (HyD, 2002a; DSD, 2002; CEDD, 2009).

Village/ Orchard

9.4.11 Village/orchard habitat refers to areas occupied by village-type residence, which comprised houses with less than three storeys, interspersed with patches of household planting. Fruit trees such as Wampi (*Clausena lansium*), Longan (*Dimocarpus longan*) and Papaya (*Carica papaya*) were commonly recorded around villages and dwellings within the assessment area. No plant species of conservation importance have been recorded in this habitat (PlanD, 1999; HyD, 2002a, 2002b; CEDD, 2009).

Agricultural Area

9.4.12 Dry and wet agricultural lands were recorded within the assessment area. The wet agricultural lands were mainly recorded at Yick Yuen Tsuen within the Project area (PlanD, 2000; HyD, 2002a). The majority of the dry agricultural lands identified within the assessment area were highly fragmented and associated with village settlement (PlanD, 1999). No species of conservation importance were recorded either in the dry or wet agricultural areas (HyD, 2002a).

Plantation

9.4.13 Plantation was mainly recorded along roadsides, in amenity areas and on the hillside of Yuen Tau Shan in the western part of the assessment area (PlanD, 2000). The habitat lacked structural complexity and was dominated by planted tree species such as the exotic Taiwan Acacia (*Acacia confusa*), Brisbane Box (*Lophostemon confertus*) and Cadaga (*Eucalyptus torelliana*) (PlanD, 2000; HyD, 2002a, 2002b).

Modified Watercourse

9.4.14 Major modified watercourses identified within the assessment area included Tin Shui Wai Main Channel (TSW Main Channel), upstream section of Hang Hau Tsuen Stream and Fung Kong Tsuen Channel (PlanD, 2001). The modified channels were mostly void of vegetation with a few records of the exotic Water Hyacinth (*Eichhornia crassipes*) on the surface and Mile-a-minute along the bank. No freshwater aquatic fauna or any species of conservation importance has been recorded in this habitat (PlanD, 2001; CEDD, 2009).

Natural Watercourse

9.4.15 Small natural watercourses were identified on the hillsides outside the western Project area. The upstream of the natural watercourses was predominantly situated outside the assessment area since the Project area lies in lowland areas. Most of the streams were dominated by Bee Shrimp (*Caridina contonensis*) and Guppy (*Poecilia reticulate*) but the downstream areas were relatively impoverished and mainly dominated by pollution tolerant taxa like Mollusca (HyD, 2002a). The freshwater crab (*Cryptopotamon anacoluthon*), which is a species of conservation importance, was recorded in an upstream area of natural watercourse at Ha Tsuen near Ling To Monastery (Watercourse 5) (**Figure 9.3C** refers). The concerned natural watercourse is outside the Project area while its upstream is situated outside the assessment area.

Woodland

9.4.16 Woodlands with low to moderate diversity were identified outside the northern Project area at a knoll behind Ngau Hom, east of Sha Kong Tsuen and along the hillside of Ngau Hom Shek (**Figures 9.3A** and **9.3B** refer) (HyD, 2002a; CEDD, 2009), as well as at the western edge of the assessment area in a small grove behind Ling To Monastery (**Figure 9.3C** refers) (CEDD, 2009). The woodlands recorded were mainly dominated by pioneer tree species such as Chinese Hackberry (*Celtis sinensis*) and plantation species such as Taiwan Acacia (HyD, 2002a; CEDD, 2009). Two species of conservation importance, Incense Tree (*Aquilaria sinensis*) and Black Olive (*Canarium pimela*), had been recorded in the above woodlands. Incense Trees were found in the woodlands behind Ngau Hom, east of Sha Kong Tsuen and behind Ling To Monastery (HyD, 2002a; CEDD, 2009), while Black Olive was recorded in the woodland behind Ling To Monastery. A fung shui wood covering an area of 1.75 ha was also identified at Ngau Hom outside the Project area (Yip *et al.*, 2004) (**Figure 9.3A** refers).

Shrubland

9.4.17 Shrubland was mainly identified on the hillside across the northern Project area around Ngau Hom Shek and west of Sha Kong Tsuen, as well as outside the western Project area on the hillside west of Ha Tsuen (HyD, 2002a; CEDD, 2009). The shrubland areas were dominated by native, small trees and shrubs such as Elephant's Ear, Turn-in-the-wind (*Mallotus paniculatus*), Sumac (*Rhus chinensis)* and Rose Myrtle (*Rhodomyrtus tomentosa*) (HyD, 2002a; CEDD, 2009). No species of conservation importance were recorded.

Grassland

9.4.18 Grasslands within the assessment area were found scattered on hillside and around disturbed areas such as open storage areas and roadsides. The habitat was dominated by a variety of herbs such as India Duck Beak (*Ischaemum indicum*), Dianella (*Dianella ensifolia*) and Angle-stemmed Hedyotis (*Hedyotis acutangula*) (DSD, 2002; HyD, 2002b; CEDD, 2009). Monospecific stands of Linear Forked Fern (*Dicranopteris linearis*) or Bracken Fern (*Pteridium aquilinum*) were also commonly identified in this habitat. A few Cycad-fern (*Brainea insignis*), which is species of conservation importance, were recorded on the hillside near Ngau Hom Shek in the northwestern corner of the Project (HyD, 2002a) (**Figure 9.3B** refers).

Fishponds and Mitigation Ponds

9.4.19 Most fishponds identified within the assessment area were located along the shores of Deep Bay and the foot of a hillside west to Ha Tsuen. No active fishponds were found within the Project area. The pond bunds were dominated by weedy vegetation such as Blunt signal-grass (*Brachiaria mutica*), Common Reedgrass (*Phragmites australis*), Diffuse Day Flower (*Commelina nudiflora*), Mile-a-minute, and Taro (*Colocasia esculenta*) (DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009). No species of conservation importance has been found.

9.4.20 Four mitigation ponds with a total area of 0.5 ha (wetland compensation area) were constructed along KSWH on the western edge of the Project under the DBL project to compensate the loss of moderate value fishponds (**Figures 9.3H** and **Figure 10.2** refer) (ACE, 2002; HyD, 2002a). The mitigation ponds support a variety of fauna with 11 species of conservation importance, including seven avifauna – Grey Heron (*Ardea cinerea*), Chinese Pond Heron (*Ardeola bacchus*), Common Buzzard (*Buteo japonicus*), Greater Coucal (*Centropus sinensis*), Little Egret (*Egretta garzetta*), Black Kite (*Milvus migrans*) and Grey-chinned Minivet (*Pericrocotus solaris*); two dragonfly species – Blue-spotted Dusk-hawker (*Gynacantha japonica*) and Scarlet Basker (*Urothemis signata signata*); and two herpetofauna species – Many-banded Krait (*Bungarus multicinctus*) and Burmese Python (*Python bivittatus*), recorded between April 2010 and January 2016 (AFCD, 2016).

Marsh

9.4.21 Marshes were identified at Ngau Hom Shek adjacent to fishponds and mudflat outside the northern Project area (CEDD, 2009). Small patches of marsh and reed bed isolated by storage areas were also recorded north of Shek Po Tsuen in the southeastern part of Project area (PlanD, 2001). Other marshy areas, mostly abandoned fishponds with weedy vegetation, were also recorded within the assessment area (HyD, 2002b; DSD, 2002). Seasonal and permanent wetlands, including marsh habitat, in the vicinity of the assessment area were estimated to have a low to moderate ecological value (PlanD, 2000).

Mudflats/Coastal Water Body

9.4.22 Part of Deep Bay WCZ are located on the northern fringe of the assessment area and its shoreline mainly comprised mudflats, limited seagrass beds and mangroves. The intertidal mudflats had a moderate to high ecological value as they supported a diverse variety of flora and fauna such as waterfowl, Horseshoe Crab and seagrass (HyD, 2002a, 2002b; DSD, 2002; CEDD, 2009).

9.4.23 All epifauna and infauna recorded were common mudflat species such as mudskippers (*Boleophthalmus pectinirostris* and *Periophthalmus cantonensis*), crabs (*Uca lactea*) and snails (*Cerithidea djadjariensis*). Sightings of juvenile Horseshoe Crabs (*Tachypleus tridentatus* and *Carcinoscorpius rotundicauda)* were reported within the assessment area at Ngau Hom Shek, suggesting that the site could, at one time, serve as a breeding ground for the species (Fong, 1999; CEDD, 2009). Seagrass (*Halophila beccarii*) had been recorded at Sha Kong Tsuen and Ngau Hom Shek (Fong, 1998; AFCD, 2005; DSD, 2008). The mudflats were also identified as foraging ground for avifauna, including species of conservation such as Red-billed Starling (*Sturnus sericeus*), Collared Crow (*Corvus torquatus*) and Black-faced Spoonbill (*Platalea minor*) (HyD, 2002b). Oyster clutches were also recorded at Ngau Hom Shek (HyD, 2002b).

Mangroves

9.4.24 Narrow and discontinuous mangrove belts were identified on the northern fringe of the assessment area. The habitat was dominated by true mangroves, *Kandelia obovata* and Black Mangrove (*Avicennia marina*), with associate species Spiny Bears Breech (*Acanthus ilicifolius*) (DSD, 2002; HyD, 2002a). This habitat is supported a relatively low abundance of avifauna compared to the mudflats and the only avifauna species of conservation importance recorded was Red-billed Starling (HyD, 2002a).

Fauna

Avifauna

9.4.25 The majority of the avifauna species recorded within the assessment area were abundant or common in Hong Kong (DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009). Twenty-six species of conservation importance, 20 of which were wetland-dependent species, were recorded in the vicinity of the assessment area (DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009; AFCD, 2016) (**Table 9.5** refers). The species of conservation importance were recorded around the woodland, abandoned fishponds, around the mitigation ponds constructed under the DBL project (AFCD, 2016), or coastline of Ngau Hom Shek outside the northern Project area (DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009).

9.4.26 Egretries supporting nesting Little Egret and Chinese Pond Heron were found in San Sang San Tsuen in the western part of the Project, Ngau Hom Shek in the northern part of the assessment area. Breeding activities of Great Tit (*Parus major*) were also recorded at upland habitats at Ngau Hom Shek outside the northern Project area (HyD, 2002b).

Table 9.5 Avifauna of Conservation Importance Previously Recorded within the Assessment Area

Common Name[1]	Distribution in Hong Kong[2]	Level of Concern[3]	Protection Status in China[4] [5]	Red List of China's Vertebrates [6]	IUCN Red List[7]
Black Kite[8][9]	Common	(RC)	Class II	Least Concern	-
Black-faced Spoonbill[9]	Common	PGC	Class II Endangered	Endangered	Endangered
Bar-tailed Godwit[9]	Common	LC	-	Near Threatened	-
Chinese Pond Heron[9]	Common	PRC (RC)	-	Least Concern	-
Collared Crow[9]	Uncommon	LC	-	Near Threatened	Near Threatened
Common Redshank[9]	Common	RC	-	Least Concern	-
Crested Goshawk[8]	Uncommon	-	Class II Rare	Near Threatened	-
Crested Serpent Eagle[8]	Uncommon	(LC)	Class II Vulnerable	Near Threatened	-
Eastern Buzzard[8][9]	Common	-	Class II	-	-
Eurasian Curlew[9]	Abundant	RC	-	Near Threatened	Near Threatened
Great Cormorant[9]	Common	PRC	-	Least Concern	-
Greater Coucal	Common	-	Class II Vulnerable	Least Concern	-
Great Crested Grebe[9]	Common	RC	-	Least Concern	-
Great Egret[9]	Common	PRC (RC)	-	Least Concern	-
Great Knot[9]	Common	LC	-	Vulnerable	-
Grey Heron[9]	Common	PRC	-	Least Concern	-
Grey-chinned Minivet	Common	LC	-	Least Concern	-
Little Egret[9]	Common	PRC (RC)	-	Least Concern	-
Little Grebe[9]	Common	LC	-	Least Concern	-
Peregrine Falcon[8][9]	Scarce	(LC)	Class II Rare	Near Threatened	-
Red-billed Starling[9]	Common	GC	-	Least Concern	-
Western Osprey[8][9]	Common	RC	Rare	Near Threatened	-
White-cheeked Starling	Common	PRC	-	Least Concern	-

Common Name[1]	Distribution in Hong Kong[2]	Level of Concern[3]	Protection Status in China[4] [5]	Red List of China's Vertebrates [6]	IUCN Red List[7]
Wood Sandpiper[9]	Common	LC	-	Least Concern	-
Yellow Bittern[9]	Uncommon	(LC)	-	Least Concern	-
Zitting Cisticola	Common	LC	-	Least Concern	-

Note:
(1) All wild birds are protected under Wild Animals Protection Ordinance (Cap. 170).
(2) AFCD (2015a). Hong Kong Biodiversity Database.
(3) Fellowes *et al*. (2002): GC=Global Concern; LC=Local Concern; RC=Regional Concern; PRC=Potential Regional Concern; PGC: Potential Global Concern. Letters in parentheses indicate that the assessment is on the basis of restrictedness in nesting and/or roosting sites rather than in general occurrence.
(4) List of Wild Animals under State Protection (promulgated by State Forestry Administration and Ministry of Agriculture on 14 January, 1989).
(5) China Red Data Book (Zheng & Wang 1998).
(6) Red List of China's Vertebrates (Jiang *et al.* 2016)
(7) IUCN (2016).
(8) Protected under Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586).
(9) Wetland-dependent species (including wetland-dependent species and waterbirds).

Mammals

9.4.27 Ten small mammal species, six of which are of conservation importance, were recorded within the assessment area in previous field surveys (HyD, 2002a, 2002b; CEDD, 2009). Evidence of rooting activity by Eurasian Wild Pig (*Sus scrofa*) and burrows that were likely to be belong to Small-toothed Ferret Badger (*Melogale moschata*) or Chinese Pangolin (*Manis pentadactyla*) were also observed outside the Project area at the hillside of Ngau Hom Shek and near Ling To Monetary, respectively (HyD, 2002a). **Table 9.6** presents the protection status and habitat recorded of the mammals of conservation importance previously recorded within the assessment area.

Table 9.6 Terrestrial Mammals of Conservation Importance Previously Recorded within the Assessment Area

Common Name	Distribution in Hong Kong[3]	Level of Concern[4]	Protection Status in China[5] [6]	Red List of China's Vertebrates [7]	IUCN Red List[8]	Habitat Recorded
Chinese Pangolin[1][2]	Rare	RC	Class II Vulnerable	Critically Endangered	Critically Endangered	Burrows recorded on lower slopes of vegetated hillside
Greater Bandicoot Rat	Rare	LC	-	Least Concern	-	Abandoned cultivated lands
Japanese Pipistrelle[1]	Very Common	-	-	Least Concern	-	Urbanised area, abandoned cultivated lands and Plantation
Short-nosed Fruit Bat[1]	Abundant	-	Indeterminate	Vulnerable	-	Roosting in woodland
Small Asian Mongoose[1]	Uncommon	-	-	Vulnerable	-	Vegetation besides stream
Small-toothed Ferret Badger[1]	Common	-	-	Near Threatened	-	Burrows recorded on lower slopes of vegetated hillside

Notes:
(1) Protected under Wild Animals Protection Ordinance (Cap. 170).
(2) Protected under Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586).
(3) AFCD (2015a). Hong Kong Biodiversity Database.
(4) Fellowes *et al*. (2002): RC=Regional Concern; LC=Local Concern.
(5) List of Wild Animals under State Protection (promulgated by State Forestry Administration and Ministry of Agriculture on 14 January, 1989).
(6) China Red Data Book (Zheng & Wang 1998).
(7) Red List of China's Vertebrates (Jiang *et al.* 2016)
(8) IUCN (2016).

Herpetofauna

9.4.28 A total of eight amphibian and 12 reptile species were recorded within the assessment area in previous field studies, of which four reptiles and two amphibians were considered to be species of conservation importance (PlanD, 2000; DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009; AFCD, 2016). **Table 9.7** presents the protection status and habitat recorded of the herpetofauna of conservation importance found within the assessment area. All other herpetofauna recorded from the assessment area in previous studies are common and widespread species in Hong Kong (Karsen *et al*., 1998).

Table 9.7 Herpetofauna of Conservation Importance Previously Recorded within the Assessment Area

Common Name	Distribution in Hong Kong[3]	Level of Concern[4]	Protection Status in China[5] [6]	Red List of China's Vertebrates [7]	IUCN Red List[8]	Habitat Recorded
Many-Banded Krait	Widely distributed in New Territories, Hong Kong Island and Lantau Island	PRC	Vulnerable	Endangered	-	Mitigation Ponds constructed under DBL project
Burmese Python[1][2]	Widely distributed	PRC	Class I Critically Endangered	Critically Endangered	Vulnerable	Mitigation Ponds constructed under DBL project
Chinese Bullfrog	Widely distributed in Lantau Island and New Territories	PRC	Class II	Endangered	-	Fishpond bund at Ngau Hom Shek
Common Rat Snake[2]	Widely distributed	PRC	Endangered	Endangered	-	Lowland at Ngau Hom Shek
Indo-Chinese Rat Snake	Widely distributed	PRC	Endangered	Vulnerable	-	Lowland at Ngau Hom Shek
Two-striped Grass Frog	Distributed in Sai Kung, Tai Lam Country Park, Pat Sin Leng Country Park, Tai A Chau	LC	-	Near Threatened	-	Abandoned fishpond near Lowland at Ngau Hom Shek

Notes:
(1) Protected under Wild Animals Protection Ordinance (Cap. 170).
(2) Protected under Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586).
(3) AFCD (2015a). Hong Kong Biodiversity Database.
(4) Fellowes *et al*. (2002): RC=Regional Concern; LC=Local Concern.
(5) List of Wild Animals under State Protection (promulgated by State Forestry Administration and Ministry of Agriculture on 14 January, 1989).
(6) China Red Data Book (Zheng & Wang 1998).
(7) Red List of China's Vertebrates (Jiang *et al.* 2016)
(8) IUCN (2016).

Butterfly

9.4.29 A total of 37 species of butterflies were recorded in the vicinity of the assessment area in the previous field surveys (PlanD, 2000; DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009; AFCD, 2016). All species recorded are common and widespread in Hong Kong. No species of conservation interest were recorded.

Odonata

9.4.30 A total of 33 odonata species (dragonflies and damselflies), four of which are of conservation importance, were recorded in the vicinity of assessment area in the previous field surveys (PlanD, 2000; DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009; AFCD, 2016). **Table 9.8** presents the protection status and habitat recorded of the odonata of conservation importance found within the assessment area. All other odonata recorded from the assessment area in previous studies are common and widespread species in Hong Kong. The majority of the odonata were found around clean streams and well vegetated abandoned ponds.

Table 9.8 Odonata of Conservation Importance Previously Recorded within the Assessment Area

Common Name	Distribution in Hong Kong(1)	Level of Concern(2)	Protection Status in China(3)	IUCN Red List(4)	Habitat Recorded
Blue-spotted Dusk-hawker	Common	LC	-	-	Mitigation Ponds constructed under DBL project
Giant Hooktail	Common	LC	-	-	Western Ha Tsuen
Indochinese Copperwing	Common	LC	-	-	Stream habitat west of Ha Tsuen
Scarlet Basker	Common	LC	-	-	Mitigation Ponds constructed under DBL project

Notes:
(1) AFCD (2015a). Hong Kong Biodiversity Database.
(2) Fellowes *et al*. (2002): LC=Local Concern.
(3) List of Wild Animals under State Protection (promulgated by State Forestry Administration and Ministry of Agriculture on 14 January, 1989).
(4) IUCN (2016).

Freshwater Fauna

9.4.31 Twenty-six freshwater species were recorded around Ngau Hom Shek and Ling To Monastery during the field surveys for the DBL EIA Study (HyD, 2002a). Likewise, seven freshwater fish species had been recorded from the mitigation ponds constructed under the DBL project (AFCD, 2016). All freshwater fauna recorded within the assessment area were common and widespread in Hong Kong (DSD, 2002; HyD, 2002a, 2002b), except for the freshwater crab, *Cryptopotamon anacoluthon*, which is a species of conservation importance (HyD, 2002a). Guppy and Bee Shrimp were dominant in natural watercourses while Tilapia (*Oreochromis* spp.) was the major freshwater fauna found in modified watercourses and abandoned fishponds (HyD, 2002a, 2002b). **Table 9.9** shows freshwater fauna of conservation importance previously recorded within the assessment area.

Table 9.9 Freshwater Fauna of Conservation Importance Previously Recorded within the Assessment Area

Scientific Name	Distribution in Hong Kong[1]	Level of Concern[2]	Protection Status in China	IUCN Red List[3]	Habitat Recorded
Cryptopotamon anacoluthon	-	PGC	-	Vulnerable	Upstream of the watercourses near Ling To Monastery

Notes:
(1) AFCD (2015a). Hong Kong Biodiversity Database.
(2) Fellowes *et al*. (2002): PGC=Potential Global Concern.
(3) IUCN (2016).

Intertidal Fauna

9.4.32 Most of the epifauna and infauna recorded on mudflats, mangrove stands and seagrass beds within the assessment area were common and widespread in Hong Kong (PlanD, 2000; DSD, 2002; HyD, 2002a, 2002b; CEDD, 2009), except for the juvenile Horseshoe Crab *Tachypleus tridentatus* and *Carcinoscorpius rotundicauda* (PlanD, 2000; HyD, 2002a, 2002b). Their protection status is presented in **Table 9.10**. No other species of conservation importance were recorded. Common epifauna recorded include mudskippers *(Boleophthalmus pectinirostris* and *Periophthalmus cantonensis*), crabs (*Uca lactea*) and snails (*Cerithidea* spp., *Clithon* spp. and *Nerita* spp.) and the overall abundance of epifauna recorded during dry season was higher than that during wet season. Polychaetes (such as *Heteromastus filiformis* and *Sigambra robusta*) were the dominant infauna recorded and the overall abundance of infauna recorded was similar during both wet and dry season (HyD, 2002a, 2002b).

Table 9.10 Intertidal Fauna of Conservation Importance Previously Recorded within the Assessment Area

Scientific Name	Distribution in Hong Kong	IUCN Red List[2]	Habitat Recorded
Tachypleus tridentatus	Rare[1]	Data Deficient	Juvenile sightings reported offshore of Ngau Hom Shek
Carcinoscorpius rotundicauda	-	Data Deficient	Juvenile sightings reported offshore of Ngau Hom Shek

Notes:
(1) AFCD (2015a). Hong Kong Biodiversity Database.
(2) IUCN (2016).

Survey Findings

Habitat and Vegetation

9.4.33 A total of 15 habitats, namely developed area/wasteland, village/orchard, agricultural area (dry), agricultural area (wet), storm water drain/recreational pond, natural watercourse, modified watercourse, plantation, grassland, marsh, shrubland, woodland, fishponds and mitigation ponds, mangrove, and mudflat/coastal water body, were identified within the assessment area. Habitat maps and representative photographs of habitats recorded within assessment area are shown in **Figures 9.3A** to **9.3N** and **Appendix 9.1**, respectively.

9.4.34 **Appendix 9.2** presents the flora recorded during the surveys from October 2011 to September 2012, in September 2014, January 2015, December 2015 and January 2016. The habitats identified within the assessment area are described below. **Table 9.11** summarises the size and percentage coverage of each habitat type within the assessment area. A total of four flora species of conservation importance were recorded. The indicative locations and representative photographs of the species of conservation importance are presented in **Figure 9.3A** to **9.3N** and **Appendix 9.3**.

Table 9.11 Estimated Size of the Habitat Types within the Assessment Area

Habitat Type	Within Project Area		Within 500 m Assessment Area		
	Total Habitat Area (ha)	Length (km)	Total Habitat Area (ha)	Length (km)	% of Total Area
Developed Area/Wasteland	522.62	-	960.14	-	60.1
Village/Orchard	72.55	-	107.68	-	6.8
Agricultural Area (Dry)	6.55	-	21.20	-	1.3
Agricultural Area (Wet)	2.20	-	2.83	-	0.2
Plantation	38.59	-	103.92	-	6.5
Woodland	10.34	-	41.31	-	2.6
Shrubland	20.57	-	197.57	-	12.4
Grassland	17.69	-	77.40	-	4.9
Marsh	0.85	-	1.75	-	0.1
Fishponds and Mitigation Ponds	0.96	-	4.39	-	0.3
Storm Water Drain/Recreational Pond	0.92	-	2.61	-	0.2
Natural Watercourse	-	0.15	-	10.27	-
Modified Watercourse	20.36	15.64	26.28	20.68	1.6
Mangrove	-	-	11.90	-	0.8
Mudflat/Coastal Water Body	-	-	34.70	-	2.2
Total	**714.20**	**15.79**	**1593.68**	**30.95**	**100.0**

Developed Area/Wasteland

9.4.35 Developed area/wasteland was the dominant habitat within the assessment area. This habitat consisted of residential buildings, light industry, multi-storey village housing, open storage, recreational parks and transport infrastructure. Vegetation recorded was mainly roadside trees and horticultural species in landscaped areas or recreational parks. The dominant species typically included planted tree species, Taiwan Acacia, Ear-leaved Acacia (*Acacia auriculiformis*) and Flame Tree (*Delonix regia*), and native tree species, Hong Kong Orchid Tree (*Bauhinia x blakeana*) and Elephant's Ear.

9.4.36 The vegetation associated with light industry, open storage and associated wasteland was dominated by weeds and herbs including Gairo Morning Glory (*Ipomoea cairica*), *Bidens alba* and Lantana (*Lantana camara*), as well as trees such as Elephant's Ear and White Popinac. Vegetation mainly grew on the edge of or scattered across these areas. The developed area/wasteland habitat was highly disturbed due to anthropogenic activities (e.g. traffic, noise and construction activities). Two individuals of Ailanthus (*Ailanthus fordii*) were planted at the roadside plantation area at Castle Peak Road – Lam Tei (refer to **Figure 9.3J**). Wild Ailanthus was listed under Forests and Countryside Ordinance Cap. 96. However, as both individuals were artificially introduced to the habitat, they were not considered as plant species of conservation importance.

Village/Orchard

9.4.37 This habitat was composed of village housing interspersed with patches of orchard supporting ornamental plants, crops and/or fruit trees. The majority of village/orchard habitat was concentrated in large areas in the southwest of the assessment area, associated with San Sang San Tsuen, Tin Sam Tsuen and Yick Yuen Tsuen; while others were relatively smaller and scattered across the assessment area.

9.4.38 The fruit trees planted within the gardens or orchards included Wampi, Longan and Lychee (*Litchi chinensis*). Ornamental species including Kwai-fah (*Osmanthus fragrans*), Brazil Bougainvillea (*Bougainvillea spectabilis*) and Chinese Banyan (*Ficus microcarpa*) were also commonly planted in the gardens and the common area of each village. The habitat was exposed to moderate levels of anthropogenic disturbance (e.g. noise, visual disturbance). The structural complexity of the vegetation was simple and no plant species of conservation importance were recorded from this habitat.

Agricultural Area (Dry)

9.4.39 Agricultural area habitat refers to man-made habitat for crop production and is usually associated with village settlement. Rotation cropping was observed within the agricultural areas.

9.4.40 The largest patch of dry agricultural area was present at San Sang San Tsuen, west of San Sang San Tsuen and Fui Sha Wai, while the remainder was scattered throughout the assessment area and was highly fragmented (**Figure 9.3I** and **9.3M** refers). Cabbage (*Brassica oleracea*), Lettuce (*Lactuca sativa*), and Chinese White Cabbage (*Brassica chinensis*), were commonly cultivated while fruit trees such as Papaya and Banana (*Musa* x *paradisiaca*) were grown along the field bunds. This habitat was subject to a high level of disturbance (e.g. agricultural activities) and no plant species of conservation importance were recorded from this habitat.

Agricultural Area (Wet)

9.4.41 Wet agriculture contributed only a small proportion of the total agricultural habitat within the assessment area (0.2%). The wet agricultural area habitat was dominated by Water Cress (*Nasturtium officinale*), Water Spinach (*Ipomoea aquatica*) and Taro crops. Weeds, including Gairo Morning Glory and *Bidens alba*, were occasionally found. The largest patch of wet agricultural habitat was located near Tseung Kong Wai and Sik Kong Wai (**Figure 9.3H** refers). Some wet agriculture fields were recorded outside the Project area, including fields near Tsing Chuen Wai, Chung Uk Tsuen and Fui Sha Wai (**Figure 9.3E**, **9.3J** and **9.3M** refers). No plant species of conservation importance were recorded from this habitat.

Plantation

9.4.42 Plantation habitat generally consisted of reforestation on natural hillside area and urban plantation. Reforestation served to restore natural hillside area that was previously degraded by disturbance (e.g. slope works, hill fire or erosion). Plantation species were generally first established in single species stands resulting in relatively simple structure and low diversity. This habitat was young and the structural complexity was simple, with a canopy of 2-5 m in height. Plantation species included exotic tree species, Taiwan Acacia, Ear-leaved Acacia, Brisbane Box and *Eucalyptus* spp., with an understorey of native shrubs and herbs, Chinese Privet (*Ligustrum sinense*), Elephant's Ear, *Wedelia trilobata* and Hawk's Beard (*Youngia japonica*).

9.4.43 Urban plantations were located near the developed area/wasteland habitat (e.g. Shek Po Tsuen at the southern part of the assessment area) and subject to high anthropogenic disturbance. This habitat was young and the structural complexity was simple, with a canopy of 2-4 m in height. Exotic tree species dominated this habitat including Taiwan Acacia and African Tulip Tree (*Spathodea campanulata*); however, native species including Hong Kong Orchid Tree and Chinese Fan-palm (*Livistona chinensis*) were also common. The understorey tended to be sparsely vegetated with shrub, Chinese Privet, herbs, Lantana and *Bidens alba*. The invasive species, White Popinac was widely distributed within this habitat and no plant species of conservation importance were recorded.

Woodland

9.4.44 The majority of woodland was recorded at Ngau Hom Shek (i.e. at northwest of the assessment area). This woodland was relatively mature (trees of 8 - 12 m in height), with moderate floristic diversity. Recorded plants included native trees, Chinese Hackberry, Ivy Tree (*Schefflera heptaphylla*), Lance-leaved Sterculia (*Sterculia lanceolata*) and exotic tree, Taiwan Acacia; shrubs, Opposite-leaved Fig, Wild Coffee (*Psychotria asiatica*) and Aporusa (*Aporusa dioica*), Splash-of-White (*Mussaenda pubescens*) and herb, Giant Alocasia.

9.4.45 The woodland at Tung Tau Tsuen at the southeast of the assessment area was relatively young (trees of 2.5-5 m in height). This habitat was dominated by plantation species, such as Ear-leaved Acacia and Horse Tail (*Casuarina equisetifolia*); however, native Elephant's Ear and Chinese Hackberry were also recorded. Climber and herb species including Chinese Privet, Rusty-haired Raspberry (*Rubus reflexus*) and Giant Alocasia were established in the understorey. One mature Incense Tree of 5 m in height was recorded at the Tung Tau Tsuen woodland within the Project area (refer to **Figure 9.3G** and **Appendix 9.3**).

9.4.46 The woodland west of Fung Kong Tsuen was relatively young (trees of 2.5-4 m in height). This woodland was dominated by native trees, Chinese Hackberry, Ivy Tree, and exotic tree, Taiwan Acacia. Shrubs, Opposite-leaved Fig, and Round-leaved Litsea (*Litsea rotundifolia*) were recorded at the understorey. This habitat was subjected to disturbance from the road nearby.

9.4.47 A fung shui woodland was identified at Ngau Hom (Yip *et al*., 2004), which was outside the northeastern Project area (refer to **Figure 9.3A**). Several individuals of plant species of conservation importance, Incense Tree, was recorded within this fung shui woodland. The species is listed under Protection of Endangered Species of Animals and Plant Ordinance (Cap. 586), "Class II" in List of Wild Plant under State Protection, "Vulnerable" in China Plant Red Data Book and "Vulnerable" in IUCN Red List. Individuals of Luofushan Joint-fir (*Gnetum luofuense*) were recorded at the woodland near Watercourse 1. This species is listed as "Vulnerable" in IUCN Red List.

Shrubland

9.4.48 Shrubland was mainly situated on hillsides within the assessment area; while the majority of the shrubland was located outside the Project area at the west of KSWH and southeast of Yuen Long Highway. The remaining shrubland was scattered at Ngau Hom Shek, west of Kau Lee UK Tsuen and Tseung Kong Wai.

9.4.49 Short shrubland at the uphill area west of KSWH and Ngau Hom Shek was dominated by shrub and herb species, such as Rose Myrtle, Dwarf Mountain Pine (*Baeckea frutescens*), Dichotomy Forked Fern (*Dicranopteris pedata*) and Chinese Silvergrass.

9.4.50 While tall shrubland was present at the low elevation of hillside area (e.g. at Ngau Hom Shek and southeast of Yuen Long Highway). Shrubs recorded at Ngau Hom Shek shrubland including Elephant's Ear, Round-leaved Litsea, and Opposite-leaved Fig, of 2-3 m in height formed a semi-shaded understorey supporting herbaceous species, Dichotomy Forked Fern and Rusty-haired Raspberry. The tall shrubland at southeast of Yuen Long Highway was mainly comprised of Chinese Red Pine (*Pinus massoniana*), Elephant's Ear and Opposite-leaved Fig. These species formed a 3-4 m canopy with occasionally recorded Taiwan Acacia. The understorey was comprised of Chinese Silvergrass and Oriental Blechnum (*Blechnum orientale*). The shrubland at southeast of Yuen Long Highway was located near a road which connected to a construction site, and disturbance (e.g. illegal dumpling of construction rubbles/furniture, vehicle noise) was observed.

9.4.51 Four species of conservation importance, including Lamb of Tartary (*Cibotium barometz*), Luofushan Joint-fir, Incense Tree and Hong Kong Pavetta (*Pavetta hongkongensis*) were recorded from the shrubland habitat outside the Project area. Lamb of Tartary is listed under Protection of Endangered Species of Animals and Plant Ordinance (Cap. 586), "Category II" in Wild Plant under State protection and “Vulnerable” in Rare and Precious Plants of Hong Kong (Hu *et al*., 2003). This species is listed as “Near Threaten” in the IUCN Red List of Endangered Species (IUCN, 2016). Hong Kong Pavetta is protected under Forests and Countryside Ordinance (Cap. 96). These native species are common in Hong Kong.

9.4.52 One individual of Lamb of Tartary and one individual of Luofushan Joint-fir was recorded at the shrubland at the west of San Sang San Tsuen outside the Project area (refer to **Figure 9.3D**). Luofushan Joint-fir was also recorded in the shrubland near Watercourse 1 (refer to **Figure 9.3C**). One mature individual of Incense Tree was recorded at the shrubland north of Tsing Chuen Wai, southeast of Hang Hau Tsuen and west of Kau Lee Uk Tsuen, respectively (refer to **Figure 9.3A**, **9.3E** and **9.3H**). A mature Hong Kong Pavetta was recorded on the hillside near Sha Kong Wai (refer to **Figure 9.3A**).

Grassland

9.4.53 Grassland is typically the first habitat established on bare soil after disturbance. Within the assessment area, grassland was mainly located at the uphill area in the west or scattered near villages. This habitat was dominated by grass or herb species, such as native Many-flowered Silvergrass (*Miscanthus floridulus*), Hilo Grass (*Paspalum conjugatum*), exotic Carpet Grass (*Axonopus compressus*) and Blunt Signal-grass. The grassland habitats in close proximity to villages tended to have higher disturbance (e.g. illegal dumping) compared to those in uphill locations. One plant species of conservation importance, Incense Tree was recorded at the grassland west of Fung Kong Tsuen, but outside the Project area (refer to **Figure 9.3B**). Two mature individuals of Incense Tree were recorded at the grassland southeast of Ngau Hom Shek, within the “Green Belt” (“GB”) zoning that would be retained under the development.

Marsh

9.4.54 Four small areas of marsh were identified within the assessment area. Three of them were located within the Project area, i.e. two at Fung Kong Tsuen (about 0.63 ha) and one at Sik Kong Tsuen (0.23 ha) (**Figure 9.3G** refers), while the remaining one was recorded outside the Project area at Hang Hau Tsuen (0.85 ha) (**Figure 9.3A** refers). Standing water was observed during the surveys. Commonly recorded wetland dependent plant species included Interrupted Tri-vein Fern (*Cyclosorus interruptus*), *Polygonum* spp., Diffuse Day-flower (*Commelina diffusa*), Primrose Willow (*Ludwigia octovalvis*), Spiny Knotweed (*Polygonum perfoliatum*) and Taro.

9.4.55 These marshes were fragmented and isolated. This habitat was subject to anthropogenic activities (e.g. wire fences were constructed at Fung Kong Tsuen marsh bund). The establishment of exotic weedy species (e.g. Mile-a-minute, Chinese Dodder *Cuscuta chinensis*) was recorded at the disturbed area. No plant species of conservation importance were recorded.

Fishponds and Mitigation Ponds

9.4.56 A total of 19 fishponds and mitigation ponds were recorded within the assessment area, three fishponds were recorded entirely within the Project area (**Figure 9.3G** and **9.3I** refer). The three fishponds, which were near San Sang San Tsuen (0.32 ha) and San Wai (0.11 ha and 0.23 ha) were inactive where no fishing operations equipment or maintenance activities were observed. The bunds of these inactive ponds were overgrown with herb species such as Guinea Grass (*Panicum maximum*), Blunt Signal-grass and Diffuse Day-flower. No active fishponds were identified within the Project area.

9.4.57 The cluster of active fishponds located at Ngau Hom Shek (outside the Project area) forms part of the wetland ecosystem of Deep Bay. Fruit trees, including Lychee, Jackfruit (*Artocarpus heterophyllus*) and Common Banana were growing at active fishpond bunds. Herbs such as Common Reedgrass, Mile-a-minute, *Bidens alba* were occasionally recorded.

9.4.58 Four small mitigation ponds (0.5 ha in total) were located adjacent to the KSWH (two were located on the western side, and two were located on the eastern side of the KSWH). These ponds were constructed to compensate the loss of fishponds under the DBL project. The two western ponds were partially located within the Project area. Blunt Signal-grass, Guinea Grass and *Bidens alba* were commonly recorded at the disturbed pond bunds. Limited wetland dependent species such as Umbrella Plant (*Cyperus involucratus*), Short-leaved Kyllinga (*Kyllinga brevifolia*), and Plume Grass (*Pennisetum alopecuroides*) were recorded. No plant species of conservation importance were recorded in the fishponds and mitigation ponds habitat.

Storm Water Drain/Recreational Pond

9.4.59 Storm water drain/recreational pond refers to water flood retention structures or artificial ponds/lake used for recreational purposes other than fishing. The storm water drains were concentrated around Sik Kong Wai, while a recreational pond was located outside the Project area near Sha Kong Wai. The waterbodies were usually fenced and planted with vegetation, such as tree species, Paper-bark Tree (*Melaleuca cajuputi*), Weeping Willow (*Salix babylonica*) or shrubs, Mock Lime (*Aglaia odorata*). Weeds and herbs including *Bidens alba* and *Wedelia trilobata* dominated the ground layer. No plant species of conservation importance were recorded from this habitat.

Natural Watercourse

9.4.60 Most of the natural watercourses, including drainage lines which carry water only under heavy rainfall, are located outside the Project Area. Only four perennial natural watercourses were identified within the assessment area in total (including Watercourse 1, Watercourse 5, part of Watercourse 6 and Watercourse 7). All are located in the west of the assessment area. Two are located to the west of San Sang San Tsuen, on the foot of Yuen Tau Shan hillside (Watercourse 6 and Watercourse 7) (**Figure 9.3D** refers). The watercourse located at the west of Kau Lee Uk Tsuen is partly modified at the upstream section (Watercourse 5) (**Figure 9.3H** refers).

9.4.61 All of the identified natural watercourses had permanent water flow. Watercourse 1, Watercourse 7, the upper sections of Watercourse 5 and Watercourse 6 were located in the hillside of Yuen Tau Shan outside the Project area and relatively undisturbed. They ranged from 0.5 m to 2 m wide and 0.5 m to 1 m deep, with moderate water flow rate. The substrate consisted of sand and rocks. The riparian zone of these natural watercourses comprised of dense vegetation, which was dominated by tree species such as Rose Apple (*Syzygium jambos*), Lance-leaved Sterculia; shrubs Round-leaved Litsea and herb species, Oriental Blechnum and *Polygonum* spp.

9.4.62 The lower sections of Watercourse 5 and Watercourse 6 fall within the Project area (refer to **Appendix 9.1**). The width and water depth of Watercourse 5 was approximately 2 m and 0.2 m to 0.5 m, respectively, with a moderate water flow rate. The width and water depth of Watercourse 6 was 0.5 m - 1 m wide and 0.2 m deep, with a fast water flow rate. These two short sections were located near developed area and were comparatively more disturbed. Signs of human disturbance (e.g. illegal dumping, construction works) were observed during the current surveys. The riparian zone of these two lower sections was sparse with only scattered trees (Elephant's Ear, Turn-in-the-wind) and herb species (*Wedelia trilobata*) recorded.

Modified Watercourse

9.4.63 Modified watercourses include artificial structures for drainage, flood control or irrigation purposes such as channels/nullahs and agricultural ditches (and included Watercourse 2 at Lam Tei, Watercourse 3 at Ha Tsuen, Watercourse 4 at San Sang San Tsuen, part of Watercourse 5 and 6, and a range of modified watercourses throughout the assessment area).

9.4.64 TSW Main Channel, Watercourse 2, Watercourse 3 and modified watercourse at Hang Hau Tsuen were channelised watercourses made of concrete base and banks (**Figure 9.1**). TSW Main Channel is a concrete trapezoidal channel of approximate 50 m in width, with limited aquatic plant species and riparian vegetation. TSW Main Channel mainly flowed through the assessment area from north to south with a moderate flow rate.

9.4.65 Watercourse 2 and Watercourse 3 is located in the centre of the Project area (i.e. south of Nai Wai and north of Ha Tsuen respectively); whilst another modified watercourse at Hang Hau Tsuen partially fell within the Project area. Similar to TSW Main Channel, Watercourse 2, Watercourse 3 and modified watercourse at Hang Hau Tsuen were also modified with concrete bank of approximate 10-30 m in width, with limited riparian zone recorded. Within the above modified watercourses, scattered plants including trees, Taiwan Acacia, Hong Kong Orchid Tree, herbs, *Bidens alba* and Blunt Signal-grass were recorded at the embankment. Human disturbance (i.e. fishing, sewage discharge) was observed.

9.4.66 Modified watercourses under KSWH linked to the natural watercourses which flowed from upper hillside and then through agricultural areas (refer to **Figure 9.3H**). The width of these modified watercourses ranged from 10-12 m, with concrete base and slow to moderate water flow. Blunt Signal-grass, *Polygonum* spp. and Giant Alocasia were recorded commonly within this habitat. Disturbance (e.g. dust, dumping) was observed during the survey.

9.4.67 Ditches, which flowed through village/orchard and agricultural area, were dug mainly to serve irrigation purposes. The ditches (e.g. Watercourse 4 near San Sang Tsuen) were mainly composed of concrete. They were typically narrow and shallow, of less than 0.5 m in width and 0.2 m in depth. The commonly recorded species included herbs, *Imperata cylindrica* var. *major*, Chinese Knotweed (*Polygonum chinense*) and Giant Alocasia. No plant species of conservation importance were recorded within the modified watercourse habitat.

9.4.68 Near San Lee Uk Tsuen, the existing Tin Sam Channel is proposed to be realigned under the proposed development (refer to **Section 5.10.24, Figure 5.2.9** and **Figure 9.3I**). Similar to other minor, modified watercourses, this modified watercourse was made of concrete base and banks of approximately 4.8-5 m in width and with limited vegetation recorded. Plant species included herbs, Guinea Grass, Chinese Knotweed and *Wedelia trilobata* were found scattered within this area.

Mangrove

9.4.69 Mangrove community was recorded along the intertidal mudflats of the Deep Bay shoreline, which is outside the northern Project area. The maximum width of the mangrove stand was approximately 160 m. The dominant species were true mangroves, *Kandelia obovata* and Black Mangrove (*Avicennia marina*), with associate mangrove species, Spiny Bears Breech (*Acanthus ilicifolius*). Scattered individuals of exotic *Sonneratia apetala* and patches of Common Reedgrass were occasionally recorded. The majority of mangrove trees were short, of less than 2 m in height. No plant species of conservation importance were recorded in this habitat.

Mudflat/Coastal Water Body

9.4.70 Intertidal mudflats occurred along the Deep Bay shoreline which is outside the northern Project area. Although the intertidal mudflat habitat in Deep Bay is large in size, only a narrow band of mudflat falls within the assessment area. This habitat was tidally influenced and supported mangrove community such as *Kandelia obovata.* According to the survey on the distribution of seagrasses in Hong Kong conducted by AFCD (AFCD, 2005), *Halophila beccarii* was found at the coast of Sha Kong Tsuen. However, no seagrass was recorded during the recent surveys.

Fauna

9.4.71 The sections below outline the findings of the fauna surveys for the assessment area between October 2011 and September 2012, January 2015, January 2016, May 2016 and June 2016.

Avifauna

9.4.72 A total of 92 avifauna species were recorded across the assessment area. While the avifauna community was dominated by common species, twenty-three species of conservation importance were recorded, most of which are wetland-dependent (16 out of 23 species) (**Appendix 9.4**). Nesting or roosting individuals of Little Egret and Chinese Pond Heron were recorded within the assessment area. Indicative locations of species of conservation importance are presented in **Figure 9.3A** to **9.3N** and their distribution within the assessment area and their status are presented in **Appendix 9.4**.

9.4.73 The highest abundance and diversity of avifauna species were recorded in developed area/wasteland (55 species). Most of the species are common and were also recorded in other habitats. The high richness and abundance in developed area/wasteland is considered a result of the vast area (approximately 60% of total area) and proximity/connectivity to other more valuable habitat types such as watercourse and woodland. A total of 10 species of conservation importance were recorded from developed area/wasteland habitat, including raptors (e.g. Black Kite *Milvus migrans*, Crested Goshawk *Accipiter trivirgatus*, and Besra *Accipiter virgatus*), wetland-dependent birds (e.g. ardeids), and resident species (e.g. Greater Coucal *Centropus sinensis*). Developed area/wasteland also supported San Sang San Tsuen egretry.

9.4.74 Most of the recorded species of conservation importance are wetland-dependent which were typically associated with coastal and inland wetlands, watercourses, fishpond, and stormwater drain/recreational pond. Thirteen species of conservation importance were recorded in the above habitats. Five of which were recorded exclusively in these habitats, such as Black-faced Spoonbill (*Platalea minor*), various waterfowl (e.g. Northern Shoveler, *Anas clypeata*), and waders (e.g. Wood Sandpiper, *Tringa glareola*). Modified watercourses, particularly TSW Main Channel, supported several wetland-dependent species of conservation importance. Little Egret was the most abundant species recorded. Species such as Black-faced Spoonbill and Wood Sandpiper were also recorded foraging in TSW Main Channel during winter; however, they were recorded outside the Project area.

9.4.75 Other important habitats for wetland-dependent species were mudflat/coastal water body and mangrove which are associated with Deep Bay. These habitats are located outside the Project area. Although the area was small (both habitats occupied no more than 3% of total area), they together supported 33 avifauna species, of which 10 species are of conservation importance. All of the species of conservation importance are wetland-dependent, and were observed foraging/resting on mudflats or perching on mangroves. Pied Avocet (*Recurvirostra avosetta*), Eurasian Curlew (*Numenius arquata*), Marsh Sandpiper (*Tringa stagnatilis*), and Common Greenshank (*Tringa nebularia*) were recorded exclusively in these habitats.

9.4.76 Woodland birds were generally recorded in shrubland, plantation and woodland. Among these habitats, numbers of recorded species were similar in which 32, 39 and 38 species were recorded in shrubland, plantation and woodland, respectively. A total of six species of conservation importance were recorded in these habitats: Crested Serpent Eagle (*Accipiter trivirgatus*), Common Emerald Dove (*Chalcophaps indica*), Greater Coucal (*Centropus sinensis*), Rufous-capped Babbler (*Stachyris ruficeps*) and two ardeids (Chinese Pond Heron and Little Egret). Common Emerald Dove was recorded once on the ground amongst dense vegetation, whereas Greater Coucal was commonly seen in the shrub layer. Rufous-capped Babbler was typically recorded in the understory layer of woodland habitat.

9.4.77 Five raptors, including Besra, Crested Goshawk, Black Kite, Eastern Buzzard and Crested Serpent Eagle, were observed perching on trees. Raptors usually nest and roost within a defined territory. Whilst there was no evidence of other raptors holding territory, six and two repeated sightings of Crested Serpent Eagle hovering over woodland/plantation habitat and perching in the trees in developed area/wasteland habitat close to Fung Kong Tsuen in the northern Project area were recorded during the dry season respectively; while similar observations during surveys for the Shenzhen Western Corridor EIA (HyD, 2002b) were made. Crested Serpent Eagle is an uncommon resident and spring and autumn migrant (Carey *et al.,* 2001). This species is known to utilise a range of habitats such as woodland, rocky hilltops, marsh or abandoned wet agricultural area. The repeated records of this species may suggest that the area including developed area, woodland, plantation, marsh at west of Fung Kong Tsuen may form an important part of this species' territory and utilise by the species as foraging or roosting grounds. Other frequent sighting of this species occurred in the developed area/wasteland next to the hillside shrubland southwest of the Project area near San Sang San Tsuen. Several individuals were often recorded perching on trees during dry season where only one sighting was recorded during wet season. The two sites were around 3 km apart and were suspected to be an important habitat for this species.

Egretry

9.4.78 San Sang San Tsuen egretry was discovered during the survey programme in early July 2012. More than 10 nests were recorded at that time. The egretry was located in open storage next to a road within the Project area at the western edge of San Sang San Tsuen, next to KSWH (**Figure 9.3l**). It consisted of two bamboo clumps about 10 m apart. Both adult and juvenile ardeids were present. The breeding ardeids flight path survey undertaken in mid-July 2012 indicated that the egretry supported Chinese Pond Heron and Little Egret (refer to **Figure 9.4A**, **Appendix 9.5**). Almost all of the ardeids were recorded flying northeast of the egretry (refer to **Figure 9.4A** and **Appendix 9.13**). Only a Chinese Pond Heron was observed flying south along KSWH. The landing locations of the ardeids could not be observed from the vantage point; however, wetland habitats such as pond and modified watercourses in which ardeids were recorded during the avifauna survey occurred 300 m - 550 m east of the egretry. Most of the egrets flew at an altitude more than 10 m, which might be a behaviour to fly over nearby structures (e.g. residential dwelling in the village/orchard habitat) in the vicinity (**Appendix 9.5**).

9.4.79 The number of nests dropped from 15 in 2013, to six in 2014, and to four in 2015 (Anon, 2015), which might be attributed to disturbance around the egretry (refer to **Appendix 9.12**). During the site visit in January 2015, it was observed that one of the two bamboo clumps comprising the egretry had been removed. Additional breeding ardeids flight path surveys were undertaken in May 2016 and June 2016. A maximum count of five nests including three nests of Little Egret and two nests of Chinese Pond Heron were recorded, which indicated the egretry still supported these two species. The flight paths recorded were similar when compared to the results in 2012 (**Figure 9.4B** and **9.4C**).

9.4.80 TSW Main Channel and associated modified watercourses were located close to San Sang San Tsuen egretry, and may serve as potential foraging grounds. Between May and July 2012, two Chinese Pond Herons were recorded in TSW Main Channel and two individuals were recorded in nearby modified watercourses. A total of ten Little Egrets were recorded in TSW Main Channel in May and July 2012, and 12 individuals were recorded in other nearby modified watercourses during the same period. Two Little Egrets were also recorded in the pond east of the egretry (i.e. fishpond near San Sang San Tsuen).

9.4.81 Ngau Hom Shek egretry was located in bamboo clumps along Deep Bay Road (**Figure 9.3B**). No breeding ardeids were identified during the surveys; however, one nest of Chinese Pond Heron was recorded at the site in 2012 according to Anon (2012a). No breeding ardeids were recorded at Ngau Hom Shek egretry during breeding season in 2013 (Anon, 2013), but this egretry re-established in 2014 with four nests recorded (Anon, 2014). The nest number increased to 8 in 2015 (Anon, 2015).

9.4.82 .Shenzhen Bay Bridge egretry is located in bamboo clumps adjacent to Shenzhen Bay Bridge outside the northern Project area (**Figure 9.3B** refers). This egretry was discovered in 2014 by HKBWS (Anon, 2014). Two ardeid species, Chinese Pond Heron and Little Egret, were recorded and the number of nests increased from 19 to 30 from 2014 to 2015 (Anon, 2015).

Mammals

9.4.83 Seven mammal species were recorded within the assessment area, five of which are species of conservation importance, namely Short-nosed Fruit Bat (*Cynopterus sphinx*), Japanese Pipistrelle (*Pipistrellus abramus*), Chinese Noctule (*Nyctalus plancyi*), *Myotis* spp., and Small Indian Civet (*Viverricula indica*) (refer to **Figures 9.3A** to **9.3N**). Most of the recorded mammal species are either “abundant” or “common” in Hong Kong. (AFCD, 2014). A comprehensive list of mammals recorded during the surveys is provided in **Appendix 9.6**.

9.4.84 No roosting bats were recorded during the surveys. The bat species recorded within the assessment area are considered “common/abundant” in Hong Kong. The most abundant bat species recorded was Japanese Pipistrelle, which is considered to be the most common bat species found in local urban areas. This species becomes active before sunset and predominantly forages or commutes over modified watercourses. It is able to roost in various types of buildings in village and urban areas (Shek, 2006). Over 100 individuals were recorded foraging/commuting along the modified watercourse adjacent to West Rail Line, at and to the north of Tin Sam Tsuen. Smaller numbers were also recorded commuting along small modified watercourses flowing through grassland and village/orchard habitat south-west of San Sang San Tsuen. A small number were also recorded foraging over a large grassland in this area.

9.4.85 Chinese Noctule was recorded foraging over a pond habitat in San Sang San Tsuen. This is a “common” species in Hong Kong and is of Potential Regional Concern (Fellowes *et al*., 2002). It roosts under the roofs of old buildings (Shek, 2006). One individual was recorded foraging over a pool along the watercourse which flows from Hung Shui Hang Reservoir to TSW Main Channel in the assessment area. Another individual was recorded at the marsh habitat near Fung Kong Tsuen.

9.4.86 An unidentified individual of *Myotis* spp. was recorded commuting along the modified watercourse south of the Project area, adjacent to stream near Tan Kwai Tsuen. Another individual was recorded in developed area/wasteland next to a modified watercourse south of Tung Tau Tsuen. The six *Myotis* species known to occur in Hong Kong are regarded as “rare”, “uncommon”, or “data deficient” (AFCD, 2015a). Most of the *Myotis* species are listed as Local Concern, Potential Regional Concern or Regional Concern due to restrictedness of roosting site (Fellowes *et al*., 2002).

9.4.87 Short-nosed Fruit Bat was recorded in flight in the village/orchard habitat near Tung Tau Tsuen. This species is locally common and widely distributed in areas with human disturbance such as parks and gardens. It favours Chinese Fan Palm (*Livistona chinensis*) and Petticoat Palm (*Washingtonia robusta*) as roosting sites (Shek, 2006).

9.4.88 Scats, suspected to be from Small Indian Civet (*Viverricula indica*), were recorded on grave mounds within grassland habitat near KSWH (refer to **Figure 9.3C**), and outside the Project area. Small Indian Civet feeds on insects and fruit seeds, and their droppings are usually found in open areas (Shek, 2006).

Herpetofauna

9.4.89 Seven amphibian species were recorded within the assessment area. All of the amphibian species recorded are widely distributed locally for example Common Toad (*Bufo melanostictus*), Asiatic Painted Frog (*Kaloula pulchra*) and Gunther's Frog (*Rana guentheri*). No amphibian species of conservation importance was recorded.

9.4.90 Ten reptile species were recorded within the assessment area. Most of the recorded reptile species are widely distributed in Hong Kong such as Chinese Gecko (*Gekko chinensis*), Changeable Lizard (*Calotes versicolor*) and Long-tailed Skink (*Mabuya longicaudata*). No reptile species of conservation importance were recorded. A comprehensive list of herpetofauna observed during the surveys is provided in **Appendix 9.7**.

Butterfly

9.4.91 A total of 66 butterfly species were recorded within the assessment area. Most of the species are common and widespread in Hong Kong. Six species of conservation importance were recorded. Indicative locations of species of conservation importance are presented in **Figures 9.3A** to **9.3N**. A comprehensive list of butterflies observed during the surveys is provided in **Appendix 9.8**.

9.4.92 Butterfly species richness was highest in village/orchard, while species abundance was the highest in agricultural area. Butterflies were frequently observed associated with fruit trees, crops and horticultural plants.

9.4.93 Recorded species of conservation importance included: Dark Brown Ace (*Halpe porus*), Grass Demon (*Udaspes folus*), Common Rose (*Pachliopta aristolochiae goniopeltis*), Cornelian (*Deudorix epijarbas menesicles*), Common Cerulean (*Jamides celeno celeno*) and Danaid Egg-fly (*Hypolimnas misippus*). One sighting was made for each of these species. All are either "uncommon", "rare" or "very rare" in Hong Kong (AFCD, 2014). Among them, Dark Brown Ace and Danaid Egg-fly are of Local Concern (Fellowes *et al*., 2002).

9.4.94 Dark Brown Ace, Grass Demon and Cornelian were recorded in village/orchard habitat within the assessment area, but only Cornelian was recorded within the Project area. Dark Brown Ace was recorded in Tai Tao Tsuen in the south of the assessment area (**Figure 9.3M**). It is associated with bamboo clumps and feeds on bird droppings and nectar (Lo and Hui, 2005). Grass Demon was recorded in Tung Tau Tsuen in the middle of the Project area (**Figure 9.3G**).

9.4.95 Cornelian was the only butterfly species of conservation importance recorded within the Project area. One individual was observed flying across vegetation next to the orchard (**Figure 9.3G**). Two larval food plant species for Cornelian were recorded: Longan (in developed area/wasteland, village/orchard, agricultural area, plantation and woodland habitat) and Lychee (in developed area/wasteland, village/orchard, plantation and fishpond habitat).

9.4.96 Common Rose was recorded in developed area/wasteland habitat (**Figure 9.3A**). Common Cerulean was recorded flying over tall grass in the grassland near Sha Kong Wai (**Figure 9.3A**) whereas Danaid Egg-fly was recorded in agricultural area close to Fui Sha Wai (**Figure 9.3M**). These three species were recorded outside the Project area.

Odonate

9.4.97 A total of 29 species of odonate were recorded. Most of them are common and widespread in Hong Kong. Scarlet Basker (*Urothemis signata signata*) was the only species of conservation importance recorded. Indicative locations of species of conservation importance are presented in **Figures 9.3A** to **9.3N.** A comprehensive list of dragonflies observed during the surveys is provided in **Appendix 9.9**.

9.4.98 Most of the dragonfly and damselfly species were associated with water and seen flying over the natural and modified watercourse, fishponds, agricultural area, and village/orchard. Breeding behaviour was often observed in these habitats. Wandering Glider (*Pantala flavascens*) and Common Red Skimmer (*Orthetrum pruinosum neglectum*) were the most abundant dragonfly species recorded within the assessment area, they are common and abundant respectively in Hong Kong (AFCD, 2014).

9.4.99 Scarlet Basker is of "Local Concern" (Fellowes *et al*., 2002). This species is locally common and occurs in abandoned fishponds throughout Hong Kong (AFCD, 2014). One individual was recorded flying over a fishpond located in the north-west region of the assessment area, just west of KSWH (**Figure 9.3B**).

Freshwater Community in Natural Watercourse

9.4.100 Natural watercourses, including Watercourse 1, Watercourse 5 and part of Watercourse 6 were surveyed. A total of eight freshwater species were recorded in the natural watercourses in the dry season and during the wet season (**Figures 9.3C**, **9.3D** and **9.3H**). Most of the species recorded were common and widespread in Hong Kong. A comprehensive list of species recorded in freshwater community in natural watercourses during the surveys in dry and wet season is provided in **Appendix 9.10a** and **9.10b** respectively.

9.4.101 Watercourse 1 was located outside the Project area at Yuen Tau Shan and relatively undisturbed. Golden Bee Shrimp (*Caridina cantonensis*) was abundant in both seasons. Nile Tilapia (*Oreochromis niloticus*) and Predaceous Chub (*Parazacco spilurus*) were recorded in lower course of Watercourse 1 (W1-L) in dry and wet season, respectively. Predaceous Chub is fish species of conservation importance, which is common in Hong Kong but is listed as "Vulnerable" in China Red Data Book. However, this species is common and widespread in Hong Kong and, occurring in most unpolluted hill streams in both upper and lower courses. This species is considered as "Least Concern" in China by the latest Red List of China's Vertebrates (Jiang *et al*., 2016).

9.4.102 The lower sections of Watercourse 5 and part of Watercourse 6 were located within the Project area, and either subjected to disturbance from nearby village housing or construction site. Only low abundance of aquatic species were recorded at the lower sections of Watercourse 5 (W5-M and W5-L). Nile Tilapia and Chinese Barb (*Puntius semifasciolatus*) were recorded at the lower sections of Watercourse 5. No aquatic species were recorded at Watercourse 6 (W6-M).

Freshwater Community in Modified Watercourse

9.4.103 Four modified watercourses (Watercourse 2, Watercourse 3, Watercourse 4 and part of Watercourse 6) were surveyed. Watercourse 2 and Watercourse 3 were open nullahs while Watercourse 4 was a ditch in village/orchard habitat (**Figures 9.3A** to **9.3N**). Watercourse 2 is situated outside the Project area. A total of seven species were

recorded in dry season whereas five species were recorded in wet season. Most of the species recorded were common and widespread in Hong Kong. Modified watercourses were dominated by exotic species such as Nile Tilapia and Variable Platyfish (*Xiphophorus variatus*). A comprehensive list of species recorded in freshwater community in modified watercourse during the surveys is provided in dry and wet season **Appendix 9.10a** and **9.10b** respectively.

9.4.104 Predaceous Chub was recorded from Watercourse 2, Watercourse 3 and Watercourse 4 (refer to **Appendix 9.10a, 9.10b** and **Figure 9.3H** to **9.3J**). It was abundant at the middle section of Watercourse 2 (W2-M), lower section of Watercourse 3 (W3-L) during the dry season; while it was abundant at the middle section of Watercourse 2 (W2-M), occasional at upper and middle section of Watercourse 3 (W3-U and W3-M) and lower section of Watercourse 4 (W4-L) during the wet season.

9.4.105 Channeled Applesnail (*Pomacea canaliculata*), an invasive species in freshwater habitats in Hong Kong, was the most dominant species in both dry and wet season in Watercourse 4. In addition to the snail, unidentified frog tadpoles were observed in the upper course of Watercourse 4.

Intertidal and Coastal Water Fauna

9.4.106 A total of 32 intertidal fauna species were recorded from both walk-through and transect surveys, of which 20 species were recorded during dry season and 25 species were recorded during wet season. All species found are common mudflat species, except a Horseshoe Crab species (*Carcinoscorpius rotundicauda*) is of conservation importance. A comprehensive list of intertidal fauna observed during the surveys in dry and wet season is provided in **Appendix 9.11a** and **9.11b** respectively.

9.4.107 Horseshoe Crab (*Carcinoscorpius rotundicauda*) was observed during the walk through survey carried out in the wet season. Two adults of approximately 20 cm in length (excluding tail) were recorded lying beneath the mudflat surface next to the transect at P1 (**Figure 9.3B**). Juveniles (approximately 1.5–2 cm in length, excluding tail) were also recorded further seaward side from the location where the adults were found. This species is considered a species of conservation importance given its restricted distribution. It is also threatened by human exploitation, pollution and habitat loss (Li, 2008; Shin *et al*., 2009).

Ecological Importance

9.4.108 In accordance with the EIAO-TM Annex 8 criteria, the ecological importance of recorded habitats and species of conservation importance identified during the field surveys has been evaluated and is presented in **Table 9.12** to **Table 9.26**. A summary of species of conservation importance recorded in the assessment area is presented in **Table 9.27**.

Developed Area/Wasteland

9.4.109 Developed area/wasteland habitat recorded within the assessment area supported moderate floral and faunal diversity, but most of the recorded species were introduced. No plant species of conservation importance were recorded from this habitat. Eleven avifauna (Little Grebe, Chinese Pond Heron, Little Egret, Black Kite, Crested Serpent Eagle, Besra, Eastern Buzzard, Greater Coucal, Collared Crow, Red-billed Starling and White-cheeked Starling), three mammal (Japanese Pipistrelle, *Myotis* spp. and Greater Bandicoot Rat), one odonate (Giant Hooktail) and one butterfly (Grass Demon) species of conservation importance were recorded during the current surveys and literature review. Being a man-made habitat which is highly disturbed by human activities (e.g. traffic, noise), the ecological value of the majority of this habitat within the assessment area was low, except the area supporting San Sang San Tsuen egretry, Ngau Hom Shek egretry and Shenzhen Bay Bridge egretry was moderate.

Table 9.12 Ecological Evaluation of Developed Area/Wasteland

Criteria	Developed Area/Wasteland
Naturalness	Low
Size	Very large (960.14 ha)
Diversity	Moderate floral and faunal diversity, but most of the recorded flora species were planted
Rarity	A very common habitat in Hong Kong. Dominated by industry, residential building and infrastructure (e.g. roads) Current Surveys No plant species of conservation importance were recorded Ten avifauna (Chinese Pond Heron, Little Egret, Black Kite, Crested Serpent Eagle, Besra, Eastern Buzzard, Greater Coucal, Collared Crow, Red-billed Starling and White-cheeked Starling), two mammal (Japanese Pipistrelle and *Myotis* spp.) and one butterfly (Grass Demon) species of conservation importance were recorded. Literature Review Six avifauna (Little Grebe, Chinese Pond Heron, Black Kite, Greater Coucal, Collared Crow and Red-billed Starling), two mammal (Greater Bandicoot Rat and Japanese Pipistrelle) and one odonate (Giant Hooktail) species of conservation importance were recorded
Re-creatability	Easily re-creatable
Fragmentation	None
Ecological linkage	The habitat is structurally linked to woodland and natural watercourse Small area of developed area at west of San Sang San Tsuen is located within "CA" The developed area at Deep Bay, southwest of the KSWH and southeast of Yuen Long Highway are located adjacent to "CA" and "CPA"
Potential value	Low
Nursery/ breeding ground	San Sang San Tsuen egretry, Shenzhen Bay Bridge egretry and Ngau Hom Shek egretry were recorded within this habitat
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**Moderate** – for areas supporting the three egretries **Low** – for all other developed area/wasteland

Village/Orchard

9.4.110 This habitat was composed of village housing interspersed with patches of orchard. This habitat supported low to moderate floral and faunal diversity, of predominantly introduced species. No plant species of conservation importance were recorded. Nine avifauna (Chinese Pond Heron, Little Egret, Crested Goshawk, Greater Coucal, Black Kite, Eastern Buzzard, Collared Crow, Red-billed Starling and White-cheeked Starling), three mammal (Short-nosed Fruit Bat, Japanese Pipistrelle and Chinese Noctule) and three butterfly (Cornelian, Dark Brown Ace and Grass Demon) species of conservation importance were recorded during the current surveys and literature review. Being a man-made habitat which was highly disturbed by human activities (e.g. noise, farming), the ecological value of this habitat was low.

Table 9.13 Ecological Evaluation of Village/Orchard

Criteria	Village/Orchard
Naturalness	Low
Size	Large (107.68 ha)
Diversity	Low to moderate floral and faunal diversity
Rarity	Common habitat in Hong Kong. Dominated by village housing interspersed with patches of orchard Current Survey No plant species of conservation importance were recorded Six avifauna (Chinese Pond Heron, Little Egret, Crested Goshawk, Greater Coucal, Red-billed Starling and White-cheeked Starling), three mammal (Short-nosed Fruit Bat, Japanese Pipistrelle and Chinese Noctule) and three butterfly (Cornelian, Dark Brown Ace and Grass Demon) species of conservation importance were recorded Literature Review Seven avifauna species of conservation importance (Chinese Pond Heron, Little Egret, Black Kite, Eastern Buzzard, Greater Coucal, Collared Crow, Red-billed Starling) were recorded
Re-creatability	Re-creatable
Fragmentation	Low to moderate
Ecological linkage	The village/orchard at Deep Bay is located within "CPA"; while that at Tsing Chuen Wai is located adjacent to "CA"
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	N/A
Abundance/ richness of wildlife	Low to moderate
Ecological value	**Low**

Agricultural Area (Dry)

9.4.111 Dry agricultural area habitat recorded within the assessment area supported low floral and faunal diversity. Eight avifauna (Chinese Pond Heron, Little Egret, Black Kite, Eastern Buzzard, Peregrine Falcon, Greater Coucal, Red-billed Starling and White-cheeked Starling) and two mammal (Japanese Pipistrelle and Small Asian Mongoose) species of conservation importance were recorded during the current surveys and literature review. This habitat was small in size and highly disturbed by human activities (i.e. crop production). No plant species of conservation importance were recorded. The ecological value of the dry agricultural area within the assessment area was low.

Table 9.14 Ecological Evaluation of Agricultural Area (Dry)

Criteria	**Agricultural Area (Dry)**
Naturalness	Low. A man-made habitat for the purpose of crop production.
Size	Small (21.20 ha)
Diversity	Low floral and faunal diversity
Rarity	A common habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded Five avifauna (Chinese Pond Heron, Little Egret, Greater Coucal, Red-billed Starling and White-cheeked Starling) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded Literature Review Three avifauna (Black Kite, Eastern Buzzard and Peregrine Falcon) and one mammal species of conservation importance (Small Asian Mongoose) was recorded
Re-creatability	Re-creatable
Fragmentation	Moderate to high. The agricultural lands were scattered throughout the assessment area
Ecological linkage	This habitat was not structurally or functionally linked to any high ecological value resources
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**Low**

Agricultural Area (Wet)

9.4.112 Wet agricultural area habitat recorded within the assessment area supported low floral and faunal diversity. This habitat was also small in size and highly disturbed by farming activities. No plant species of conservation importance were recorded. Seven avifauna (Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Greater Coucal, Red-billed Starling and White-cheeked Starling) and one butterfly (Danaid Eggfly) species of conservation importance were recorded during the current surveys and literature review. The ecological value of the wet agricultural area within the assessment area was low.

Table 9.15 Ecological Evaluation of Agricultural Area (Wet)

Criteria	Agricultural Area (Wet)
Naturalness	Low. A man-made habitat for the purpose of crop production.
Size	Small (2.83 ha)
Diversity	Low floral and faunal diversity
Rarity	A fairly common habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded Three avifauna (Greater Coucal, Red-billed Starling and White-cheeked Starling) and one butterfly (Danaid Eggfly) species of conservation importance were recorded Literature Review Four avifauna species of conservation importance (Chinese Pond Heron, Grey Heron, Eastern Great Egret and Little Egret) were recorded
Re-creatability	Re-creatable
Fragmentation	High. This habitat was scattered throughout the assessment area
Ecological linkage	This habitat was not structurally or functionally linked to any high ecological value resources
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**Low**

Plantation

9.4.113 The plantation areas were highly fragmented within the assessment area, as they were either surrounded by developed area within the Project area, or located on the hillside area outside the Project area. No plant species of conservation importance were recorded. Five avifauna (Chinese Pond Heron, Grey Heron, Little Egret, Crested Serpent Eagle and Greater Coucal) and five mammal (Chinese Pangolin, Small-toothed Ferret Badger, Japanese Pipistrelle, Chinese Noctule and Short-nosed Fruit Bat) species of conservation importance were recorded during the current surveys and literature review.

9.4.114 Hillside plantation supported low to moderate floral diversity dominated by exotic plantation trees. The faunal diversity was low. Although this habitat was young and had simple structural complexity, given its proximity to woodland and shrubland habitat, regeneration of native species is possible in the long-term. Urban plantation was fragmented and subject to edge-effects (e.g. disturbance from traffic noise, air quality/dust, road lighting/glare), and supported low to moderate floral diversity and low faunal diversity. The ecological value for hillside plantation was low to moderate, while that of the urban plantation was low.

Table 9.16 Ecological Evaluation of Plantation

Criteria	Plantation
Naturalness	Low
Size	Large (103.92 ha)
Diversity	Low to moderate floral diversity Low faunal diversity
Rarity	A common habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded Five avifauna (Chinese Pond Heron, Grey Heron, Little Egret, Crested Serpent Eagle and Greater Coucal) and two mammal (Japanese Pipistrelle and Chinese Noctule) species of conservation importance were recorded Literature Review Four mammal species of conservation importance (Chinese Pangolin, Small-toothed Ferret Badger, Japanese Pipistrelle, Short-nosed Fruit Bat) were recorded
Re-creatability	Re-creatable
Fragmentation	Moderate
Ecological linkage	Hillside plantation is structurally linked to woodland and shrubland Plantation near Watercourse 1 and southwest of KSWH are located within "CA"; while that at Deep Bay is located within "CPA"
Potential value	Low to moderate – for hillside plantation as long-term regeneration of native species is possible if time is given Low – for urban plantation

Criteria	Plantation
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Young
Abundance/ richness of wildlife	Low to moderate
Ecological value	**Low to Moderate** – for hillside plantation **Low** – for urban plantation

Woodland

9.4.115 A number of woodland areas/patches were present across the assessment area. This habitat tended to support relatively mature trees and most of the recorded species were native. This habitat typically supported moderate floral diversity and low to moderate faunal diversity. Three plant species of conservation importance (Incense Tree, Luofushan Joint fir and Black Olive) were recorded; however, only one individual of Incense Tree at Tung Tau Tsuen would be impacted. Six avifauna (Crested Serpent Eagle, Black Kite, Crested Goshawk, Common Emerald Dove, Greater Coucal and Rufous-capped Babbler) and two mammal (Japanese Pipistrelle and Short-nosed Fruit Bat) and two herpetofauna (Indo-Chinese Rat Snake and Common Rat Snake) species of conservation importance were recorded during the current surveys and literature review. The woodland at the hillside area and fung shui woodland at Ngau Hom were subjected to less disturbance and therefore had moderate ecological value. The woodland at Tung Tau Tsuen and woodland at low elevation between the Kai Pak Ling Road and Fung Kong Tsuen marsh (i.e. west of Fung Kong Tsuen) were subjected to disturbance from nearby roads, therefore the ecological value was considered as low to moderate.

Table 9.17 Ecological Evaluation of Woodland

Criteria	Woodland
Naturalness	Moderate – for woodland at the hillside area and fung shui woodland at Ngau Hom Low to moderate – for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen
Size	Medium (41.31 ha)
Diversity	Typically supported moderate floral diversity and higher structural complexity Low to moderate faunal diversity

Criteria	Woodland
Rarity	A fairly common habitat in Hong Kong <u>Current Surveys</u> Two plant species of conservation importance (Incense Tree and Luofushan Joint fir) was recorded Four avifauna (Crested Serpent Eagle, Common Emerald Dove, Greater Coucal and Rufous-capped Babbler) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded <u>Literature Review</u> Two plant species of conservation importance (Incense Tree and Black Olive) were recorded Three avifauna (Black Kite, Crested Serpent Eagle and Crested Goshawk), one mammal (Short-nosed Fruit Bat) and two herpetofauna (Indo-Chinese Rat Snake and Common Rat Snake) species of conservation importance were recorded
Re-creatability	Re-creatable if time is given for natural succession
Fragmentation	Low
Ecological linkage	The habitat is functionally linked to shrubland and natural watercourse outside the Project area Part of the woodland near Watercourse 1 is located within "CA"
Potential value	Moderate – for woodland at the hillside area and fung shui woodland at Ngau Hom. These woodlands can become mature if time is allowed Low to moderate – for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen. These woodlands were subjected to disturbance from nearby roads
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Semi-mature to mature – for woodland at the hillside area and fung shui woodland at Ngau Hom Relatively young – for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen
Abundance/ richness of wildlife	Moderate – for woodland at the hillside area and fung shui woodland at Ngau Hom Low to moderate – for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen
Ecological value	**Moderate** – for woodland at the hillside area and fung shui woodland at Ngau Hom **Low to Moderate** – for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen

Shrubland

9.4.116 Shrubland supported low to moderate floral and faunal diversity. Four plant species of conservation importance (Lamb of Tartary, Incense Tree, Luofushan Joint-fir and Hong Kong Pavetta) were recorded, but all of them were located outside the Project area. Six avifauna (Chinese Pond Heron, Black Kite, Crested Serpent Eagle, Crested Goshawk, Peregrine Falcon and Greater Coucal) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded during the current surveys and literature review. The shrubland is linked with woodland at Yuen Tau Shan and southeast of Yuen Long Highway outside the Project area. Parts of shrubland at west and southwest of KSWH, and southeast of Yuen Long Highway are located within "CA". This habitat has the potential to succeed to woodland if time is allowed for maturation and natural succession. The shrublands at low elevation near developed area were subjected to more frequent disturbance such as construction noise and traffic than the shrublands at uphill areas. Therefore, the ecological value of shrubland at hillside areas at southeast of Yuen Long Highway and Ngau Hom Shek knoll within the Project area, and other shrublands outside the Project area was low to moderate; while that of the remaining shrublands within Project area near developed area was low.

Table 9.18 Ecological Evaluation of Shrubland

Criteria	Shrubland
Naturalness	Low to moderate
Size	Large (197.57 ha)
Diversity	Low to moderate floral and faunal diversity
Rarity	A common habitat in Hong Kong Current Surveys Four plant species of conservation importance (Lamb of Tartary, Incense Tree, Luofushan Joint-fir and Hong Kong Pavetta) were recorded One avifauna (Greater Coucal) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded Literature Review Five avifauna (Chinese Pond Heron, Black Kite, Crested Serpent Eagle, Crested Goshawk and Peregrine Falcon) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded
Re-creatability	Re-creatable if time is allowed for maturation and natural succession
Fragmentation	Low to moderate – for shrubland at southeast of Yuen Long Highway and Ngau Hom Shek knoll within the Project area, and other shrublands outside the Project area High – for the remaining shrubland within the Project area
Ecological linkage	This habitat is structurally linked with woodland at Yuen Tau Shan and natural watercourse habitats Parts of shrubland at west and southwest of KSWH, and southwest of the assessment area are located within "CA"
Potential value	Moderate – for the shrublands linked with woodlands at Ngau Hom Shek knoll and Yuen Tau Shan outside the Project area Low to moderate – for other hillside shrubland or patches located near developed area with frequent human disturbance

Criteria	Shrubland
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Young for short shrubland Relatively mature for tall shrubland
Abundance/ richness of wildlife	Low to moderate
Ecological value	**Low to Moderate** – for shrubland at hillside areas at southeast of Yuen Long Highway and Ngau Hom Shek knoll within the Project area, and other shrublands outside the Project area **Low** – for other remaining shrublands within the Project area near developed area

Grassland

9.4.117 The majority of this habitat is linked with shrubland and plantation on hillside areas, while the remaining grassland was fragmented and close to developed area. Some of this habitat is located within "CA". This habitat supported low floral and faunal diversity. A small number and low abundance of species of conservation importance including two plant (Incense Tree, Cycad-fern), six avifauna (Chinese Pond Heron, Little Egret, Besra, Greater Coucal, Red-billed Starling and White-cheeked Starling), two mammal (Japanese Pipistrelle and Small Indian Civet) and one butterfly (Common Cerulean) species were recorded during the current surveys and literature review. Therefore, the ecological value of the grassland within the assessment area was low.

Table 9.19 Ecological Evaluation of Grassland

Criteria	Grassland
Naturalness	Low
Size	Medium (77.40 ha)
Diversity	Low floral and faunal diversity
Rarity	A very common habitat in Hong Kong Current Surveys One plant (Incense Tree) species of conservation importance was recorded Six avifauna (Chinese Pond Heron, Little Egret, Besra, Greater Coucal, Red-billed Starling and White-cheeked Starling), two mammal (Japanese Pipistrelle and Small Indian Civet) and one butterfly (Common Cerulean) species of conservation importance were recorded Literature Review One plant species of conservation importance (Cycad-fern) was recorded
Re-creatability	Re-creatable
Fragmentation	Moderate

Criteria	Grassland
Ecological linkage	This habitat is structurally linked to woodland and shrubland Parts of grassland at southwest of KSWH are located within "CA"
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Young
Abundance/ richness of wildlife	Low
Ecological value	**Low**

Marsh

9.4.118 Marsh habitat is a relatively uncommon habitat in Hong Kong. However, the marshes recorded within the assessment area were highly fragmented, small in size and disturbed. They supported low floral and faunal diversity and were subject to disturbance from adjacent land use. No plant species of conservation importance were recorded. One avifauna (Chinese Pond Heron) and two mammal (Japanese Pipistrelle and Chinese Noctule) and one herpetofauna (Two-striped Grass Frog) species of conservation importance were recorded during the current surveys and literature review. Therefore, the ecological value of this habitat was low.

Table 9.20 Ecological Evaluation of Marsh

Criteria	Marsh
Naturalness	Low to Moderate
Size	Small (1.75 ha)
Diversity	Low floral and faunal diversity
Rarity	Uncommon habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded One avifauna (Chinese Pond Heron) and one mammal (Chinese Noctule) species of conservation importance were recorded Literature Review One mammal (Japanese Pipistrelle) and one herpetofauna (Two-striped Grass Frog) species of conservation importance were recorded
Re-creatability	Re-creatable
Fragmentation	High

Criteria	Marsh
Ecological linkage	The marsh near Hang Hau Tsuen (outside the Project area) is structurally and functionally linked to mudflat/coastal water body at Deep Bay The other two patches (inside the Project area) are not structurally or functionally linked to any high ecological value resources
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**Low**

Fishponds and Mitigation Ponds

9.4.119 The fishponds at Deep Bay (outside of the Project area) were located in the "CPA", ecologically linked to the wetlands of Deep Bay, and potentially provided a foraging resource for the Deep Bay avifauna. Therefore, the ecological value for the fishponds at Deep Bay was moderate.

9.4.120 Four avifauna (Grey Heron, Eastern Great Egret, Little Egret and Besra) and one mammal (Chinese Noctule) species of conservation importance were recorded at the abandoned fishpond near San Sang San Tsuen during the current surveys. The maximum count of each avifauna species of conservation importance was two individuals of Grey Heron, one individual of Eastern Great Egret, three individuals of Little Egret and one individual of Besra over the 12-month monitoring. According to the ardeid flight path survey conducted at San Sang San Tsuen egretry, a small number of ardeids flew towards the east i.e. the direction of the abandoned fishpond near San Sang San Tsuen and other modified watercourses (approximately 15% of the total recorded ardeids), while the majority flew towards the direction of TSW Main Channel (approximately 80% of the total recorded ardeids). The results above indicated that the fishpond near San Sang San Tsuen may contribute a foraging resource for the ardeids, but was relatively less important than TSW Main Channel, which supported a greater avifauna diversity and abundance. This is further supported by the low number of ardeids recorded in this fishpond during the breeding season (i.e. only two Little Egret, **Section 9.4.80** refers), possibly because this inactive fishpond was overgrown with herb species and thus the condition was unfavorable for ardeids. Therefore, the ecological value of abandoned fishpond near San Sang San Tsuen was considered as low.

9.4.121 The other fishponds, which were located at Sha Kong Wai, west of KSWH, San Wai, Tsing Chuen Wai and Fui Sha Wai, were highly fragmented across the assessment area and small in size. No plant species of conservation importance were recorded. Five avifauna (Chinese Pond Heron, Grey Heron, Little Egret, Greater Coucal and Red-billed Starling), one mammal (Japanese Pipistrelle) and one odonate (Scarlet Basker) species of conservation importance were recorded. The maximum count of each avifauna species of conservation importance found was six individuals of Chinese Pond Heron, five individuals of Grey Heron, eight individuals of Little Egret, two individuals of Greater Coucal, and 22 individuals of Red-billed Starling. These fishponds were considered to be of low ecological value.

9.4.122 The four small mitigation ponds northwest of San Sang San Tsuen were constructed to compensate the loss of fishponds under the DBL project (ACE, 2002; HyD, 2002a). The mitigation ponds were located adjacent to KSWH (two were located on the western side and connected to agricultural area and grassland habitat, while the other two were located on the eastern side of the KSWH and adjacent to an open storage area). Two avifauna species of conservation importance (Grey Heron and Little Egret) were recorded during the survey programme. A range of species were recorded at these ponds, including 42 avifauna (seven species of conservation importance), 23 odonate, 17 herpetofauna and seven freshwater fish species, according to the regular monitoring conducted from 2010 to 2016 (AFCD, 2016). The mitigation ponds were considered to be of low to moderate ecological value.

Table 9.21 Ecological Evaluation of Fishpond and Mitigation Ponds

Criteria	Fishpond and Mitigation Ponds
Naturalness	Low
Size	Small (4.39 ha)
Diversity	Low floral and faunal diversity
Rarity	An uncommon habitat in Hong Kong <u>Current Survey</u> No plant species of conservation importance were recorded Seven avifauna (Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Besra, Greater Coucal and Red-billed Starling), two mammal (Japanese Pipistrelle and Chinese Noctule) and one odonate (Scarlet Basker) species of conservation importance were recorded <u>Literature Review</u> Eleven avifauna (Little Grebe, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Black Kite, Greater Coucal, Red-billed Starling, White-cheeked Starling, Grey-chinned Minivet, Zitting Cisticola), one mammal (Japanese Pipistrelle), two herpetofauna (Burmese Python, Chinese Bullfrog) and two odonate (Blue-spotted Dusk-hawker, Scarlet Basker) species of conservation importance were recorded
Re-creatability	Easily re-creatable
Fragmentation	High
Ecological linkage	Fishpond near San Sang San Tsuen - potentially acted as a foraging ground for the San Sang San Tsuen egretry, but relatively less important than TSW Main Channel Fishponds at Deep Bay – located within "CPA", linked to the Deep Bay wetlands and potentially acted as a foraging ground for waterbirds at Deep Bay area Mitigation ponds – not structurally or functionally linked to any high ecological value resources Other fishponds – not structurally or functionally linked to any high ecological value resources
Potential value	High – for fishponds at Deep Bay Low to Moderate – for mitigation ponds Low – for fishponds near San Sang San Tsuen and other fishponds

Criteria	Fishpond and Mitigation Ponds
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Young
Abundance/ richness of wildlife	Low to Moderate
Ecological value	**Moderate** – for fishponds at Deep Bay **Low to Moderate** – for mitigation ponds **Low** – for fishpond near San Sang San Tsuen and other fishponds

Storm Water Drain/Recreational Pond

9.4.123 This habitat was a man-made habitat for water flood retention or recreational purposes. As such, this habitat supported low floral and faunal diversity. This habitat was also small in size and highly disturbed by human activities. No plant species of conservation importance were recorded. Three avifauna (Chinese Pond Heron, Grey Heron and Little Egret) species of conservation importance were recorded in low numbers during the current surveys and literature review. Therefore, the ecological value of this habitat was low.

Table 9.22 Ecological Evaluation of Storm Water Drain/Recreational Pond

Criteria	Storm Water Drain/Recreational Pond
Naturalness	Low. A man-made habitat for water flood retention or recreational purposes other than fishing
Size	Small (2.61 ha)
Diversity	Low floral and faunal diversity
Rarity	A fairly common habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded Three avifauna (Chinese Pond Heron, Grey Heron and Little Egret) species of conservation importance were recorded Literature Review No flora and fauna species of conservation importance were recorded
Re-creatability	Easily re-creatable
Fragmentation	None
Ecological linkage	This habitat is not structurally or functionally linked to any high ecological value resources
Potential value	Low
Nursery/ breeding ground	No record of significant nursery or breeding ground

Criteria	Storm Water Drain/Recreational Pond
Age	Young
Abundance/ richness of wildlife	Low
Ecological value	**Low**

Natural Watercourse

9.4.124 Natural watercourse habitat identified within the assessment area tended to support low to moderate floral and faunal diversity. One freshwater fish species of conservation importance (Predaceous Chub) was recorded at Watercourse 1. Watercourse 1, Watercourse 7, the upper sections of Watercourse 5 and Watercourse 6 were relatively undisturbed. Signs of human disturbance (e.g. illegal dumping, construction works) were observed at lower sections of Watercourse 5 and Watercourse 6. The ecological value of the Watercourse 1, Watercourse 7 and undisturbed upper sections of Watercourse 5 and Watercourse 6 was low to moderate, while that of the disturbed lower sections of Watercourse 5 and Watercourse 6 of was low.

Table 9.23 Ecological Evaluation of Natural Watercourse

Criteria	Natural Watercourse
Naturalness	Low
Size	Small (<0.1 ha, 10.27 km long in total)
Diversity	Low floral and faunal diversity
Rarity	A common habitat in Hong Kong Current Survey No plant species of conservation importance were recorded One freshwater fish species of conservation importance (Predaceous Chub) was recorded at Watercourse 1. However, Predaceous Chub is a common species in Hong Kong and considered as "Least Concern" in China by the latest Red List of China's Vertebrates Literature Review One mammal (Japanese Pipistrelle) and one odonate (Indochinese Copperwing) species of conservation importance were recorded
Re-creatability	Hard to re-create
Fragmentation	None
Ecological linkage	This habitat is structurally linked to woodland and shrubland The lower section of Watercourse 1, Watercourse 7, the upper of Watercourse 5 and Watercourse 6 are located within "CA"
Potential value	Low to moderate
Nursery/ breeding ground	No record of significant nursery or breeding ground

Criteria	Natural Watercourse
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**Low to Moderate** – for Watercourse 1, Watercourse 7, the undisturbed upper sections of Watercourse 5 and Watercourse 6 **Low** – for the disturbed lower sections of Watercourse 5 and Watercourse 6

Modified Watercourse

9.4.125 Modified watercourses within the assessment area comprised man-made concrete embankment and base, with sparse floral coverage. The floral and faunal diversity tended to be low. Nine avifauna (Northern Shoveler, Black-faced Spoonbill, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Wood Sandpiper, Collared Crow and Red-billed Starling), one mammal (Japanese Pipistrelle) and one freshwater fish (Predaceous Chub) species of conservation importance were recorded. However, Predaceous Chub is a common species in Hong Kong and considered as "Least Concern" in China by the latest Red List of China's Vertebrates (Jiang *et al*., 2016). The ecological value was moderate for TSW Main Channel as it provided foraging resources for some bird species of conservation importance (e.g. Black-faced Spoonbill, Little Egret); whereas the ecological value was low for other modified watercourses.

Table 9.24 Ecological Evaluation of Modified Watercourse

Criteria	Modified Watercourse
Naturalness	Low
Size	Small (26.28 ha, 20.68 km long in total)
Diversity	Low floral and faunal diversity
Rarity	A fairly common habitat in Hong Kong Current Survey No plant species of conservation importance were recorded Nine avifauna (Northern Shoveler, Black-faced Spoonbill, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Wood Sandpiper, Collared Crow and Red-billed Starling), one mammal (Japanese Pipistrelle) and one freshwater fish (Predaceous Chub) species of conservation importance were recorded. However, Predaceous Chub is a common species in Hong Kong and considered as "Least Concern" in China by the latest Red List of China's Vertebrates Literature Review No flora and fauna species of conservation importance were recorded
Re-creatability	Easily re-creatable
Fragmentation	None

Criteria	Modified Watercourse
Ecological linkage	This habitat was structurally linked to natural watercourse from uphill area TSW Main Channel and its associated modified watercourses connected to intertidal mudflat and mangrove downstream at Deep Bay
Potential value	Moderate – for TSW Main Channel, ecological enhancement (e.g. bioengineering) would be considered to improve the habitat Low – for other modified watercourses, small channelised watercourse have very limited potential
Nursery/ breeding ground	No record of significant nursery or breeding ground.
Age	Young
Abundance/ richness of wildlife	Low to moderate
Ecological value	**Moderate** – for TSW Main Channel **Low** – for other modified watercourses

Mangrove

9.4.126 Mangrove is an important habitat in Hong Kong. No plant species of conservation importance were recorded. Six avifauna (Chinese Pond Heron, Eastern Great Egret, Little Egret, Pied Avocet, Common Greenshank and Red-billed Starling) species of conservation importance were recorded. This habitat was located within "CPA" and structurally linked to mudflat/coastal water body. It acted as a foraging ground for intertidal species and waterbirds. Therefore, the ecological value for this habitat was high to moderate.

Table 9.25 Ecological Evaluation of Mangrove

Criteria	Mangrove
Naturalness	Moderate
Size	Small (11.90 ha)
Diversity	Low floral and faunal diversity
Rarity	An important habitat in Hong Kong Current Survey No plant species of conservation importance were recorded Six avifauna (Chinese Pond Heron, Eastern Great Egret, Little Egret, Pied Avocet, Common Greenshank and Red-billed Starling) species of conservation importance were recorded Literature Review No flora and fauna species of conservation importance were recorded
Re-creatability	Re-creatable
Fragmentation	N/A

Criteria	Mangrove
Ecological linkage	It was structurally and functionally linked to mudflat/coastal water body. It acted as a foraging ground for intertidal fauna species and waterbirds This habitat is located within "CPA"
Potential value	Moderate to high
Nursery/ breeding ground	No record of significant nursery or breeding ground
Age	Semi-mature to mature
Abundance/ richness of wildlife	Low to Moderate
Ecological value	**Moderate to High**

Mudflat/Coastal Water Body

9.4.127 This habitat is an important habitat in Hong Kong. Only a small area of the mudflat/coastal waterbody falls within the assessment area. One plant species of conservation importance (*Halophila beccarii*), twenty avifauna (Great Crested Grebe, Black-faced Spoonbill, Great Cormorant, Western Osprey, Crested Goshawk, Bar-tailed Godwit, Eurasian Curlew, Common Redshank, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Pied Avocet, Great Knot, Wood Sandpiper, Marsh Sandpiper, Collared Crow, Greater Coucal, Red-billed Starling and White-cheeked Starling) and two intertidal (*Tachypleus tridentatus* and *Carcinoscorpius rotundicauda*) species of conservation importance were recorded during the current surveys and literature review. This habitat was located adjacent to "CPA" and structurally linked to marine environment. It acted as a foraging ground for wetland dependent species (e.g. waterbirds). It was also a potential breeding/nursery ground for juvenile Horseshoe Crabs. Moreover, although no Chinese White Dolphin were recorded within the assessment area, this small area of mudflat/coastal waterbody was linked to the Sha Chau and Lung Kwu Chau Marine Park located outside the 500 m assessment area at the western edge of the North Western WCZ, which supports a Chinese White Dolphin breeding ground. Therefore, the ecological value for this habitat was high.

Table 9.26 Ecological Evaluation of Mudflat/Coastal Water Body

Criteria	Mudflat/Coastal Water Body
Naturalness	Moderate to High
Size	Small (34.70 ha)
Diversity	Low flora and faunal diversity
Rarity	An important habitat in Hong Kong Current Surveys No plant species of conservation importance were recorded Nine avifauna (Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Pied Avocet, Eurasian Curlew, Marsh Sandpiper, Collared Crow and Collared Crow) and one intertidal (*Carcinoscorpius rotundicauda*) species of conservation importance were recorded

Criteria	Mudflat/Coastal Water Body
	Literature Review One plant species of conservation importance (*Halophila beccarii*) was recorded Eighteen avifauna species of conservation importance (Great Crested Grebe, Black-faced Spoonbill, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Great Cormorant, Western Osprey, Crested Goshawk, Bar-tailed Godwit, Eurasian Curlew, Common Redshank, Wood Sandpiper, Great Knot, Greater Coucal, Collared Crow, Red-billed Starling, White-cheeked Starling) and two intertidal (*Tachypleus tridentatus*, *Carcinoscorpius rotundicauda*) species of conservation importance were recorded
Re-creatability	Hard to re-create
Fragmentation	None
Ecological linkage	It was structurally and functionally linked to marine environment and mangrove This habitat is located adjacent to "CPA"
Potential value	High
Nursery/ breeding ground	Potentially be a breeding/nursery ground for juvenile Horseshoe Crabs
Age	N/A
Abundance/ richness of wildlife	Low
Ecological value	**High**

Table 9.27 Species of Conservation Importance Recorded in the Assessment Area

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Flora						
Incense Tree	*Aquilaria sinensis*	Ling To Monastery Woodland[4]; Woodland near Ngau Hom Tsuen[7]	Within the Project area Woodland Outside the Project area Shrubland; Woodland; Grassland	Cap. 586; "Class II" in List of Wild Plant under State Protection; "Vulnerable" in China Plant Red Data Book; IUCN Red List (Vulnerable)	Lowland forests and fung shui woods	Common
Cycad-fern	*Brainea insignis*	Hillside grassland near Ngau Hom Shek [4]	Not recorded	Status in China: Vulnerable; Category II in List of Wild Plant under State Protection	On open hillsides, margin of forests and sometimes in secondary forests	Common
Black Olive	*Canarium pimela*	Ling To Monastery Woodland[4]	Not recorded	-	Fung shui woods	Rare[8]
Lamb of Tartary	*Cibotium barometz*	Not recorded	Shrubland outside the Project area	Cap. 586; "Category II" in Wild Plant under State protection; "Vulnerable" in Rare and Precious Plants of Hong Kong	Forest and shrubland	Common
Luofushan Joint-fir	*Gnetum luofuense*	Not recorded	Shrubland and Woodland outside the Project area	IUCN Red List (Near Threatened)	Woodland	Very common
Hong Kong Pavetta	*Pavetta hongkongensis*	Not recorded	Shrubland outside the Project area	Cap. 96	Fung shui woods and lowland forest	Common
Seagrass	*Halophila beccarii*	Coast of Sha Kong Tsuen[2]	Not recorded	IUCN Red List (Vulnerable)	Recorded in Ngau Hom Shek, Sha Kwong Tsuen, Sheung Pak Nai and Nam Chung Yeung Uk and Tai Ho Wan	Rare

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Avifauna						
Northern Shoveler	*Anas clypeata*	Not recorded	Modified Watercourse outside the Project area	Cap. 170; RC by Fellowes *et al.* (2002)	Found in Deep Bay area	Abundant
Little Grebe	*Tachybaptus ruficollis*	Ling To Monastery Road[4]; Ngau Hom Shek Fishpond[5]	Storm Water Drain/Recreational Pond outside the Project area	Cap. 170; LC by Fellowes *et al*. (2002)	Found in Deep Bay area	Common
Great Crested Grebe	*Podiceps cristatus*	Ngau Hom Shek shoreline[4]	Not recorded	Cap. 170; RC by Fellowes *et al*. (2002)	Found in Tsim Bei Tsui, Starling Inlet	Common
Black-faced Spoonbill	*Platalea minor*	Mudflat at Ngau Hom Shek[5]	Modified Watercourse outside the Project area	Cap. 170; PGC by Fellowes *et al*. (2002); Class II Protection Status in China; China Red Data Book (Endangered); Red List of China's Vertebrates (Endangered); IUCN Red List (Endangered);	Found in Deep Bay area	Common
Chinese Pond Heron	*Ardeola bacchus*	Ngau Hom Shek Shoreline, hillside and shoreline Farm; Ling To Monastery Road; San Sang San Tsuen (wet cultivation); Yick Yuen Tsuen [4]; Ngau Hom Shek fishpond and mudflat[5]; Fishpond near San Wai sewage treatment works[6]; Coastal shrubland and fishpond[7]; Mitigation ponds constructed under DBL project[10]	<u>Within the Project Area</u> Developed Area/Wasteland; Village/Orchard; Grassland; Modified Watercourse <u>Outside the Project Area</u> Village/Orchard; Agricultural Area (Dry); Storm Water Drain/Recreational Pond; Natural Watercourse; Modified Watercourse; Plantation; Grassland; Fishponds and Mitigation Ponds; Mangrove; Mudflat/Coastal Water Body	Cap. 170; PRC (RC) by Fellowes *et al*. (2002)	Widely distributed in Hong Kong	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Grey Heron	*Ardea cinerea*	Ngau Hom Shek shoreline, hillside and shoreline farm[4]; Ngau Hom Shek fishpond and mudflat[5]; Mitigation ponds constructed under DBL project[10]	Within the Project Area Modified Watercourse; Fishponds and Mitigation Ponds Outside the Project Area Storm Water Drain/Recreational Pond; Modified Watercourse; Plantation; Mudflat/Coastal Water Body	Cap. 170; PRC by Fellowes *et al.* (2002)	Found in Deep Bay area, Starling Inlet, Kowloon Park, Cape D'Aguilar	Common
Eastern Great Egret	*Ardea modesta*	Ngau Hom Shek shoreline (pond); Ngau Hom Shek hillside and shoreline farm[4]; Ngau Hom Shek fishpond and mudflat[5]; Fishpond and agricultural land[6]	Within the Project Area Modified Watercourse; Fishponds and Mitigation Ponds Outside the Project Area Modified Watercourse; Mangrove; Mudflat/Coastal Water Body	Cap. 170; PRC (RC) by Fellowes *et al.* (2002)	Widely distributed in Hong Kong	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Little Egret	*Egretta garzetta*	Ngau Hom Shek shoreline, hillside and shoreline farm; San Sang San Tsuen[4]; Ngau Hom Shek fishpond and mudflat[5]; coastal area[7]; Mitigation ponds constructed under DBL project[10]	Within the Project Area Developed Area/Wasteland Agricultural Area (Dry); Modified Watercourse; Grassland; Fishponds and Mitigation Ponds Outside the Project Area Developed Area/Wasteland; Village/Orchard; Agricultural Area (Dry); Storm Water Drain/Recreational Pond; Natural Watercourse; Modified Watercourse; Plantation; Fishponds and Mitigation Ponds; Marsh; Mangrove; Mudflat/Coastal Water Body	Cap. 170; PRC (RC) by Fellowes *et al.* (2002)	Widely distributed in coastal area throughout Hong Kong	Common
Great Cormorant	*Phalacrocorax carbo*	Off-shore at Ngau Hom Shek[4]	Not recorded	Cap. 170; PRC by Fellowes *et al*. (2002)	Widely distributed in coastal areas throughout Hong Kong	Common
Western Osprey	*Pandion haliaetus*	Coastal habitats at Ngau Hom Shek[4]	Not recorded	Cap. 170; Cap. 586; RC by Fellowes *et al*. (2002); Class II Protection Status in China; China Red Data Book (Rare); Red List of China's Vertebrates (Near Threatened)	Widely distributed in coastal areas throughout Hong Kong	Rare

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Black Kite	*Milvus migrans*	Ngau Hom Shek hillside and shoreline farm; Ling to Monastery Valley; Ling To Monastery Road; San Sang San Tsuen[4]; Ngau Hom Shek shrublands and stream valley[5]; Mitigation ponds constructed under DBL project[10]	Developed Area/Wasteland outside the Project area	Cap. 170; Cap. 586; Class II Protection Status in China	Widely distributed in Hong Kong	Common
Crested Serpent Eagle	*Spilornis cheela*	Hillside of Ling To Monastery Valley[4]; Ngau Hom Shek shrublands and stream valley[5]	Developed Area/Wasteland; Woodland; Plantation within the Project area	Cap. 170; Cap. 586; Class II Protection Status in China; China Red Data Book (Vulnerable); Red List of China's Vertebrates (Near Threatened)	Widely distributed in shrubland on hillsides throughout Hong Kong	Uncommon
Besra	*Accipiter virgatus*	Not recorded	Developed Area/Wasteland; Grassland; Fishponds and Mitigation Ponds within the Project area	Cap. 170; Cap. 586; Class II Protection Status in China	Found in Tai Po Kau, Deep Bay area, Chek Lap Kok, Cheung Chau, Soko Islands	Scarce
Crested Goshawk	*Accipiter trivirgatus*	Hillside of Ling To Monastery Valley and San Sang San Tsuen[4]; Ngau Hom Shek shrublands and stream valley[5]; Coastal area[7]	Village/Orchard within the Project area	Cap. 170; Cap. 586; Class II Protection Status in China; China Red Data Book (Rare); Red List of China's Vertebrates (Near Threatened)	Widely distributed in woodlands and shrublands throughout Hong Kong	Uncommon
Eastern Buzzard	*Buteo japonicus*	Ngau Hom Shek hillside and shoreline farm; Yick Yuen Tsuen[4]	Developed Area/Wasteland within the Project area	Cap. 170; Cap. 586; Class II Protection Status in China	Widely distributed in Hong Kong	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Peregrine Falcon	*Falco peregrinus*	Ngau Hom Shek hillside and shoreline farm[4]; upland habitats at shrubland[5]	Not recorded	Cap. 170; Cap. 586; Class II Protection Status in China; China Red Data Book (Rare); Red List of China's Vertebrates (Near Threatened)	Widely distributed in Hong Kong	Scarce
Pied Avocet	*Recurvirostra avosetta*	Not recorded	Mangrove; Mudflat/Coastal Water Body outside the Project area	Cap. 170; RC by Fellowes *et al*. (2002)	Found in Deep Bay area	Abundant
Bar-tailed Godwit	*Limosa lapponica*	Ngau Hom Shek shoreline[4]	-	Cap. 170; LC by Fellowes *et al*. (2002); Red List of China's Vertebrates (Near Threatened)	Found in Deep Bay area	Common
Eurasian Curlew	*Numenius arquata*	Ngau Hom Shek shoreline[4]; Ngau Hom Shek mudflat[5]	Mudflat/Coastal Water Body outside the Project area	Cap. 170; RC by Fellowes *et al*. (2002); Red List of China's Vertebrates (Near Threatened); IUCN Red List (Near Threatened)	Found in Deep Bay area	Abundant
Common Redshank	*Tringa totanus*	Ngau Hom Shek shoreline[4]; Coastal area[7]	Not recorded	Cap. 170; RC by Fellowes *et al*. (2002)	Found in Deep Bay area	Common
Marsh Sandpiper	*Tringa stagnatilis*	Not recorded	Mudflat/Coastal Water Body outside the Project area	Cap. 170; RC by Fellowes *et al*. (2002)	Found in Deep Bay area, Shuen Wan, Long Valley, Kam Tin, Sai Kung	Common
Common Greenshank	*Tringa nebularia*	Not recorded	Mangrove outside the Project area	Cap. 170; RC by Fellowes *et al*. (2002)	Found in Deep Bay area	Abundant
Wood Sandpiper	*Tringa glareola*	Ngau Hom Shek shoreline[4]	Modified Watercourse outside the Project area	Cap. 170; LC by Fellowes *et al*. (2002)	Widely distributed in Hong Kong	Common
Great Knot	*Calidris tenuirostris*	Ngau Hom Shek shoreline[4]	Not recorded	Cap. 170; LC by Fellowes *et al*. (2002); Red List of China's Vertebrates (Vulnerable); IUCN Red List (Endangered)	Found in Deep Bay area	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Common Emerald Dove	*Chalcophaps indica*	Not recorded	Woodland outside the Project area	Cap. 170; China Red Data Book (Vulnerable)	Widely distributed in woodland throughout Hong Kong	Scarce
Greater Coucal	*Centropus sinensis*	Ngau Hom Shek shoreline; Ling To Monastery Road; San Sang San Tsuen[4]; fishpond[7]; Mitigation ponds constructed under DBL project[10]	Within the Project Area Developed Area/Wasteland; Village/Orchard; Agricultural Area (Wet) Outside the Project Area Agricultural Area (Dry); Plantation; Grassland; Shrubland; Woodland; Fishponds and Mitigation Ponds	Cap. 170; Class II Protection Status in China; China Red Data Book (Vulnerable)	Widely distributed in Hong Kong	Common
Lesser Coucal	*Centropus bengalensis*	Not recorded	Not recorded	Cap. 170; Class II Protection Status in China; China Red Data Book (Vulnerable)	Widely distributed in Hong Kong	Common
Asian Barred Owlet	*Glaucidium cuculoides*	Not recorded	Not recorded	Cap. 170; Cap. 586; Class II Protection Status in China	Widely distributed in woodland throughout Hong Kong	Uncommon
Collared Crow	*Corvus torquatus*	Ngau Hom Shek shoreline; Yick Yuen Tsuen[4]; urbanised area[7]	Within the Project Area Developed Area/Wasteland Outside the Project Area Modified Watercourse; Mudflat/Coastal Water Body	Cap. 170; LC by Fellowes *et al*. (2002); Red List of China's Vertebrates (Near Threatened); IUCN Red List (Near Threatened)	Found in Inner Deep Bay area, Nam Chung, Kei Ling Ha, Tai Mei Tuk, Pok Fu Lam, Chek lap Kok, Shuen Wan, Lam Tsuen	Uncommon
Rufous-capped Babbler	*Stachyris ruficeps*	Not recorded	Woodland outside the Project area	Cap. 170; LC by Fellowes *et al*. (2002)	Found in Shing Mun, Tai Po Kau, Tai Mei Tuk, Ng Tung Chai, Fo Tan, Tai Mo Shan, The Peak and Kadoorie Agricultural Research Centre	Uncommon

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Red-billed Starling	*Spodiopsar sericeus*	Ngau Hom Shek Shoreline; Yick Yuen Tsuen; ponds by the road to Ling To Monastery[4] ; fishpond near Ngau Hom Shek[5]; Urbanised area and abandoned agricultural land[7]	Within the Project Area Developed Area/Wasteland Village/Orchard; Agricultural Area (Dry); Agricultural Area (Wet); Modified Watercourse; Grassland; Fishponds and Mitigation Ponds Outside the Project Area Developed Area/Wasteland Village/Orchard; Grassland; Fishponds and Mitigation Ponds; Mangrove; Mudflat/Coastal Water Body	Cap. 170; GC by Fellowes *et al*. (2002)	Widely distributed in Hong Kong	Common
White-cheeked Starling	*Spodiopsar cineraceus*	Ngau Hom Shek shoreline[4]; fishpond bunds in Ngau Hom Shek[5]	Within the Project Area Agricultural Area (Wet); Grassland Outside the Project Area Developed Area/Wasteland; Village/Orchard	Cap. 170; PRC by Fellowes *et al*. (2002)	Found in Deep Bay area, Kam Tin, Long Valley	Common
Grey-chinned Minivet	*Pericrocotus solaris*	Mitigation ponds constructed under DBL project[2]	Not recorded	Cap. 170; LC by Fellowes *et al*. (2002)	Common in winter, scarce in summer. Found in Tai Po Kau, Shing Mun, Ho Chung, Kadoorie Farm and Botanic Garden, Tung Ping Chau	Common
Zitting Cisticola	*Cisticola juncidis*	Ngau Hom Shek Fishpond[3]	Not recorded	Cap. 170; LC by Fellowes *et al*. (2002)	Common passage migrant and winter visitor. Widely distributed in grassland throughout Hong Kong	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Mammal						
Chinese Pangolin	*Manis pentadactyla*	Lower slopes adjacent to abandoned agricultural fields in Ngau Hom Shek and on a planted knoll to the west of Ha Tsuen near the hillside of Yuen Tau Shan [4]	Not recorded	Cap. 170; Cap. 586; RC by Fellowes *et al*. (2002); Class II Protection Status in China; China Red Data Book (Vulnerable); Red List of China's Vertebrates (Critically Endangered); IUCN Red List (Critically Endangered)	Thinly distributed in forested areas throughout Hong Kong	Rare
Small-toothed Ferret Badger	*Melogale moschata*	Lower slopes adjacent to abandoned agricultural fields in Ngau Hom Shek[4]	Not recorded	Cap. 170; Red List of China's Vertebrates (Near Threatened)	Widely distributed in forested areas throughout Hong Kong	Common
Greater Bandicoot Rat	*Bandicota indica*	Abandoned cultivated lands[7]	Not recorded	LC by Fellowes *et al*. (2002)	Records were found in Mai Po, Lantau and Pat Sin Leng Country Park	Rare
Japanese Pipistrelle	*Pipistrellus abramus*	Stream in Ngau Hom Shek[4]; urbanised area, abandoned cultivated lands; shrubs; fishpond; stream and plantation[7]	<u>Within the Project Area</u> Developed Area/Wasteland; Village/Orchard; Modified Watercourse; Plantation; Grassland <u>Outside the Project Area</u> Developed Area/Wasteland; Village/Orchard; Agricultural Area (Dry); Modified Watercourse; Plantation; Grassland; Woodland; Fishponds and Mitigation Ponds	Cap. 170; LC by Fellowes *et al*. (2002)	Widely distributed in Hong Kong	Abundant

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Short-nosed Fruit Bat	*Cynopterus sphinx*	Roosting beneath Chinese Fan-palm leaves on the edge of a small Ling To Monastery woodland at Ha Tsuen and on a planted knoll to the west of Ha Tsuen near the hillside of Yuen Tau Shan[4]	Village/Orchard within the Project area	Cap. 170; LC by Fellowes *et* al. (2002); China Red Data Book (Indeterminate); Red List of China's Vertebrates (Near Threatened)	Widely distributed in urban and forested areas throughout Hong Kong	Abundant
Chinese Noctule	*Nyctalus plancyi*	Not recorded	Within the Project Area Marsh; Fishponds and Mitigation Ponds Outside the Project Area Village/Orchard; Plantation	Cap. 170; PRC by Fellowes *et al.* (2002)	Widely distributed in forested areas throughout Hong Kong	Common
Myotis spp.	*Myotis* spp.	Not recorded	Developed Area/Wasteland outside the Project area	Cap. 170	-	-
Small Indian Civet	*Viverricula indica*	Not recorded	Grassland outside the Project area	Cap. 170; Class II Protection Status in China; Red List of China's Vertebrates (Vulnerable)	Widely distributed in forested areas throughout Hong Kong, except Lantau Island	Abundant
Small Asian Mongoose	*Herpestes javanicus*	Agricultural/village habitat along Deep Bay road near Ngau Hom Sha[4]	Not recorded	Cap. 170; Red List of China's Vertebrates (Vulnerable)	Found in the New Territories countryside area	Uncommon
Herpetofauna						
Two-striped Grass Frog	*Rana taipehensis*	Ngau Hom Shek in abandoned pond [4]	Not recorded	LC by Fellowes *et al.* (2002); Red List of China's Vertebrates (Near Threatened)	A few scattered sites in the New Territories and from several islands	Uncommon

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Burmese Python	*Python bivittatus*	Fishpond bund in Ngau Hom Shek[5]; Mitigation ponds constructed under DBL project[10]	Not recorded	Cap. 170; Cap. 586; PRC by Fellowes *et al*. (2002); Class I Protection Status in China; China Red Data Book (Critically Endangered); Red List of China's Vertebrates (Critically Endangered); IUCN Red List (Vulnerable)	Widely distributed throughout Hong Kong	-
Indo-Chinese Rat Snake	*Ptyas korros*	Ngau Hom Shek[4]	Not recorded	PRC by Fellowes *et al*. (2002); Red List of China's Vertebrates (Near Threatened)	Common and occurs in many types of habitats	Common
Common Rat Snake	*Ptyas mucosus*	Ngau Hom Shek[4]	Not recorded	Cap. 586; PRC by Fellowes *et al*. (2002); China Red Data Book (Endangered)	Widely distributed throughout Hong Kong	Common
Many-banded Krait	*Bungarus multicinctus*	Mitigation ponds constructed under DBL project[2]	Not recorded	PRC by Fellowes *et al*. (2002); China Red Data Book (Vulnerable); Red List of China's Vertebrates (Endangered)	Widely distributed in New Territories, Hong Kong Island and Lantau Island.	-
Chinese Bullfrog	*Hoplobatrachus rugulosus*	Fishpond bund at Ngau Hom Shek[13]	Not recorded	PRC by Fellowes *et al*. (2002); Class II Protection Status in China	Widely distributed in Lantau Island and New Territories	
Butterfly						
Dark Brown Ace	*Halpe porus*	Not recorded	Village/Orchard outside the Project area	LC by Fellowes *et al*. (2002)	Lam Tsuen, Tsiu Hang, Lung Kwu Tan	Very rare
Cornelian	*Deudorix epijarbas menesicles*	Not recorded	Village/Orchard within the Project area	-	Lam Tsuen, Shan Liu, Wu Kau Tang, Pak Sha O, Fung Yuen	Rare

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Danaid Eggfly	*Hypolimnas misippus*	Not recorded	Agricultural Area (Wet) outside the Project area	LC by Fellowes *et al*. (2002)	Widely distributed in Hong Kong	Uncommon
Common Cerulean	*Jamides celeno*	Not recorded	Grassland outside the Project area	-	Shek Pik, High Junk Peak, Shek Mun Kap, Fung Yuen, Pui O, Ma On Shan	Rare
Dragonfly						
Blue-spotted Dusk-hawker	*Gynacantha japonica*	Mitigation Ponds constructed under DBL project[4]	Not recorded	LC by Fellowes *et al*. (2002)	Hoi Ha, Ngong Ping, Sha Lo Tung and Wu Kau Tang	Common
Giant Hooktail	*Megalogomphus sommeri*	Western Ha Tsuen[4]	Not recorded	LC by Fellowes *et al*. (2002)	Sha Lo Tung, Sham Tseng, Tai Tong, Tsing Fai Tong, Upstream Tai Lam Chung Reservoir and Yeung Ka Tsuen	Common
Indochinese Copperwing	*Mnais mneme*	Stream habitat west of Ha Tsuen[3]	Not recorded	LC by Fellowes *et al*. (2002)	Widely distribute in woodland streams throughout the New Territories	Common
Scarlet Basker	*Urothemis signata*	Mitigation ponds constructed under DBL project[10]	Fishponds and Mitigation Ponds outside the Project area	LC by Fellowes *et al*. (2002)	Common in areas containing abandoned fishponds throughout Hong Kong	Common
Freshwater Community						
Predaceous Chub	*Parazacco spilurus*	Not Recorded	<u>Within the Project Area</u> Modified Watercourse <u>Outside the Project Area</u> Natural Watercourse; Modified Watercourse	China Red Data Book (Vulnerable); Red List of China's Vertebrates (Least Concern)	A widespread species occurring in most unpolluted hill streams in both upper and lower courses	Common

Common Name	Scientific Name	Previous Studies' Locations	Current Study's Location	Protection Status[1][2][14]	Distribution[2]	Rarity[2]
Intertidal						
Horseshoe Crab	*Tachypleus tridentatus*	Intertidal area at Deep Bay [3]	Not recorded	IUCN Red List (Data Deficient)	Restricted distribution in Hong Kong, mainly occur on coastline along Deep Bay and Lantau Island. Also found in northeastern New Territories and Lamma Island[12]	Rare
Horseshoe Crab	*Carcinoscorpius rotundicauda*	Intertidal area at Ngau Hom Shek[4]	Mudflat/Coastal Water Body outside the Project area	IUCN Red List (Data Deficient)	-	-

Notes:
(1) Fellowes *et al*. (2002): LC=Local Concern; RC=Regional Concern; PRC=Potential Regional Concern; PGC: Potential Global Concern. Letters in parentheses indicate that the assessment is on the basis of restrictedness in nesting and/or roosting sites rather than in general occurrence.
(Cap. 96) Forests and Countryside Ordinance
(Cap. 170) Wild Animals Protection Ordinance
(Cap. 586) Protection of Endangered Species of Animals and Plants Ordinance
(2) IUCN (2016).
(3) AFCD (2016).
(4) HyD (2002a).
(5) HyD (2002b).
(6) DSD (2002).
(7) CEDD (2009).
(8) Xing *et al*. (2000).
(9) AFCD (2005).
(10) AFCD (2016).
(11) AFCD (2012b).
(12) Li (2008); Shin *et al*. (2009).
(13) Chan and Caley (2003).
(14) PlanD (2000).
(15) Jiang *et al*. (2016).

9.5 Identification and Evaluation of Environmental Impacts

9.5.1 No marine works are required for the Project and the proposed development would only be undertaken within the Project area. Under the Revised RODP, the land zoning of most "Village Type Development" ("V") and "GB", and existing / planned developments under other projects within the Project area would be retained and remain the same as those on the existing OZPs. These areas would be excluded from the proposed phasing development under the current project as shown in **Figure 2.19**.

9.5.2 In addition to the Schedule 3 (Item 1) Designated Project (DP), the details of following DPs were presented in **Section 1.4** and **Figure 1.2** and **Figure 1.3**. A summary of the potential impacts of these DPs that have been included in the ecological impact assessment are presented in **Section 9.9** and **Table 9.46**.

- DP1 – Construction of new primary distributor road (Road P1)
- DP2 – Construction of eight new distributor roads (Roads D1 to D8)
- DP3 – Construction of new West Rail Hung Shui Kiu Station (Site 4-34)
- DP4 – Potential construction of Environmental Friendly Transport Services (EFTS) – subject to further review.
- DP5 – Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; junction of D8/P1 and junction of D7/P1; and KSWH connection to road D3.
- DP6 – Construction of partly depressed and partly decked-over roads located at Road D2, Road D4, and Road D6
- DP7 – Potential construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) – subject to further review.
- DP8 – Construction of new Hung Shui Kiu Sewage Treatment Works (Site 3-26 and part of existing San Wai STW)
- DP9 – Construction of four new sewage pumping stations (Sites 2-34, 3-41, 3-48 and 4-35)
- DP10 – Construction of flushing water service reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40)
- DP11 – Construction of a refuse transfer station (Site 3-12)
- DP12 – Construction of Road P1 and Slip-Road from Kong Sham Western Highway to Road D3 partly located within the "Conservation Area" of Yuen Tau Shan

9.5.3 The potential impacts of the development can be divided into the following categories:

- Direct Ecological Impacts
  - Habitat/vegetation loss;
  - Impact to species of conservation importance; and
  - Harm/mortality to wildlife.

- Indirect and Secondary Ecological Impact
  - Impacts to recognised sites of conservation importance and ecologically sensitive areas;
  - Habitat fragmentation;
  - Impacts to fauna species of conservation importance;
  - Disturbance impacts to habitats, fauna and vegetation; and
  - Water quality impacts.

Construction Phase - Direct Impacts

Habitat/Vegetation Loss

Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.5.4 Five sites of conservation importance were identified within the assessment area, namely "CA", "CPA", Ngau Hom Shek egretry, Shenzhen Bay Bridge egretry and San Sang San Tsuen egretry. As the "CPA", Shenzhen Bay Bridge egretry and Ngau Hom Shek egretry were located outside the Project area and San Sang San Tsuen egretry is located within the "GB" zoning within the Revised RODP that would be retained, no direct impact/habitat loss will arise.

9.5.5 A small section of Road P1 and a slip road at the northwest of San Sang San Tsuen are proposed within the edge of the "CA" (comprising 0.1 ha). This affected "CA" comprised developed area and no species of conservation importance were recorded. The affected "CA" was comprised of roads and reinforced concrete flood storage pond with limited plant species recorded (**Appendix 2.1.2** refers). All the recorded species are common in Hong Kong. As the construction works would be localised, the direct impact (i.e. loss of developed/wasteland habitat in "CA" zone) is anticipated to be negligible.

Other Habitats

9.5.6 No habitat loss of mangrove and mudflat/coastal water habitats would result since they were located outside of the Project area. The majority of woodland, shrubland, and plantation located within the "GB" zoning in the Revised RODP would be retained. The majority of "V" would be same as those on the existing OZPs, except the extension of boundary to the south of Tin Sam Tsuen (about 3.96 ha). The extended area is currently developed area and village / orchard habitat in "R(C)" zone in the existing OZPs. The loss of habitat arising from the proposed Project (i.e. except those to be retained within "GB" and those excluded areas in the "V" and existing / planned developments under other projects as allowed in the existing OZPs as showed in **Figure 2.19**) is summarised in **Table 9.28**.

Table 9.28 Area of Habitat Loss due to the Project

Habitat Type	Area of Permanent Habitat Loss	Area of Temporary Habitat Loss	Ecological Value of the Habitat Lost
Developed Area/Wasteland	335.27 ha	-	**Low**
Village/Orchard	61.60 ha	-	**Low**
Agricultural Area (Dry)	6.26 ha	-	**Low**
Agricultural Area (Wet)	1.33 ha	-	**Low**
Plantation	14.72 ha	-	**Low to Moderate** - for hillside plantation at Ngau Hom Shek and west of San Sang San Tsuen (2.37 ha) **Low** – for urban plantation (12.35 ha)
Woodland	0.21 ha	-	**Moderate** – for woodland at Ngau Hom Shek (0.10 ha) **Low to Moderate** – for woodland at west of Fung Kong Tsuen (0.11 ha)
Shrubland	2.84 ha	-	**Low to Moderate** – for shrubland at Ngau Hom Shek (0.25 ha) and at southeast of Yuen Long Highway (0.5 ha) **Low** - for remaining shrubland
Grassland	14.19 ha	-	**Low**
Marsh	0.56 ha	-	**Low**
Fishponds and Mitigation Ponds	0.66 ha	-	**Low** – for fishponds near San Sang San Tsuen (0.32 ha) and near San Wai (0.11 ha and 0.23 ha)
Natural Watercourse	<0.1 ha, 8 m	-	**Low** – for the disturbed lower sections of Watercourse 5
Modified Watercourse	1.36 ha, 8.52 km	Tin Sam Channel (0.2 ha, 500 m) TSW Main Channel (<0.1 ha)	**Moderate** - TSW Main Channel **Low** - for other modified watercourses
Total	**441.0 ha**	**0.2 ha**	**-**

9.5.7 Approximately 441 ha of habitat would be directly impacted during the construction phase; however, over 98% of the lost habitats are of low ecological value. No plant species of conservation importance were recorded within these habitats.

9.5.8 Only two small areas of woodland edges (comprising about 0.2 ha in total, approximately 0.4% of the total woodland within the assessment area) located at Ngau Hom Shek (0.1 ha) and at west of Fung Kong Tsuen (0.11 ha) would be lost. The ecological value of woodland at Ngau Hom Shek was moderate, while that at west of Fung Kong Tsuen was low to moderate. Most recorded species were common species. Potential impacts to Crested Serpent Eagle, which utilised the woodland at the west of Fung Kong Tsuen are discussed in **Section 9.5.23**. All the affected woodlands were subject to anthropogenic disturbance due to the presence of roads and graves. Given the relatively small proportion of habitat to be lost, the direct impact is considered to be minor. Nevertheless, an enhancement planting area with native species would be considered at the suitable area of the proposed Fung Kong Tsuen service reservoir site.

9.5.9 A small area of hillside plantation at Ngau Hom Shek (comprising 2.4 ha, approximately 2% of the total plantation within the assessment area) of low to moderate ecological value would be permanently lost; while the majority of urban plantation (comprising 12.4 ha, approximately 12% of the total plantation within the assessment area) of low ecological value. The affected urban plantations are scattered within the Project area. Within the affected area, most of the species recorded were common, with no plant but one mammal species of conservation importance (Japanese Pipistrelle) recorded in urban plantation. Due to its close proximity to developed area, this habitat was frequently subjected to human disturbance from graves, road and residential areas. Although Crested Serpent Eagle was recorded hovering over the plantation at the west of Fung Kong Tsuen, this species is known to utilise a range of habitats (refer to **Section 9.5.23**). Therefore the direct impact of plantation habitat loss is anticipated as minor.

9.5.10 Two small patches of shrubland of approximately 2.8 ha at Ngau Hom Shek knoll, east of Tseung Kong Wai and southeast of the Project area would be affected. These shrublands supported low species diversity with common plant species recorded. They were subjected to edge-effects from the road nearby (e.g. dust, noise). No species of conservation importance were recorded within the affected habitats. The loss of these two shrublands is considered to be minor.

9.5.11 A total of approximately 17.8 ha of wooded area, including woodland, plantation and shrubland habitats, would be lost under the development. Among them, only 16% of the total lost wooded area were of moderate or low to moderate ecological value, including woodland at Ngau Hom Shek (0.1 ha), woodland at west of Fung Kong Tsuen (0.11 ha), shrubland at southeast of Yuen Long Highway (0.5 ha), shrubland (0.25 ha) and plantation (2.4 ha) at Ngau Hom Shek (**Table 9.28** refers). The remaining affected wooded area were of low ecological value. These small areas of woodland/shrubland/plantation at Ngau Hom Shek, southeast of Yuen Long Highway and west of San Sang San Tsuen were linked to uphill habitats such as woodland, shrubland and plantation. However, they supported a relatively low floral and faunal diversity as they were already subjected to human disturbance from nearby roads and graves. Other affected wooded areas were either isolated/highly disturbed shrubland or urban plantation near developed area or villages, of low ecological value. No species of conservation importance were recorded within these affected areas. Given the above, the cumulative impact of habitat loss of wooded area is anticipated to be acceptable.

9.5.12 Under the Revised RODP, the land zoning use at Site 3-49 was revised to “GB” to preserve the natural watercourse (Watercourse 6) from direct loss under the development (**Figure 2.2.1** and **Figure 9.7D** refers). However, the lower sections of natural watercourse (Watercourse 5) within the Project area would be unavoidably affected under this Project (**Figure 9.7H** refers). The affected area was small in size (approximately 8 m) and subjected to a high level of disturbance from a nearby village housing and developed area. This affected stream section supported low floral and faunal diversity and no species of conservation importance were recorded. Therefore, the direct impact of habitat loss is anticipated to be minor.

9.5.13 Three fishponds located between San Wai and Fung Kong Tsuen (approximately 0.3 ha), and near San Sang San Tsuen (approximately 0.3 ha) would be lost (refer to **Figure 9.7G and 9.7I**). These fishponds were inactive and small in size. The fishpond near San Sang Tsuen contributed a potential foraging resource for the ardeids, but was relatively less important than TSW Main Channel. Low floral and faunal diversity were recorded. Five avifauna (Red-billed Starling, Grey Heron, Besra, Little Egret and Eastern Great Egret) and one mammal (Chinese Noctule) species of conservation importance were recorded. All the inactive fishponds were overgrown with herb species and the condition may not be favourable for ardeids. No plant species of conservation importance were recorded. The direct impact of habitat loss to a fishpond near San Sang San Tsuen and San Wai is considered to be minor.

9.5.14 The major modified watercourses (e.g. TSW Main Channel and its associated channels) would be retained, but minor modified watercourses and ditches would be filled, diverted, or culverted under this Project (**Figure 9.1,** and **Figure 5.2.1 to 5.2.14** refers). Approximately 7 km of modified watercourses would be directly affected. This man-made habitat supported limited vegetation. No plant species of conservation importance were recorded. A low number of species of conservation importance including ardeids, Red-billed Starling, Predaceous Chub and Japanese Pipistrelle were recorded at these modified watercourses. Given Predaceous Chub is a common and widespread species in Hong Kong, and only low abundance was recorded in the impacted Watercourse 3 and Watercourse 4, the direct impact is minor. Since the other minor modified watercourses are common habitat, the direct impact is considered as minor.

9.5.15 A section of Tin Sam Channel (approximately 500 m) which is located to the west of Tin Sam Tsuen (**Figure 9.7I** refers) would be realigned due to the Project. The newly constructed section would form the new western boundary of the "V". Opportunities for ecological enhancement (e.g. bioengineering, meandering alignment) would be considered during the detailed design stage. Loss of this section of modified watercourse would therefore be temporary in nature and the new design has the potential to result in an improvement to the ecological value; hence, the impact is minor adverse during construction, but could be minor positive at operation with an environmentally-friendly design.

9.5.16 Two footbridges and two cycle bridges are proposed across TSW Main Channel for pedestrian and cycling use (as shown in **Figure 2.2.1**). All the piers of these bridges would be land-based and located outside the watercourses during the operational phase. As these piers are located close to the bank of the watercourses, construction of these minor bridge piers would involve temporary construction activities (e.g. piling and excavation) in the watercourses. Less than 0.1 ha of the TSW Main Channel would be affected temporarily during the construction phase. Given the plantation on the riverside is of low ecological value and the removal would be confined to the work sites of the bridges only, the impact due to vegetation loss is minor. The temporary loss of modified watercourse habitat would be also minor as the channel would be resumed after the construction works.

9.5.17 The endangered Black-faced Spoonbill was recorded in TSW Main Channel outside the Project area. Only three individuals recorded in February 2012 (**Figure 9.3F** refers). The lower section of the TSW Main Channel outside the Project area is likely to provide higher quality habitat for wetland-dependent species, as it is located downstream of the inflatable dam (and hence subject to tidal influence and therefore periodically exposed mudflats) and closer proximity to the Deep Bay habitats. On the contrary, the Channel section within the Project area is less likely to provide important foraging habitat for Black-faced Spoonbill. Since the construction works of bridge piers would only involve temporary construction activities (e.g. piling and excavation), small in scale and localised, no significant impacts are anticipated for this species.

9.5.18 The majority of habitats that would also be lost under the Revised RODP were of low ecological value. These habitats mainly supported low species diversity with common species recorded. Most of the habitats were disturbed under the current condition and are described below.

9.5.19 Approximately 6 ha of dry agricultural area and 1 ha of wet agricultural area would be lost. Most of the recorded species within these small agricultural areas were common crop species. No plant species of conservation importance were recorded. A low abundance of fauna species of conservation importance were recorded within the affected dry and wet agricultural area, including Red-billed Starling, White-cheeked Starling, Little Egret and Greater Coucal. As these species also utilise other suitable habitats including watercourses and woodland available nearby, the direct impact of habitat loss is anticipated to be minor.

9.5.20 Approximately 14 ha of scattered grassland would be permanently lost within the Project area. This habitat was highly fragmented and supported low floral and faunal diversity, and most of the recorded species were very common or common in Hong Kong. No plant species of conservation importance were recorded, while five avifauna (Chinese Pond Heron, Little Egret, White-cheeked Starling, Red-billed Starling and Besra) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded in low abundance. As these mobile species are also able to use nearby habitats including watercourses and shrubland, the impact of grassland loss is considered as minor.

9.5.21 A patch of marsh (comprising 0.6 ha in total) located at Fung Kong Tsuen would be lost as a result of the Project. This habitat supported low floral and faunal diversity. No plant species of conservation importance were recorded; while one mammal species of conservation importance (Chinese Noctule) was recorded within the affected area. This habitat was subject to disturbance from village housing nearby (e.g. farming). Although this small area of mash is part of the territory of Crested Serpent Eagle, this species is known to utilise a range of habitats (refer to **Section 9.5.23**). The loss of this marsh is considered to be minor.

Impacts to Species of Conservation Importance

Plant Species of Conservation Importance

9.5.22 Seven plant species of conservation importance, including Incense Tree, Lamb of Tartary, Luofushan Joint-fir, Hong Kong Pavetta, Cycad-fern, Black Olive, and *Halophila beccarii*, were recorded within the assessment area during the current surveys and literature review. They are either located outside the Project area or within the retained land use zones (e.g. Incense Tree at the grassland southeast of Ngau Hom Shek, Hong Kong Pavetta at the hill near Sha Kong Wai, *Halophila beccarii* at coastal area near Sha Kong Tsuen) under the proposed development, except one individual of Incense Tree at Tung Tau Tsuen woodland. This affected mature individual of Incense Tree would be lost during the site formation for the Project. The impact is considered as minor to moderate, if unmitigated.

Fauna Species of Conservation Importance

Avifauna

9.5.23 Six and two repeated sightings of Crested Serpent Eagle hovering over woodland/plantation habitat, and perching in the trees in developed area/wasteland habitat at the west of Fung Kong Tsuen were recorded during the dry season, respectively. Therefore, the mosaic of plantation, woodland, marsh, developed area/wasteland habitat at the west of Fung Kong Tsuen (i.e. Sites 2-20, 3-1, 3-2, 3-3, 3-4, 3-5) is likely to form part of the Crested Serpent Eagle's territory. These habitats are currently subjected to human disturbance due to the close proximity of transitional storage areas. During the construction phase, a small area of woodland, plantation and marsh habitat would be permanently lost (covering approximately 1.6 ha in total). Moreover, construction noise, air quality deterioration and visual impact would occur. Crested Serpent Eagle is known to utilise a range of habitats such as woodland, rocky hilltops, marsh or abandoned wet agricultural area (Carey *et al*., 2002). Under the proposed development, only small parts of the potential territory would be affected, therefore; a minor impact would be anticipated.

9.5.24 Three small, isolated, abandoned fishponds would be lost. The pond near San Sang San Tsuen potentially contributed a foraging resource for nesting ardeids from San Sang San Tsuen egretry. Moreover, small area of other habitats including some sections of minor modified watercourses, marshes and wet agricultural areas, which acted as potential foraging grounds for other waterbirds, would be lost. The maximum count of waterbirds recorded at TSW Main Channel and other major modified watercourses was

467, while that recorded at the affected minor modified watercourses, marshes and wet agricultural areas was only 105 throughout the 12-month monitoring period. This may indicate that all these habitats were relatively less important than TSW Main Channel and other major modified watercourses, where the majority of waterbirds (including ardeids) were recorded. As TSW Main Channel and other major modified watercourses will be retained under the Project, the secondary impact of loss of foraging resources to waterbirds is considered to be minor.

Mammals

9.5.25 Three mammal species of conservation importance (Short-nosed Fruit Bat, Japanese Pipistrelle and Chinese Noctule) were recorded within the Project area. Some of the bat species recorded are known to roost in buildings or trees in developed area or village/orchard which may otherwise be of low ecological value. Given the large area of these potential habitats and high number of foraging/commuting bats recorded, it is likely that bats are using some buildings/trees for roosting within the Project area. All bats are protected under the Wild Animals Protection Ordinance. The demolition of buildings or removal of trees during the construction phase may impact roosting bats, if present. Given majority of "GB" and natural watercourses within the Project area would be retained, and there were no changes in the majority of existing "V" in the Revised RODP and precautionary measures stated in **Section 9.7.8**, the impact is likely to be minor.

Butterflies / Odonates

9.5.26 Only one butterfly species of conservation importance (Cornelian) at village/orchard habitat at Tung Tau Tsuen has the potential to be directly affected. As only one individual was recorded during the current survey, the impact is anticipated to be low. While, only one odonate species of conservation importance (Scarlet Basker) was recorded; however, it was located outside the Project area and no impact would be anticipated.

Aquatic Communities

9.5.27 Predaceous Chub recorded in the upper section of Watercourse 3 and lower section of Watercourse 4 would be directly impacted by the Project. Given Predaceous Chub is a common and widespread species in Hong Kong, and only low abundance was recorded in the impacted Watercourse 3 and Watercourse 4, the direct impact is minor. Moreover, although downstream of Watercourse 3 would be retained, the removal of the upper section of watercourses could also result in water quality deterioration and impact the Predaceous Chub (refer to **Section 9.5.43**).

Intertidal Communities

9.5.28 Intertidal fauna species of conservation importance (*Tachypleus tridentatus* and *Carcinoscorpius rotundicauda*) were recorded outside the Project area during the current surveys and literature review. Due to the distance between the proposed development and their recorded location, no direct impact was anticipated.

Direct Harm/Mortality to Wildlife

9.5.29 Construction phase activities (e.g. site clearance and formation) have the potential to cause direct injury/mortality to wildlife. Within the Project area, most recorded species were common and widespread in Hong Kong, in low abundance, and highly mobile. Any impacts are unlikely to have a significant effect on population numbers therefore the impact is considered as minor.

Construction Phase - Indirect and Secondary Ecological Impacts

Impacts to Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.5.30 Since Shenzhen Bay Bridge egretry, Ngau Hom Shek egretry and the "CPA" were separated from the Project area by a vegetated buffer of approximately 200 to 300 m, the site formation and construction works are unlikely to indirectly affect these sites of conservation importance. Although San Sang San Tsuen egretry would be retained, construction activities (e.g. noise, dust and visual impacts) would likely disturb any nesting ardeids including the use of flight paths to and from foraging grounds. As the egretry is small in the Hong Kong context, currently subject to disturbance, and given the transitional nature of egretries in the area, a minor to moderate impact is anticipated.

9.5.31 Given the close proximity of semi-natural/natural habitats comprising the "CA" to the proposed works areas, (e.g. west of San Sang San Tsuen and southeast of the Yuen Long Highway) the "CA" habitats and associated fauna may be subjected to indirect disturbance impacts (e.g. arising from construction vehicles, noise and vibration, air/dust). However, given the temporary nature of the construction works and that these areas are already subject to human disturbance (i.e. located near open storage areas/roads), the impact is considered as minor.

Disturbance Impacts to Habitat, Fauna and Vegetation

9.5.32 Construction activities would increase human activities, noise and vibration disturbance from traffic and construction machinery, and result in indirect and secondary impacts to nearby habitats (i.e. woodland, shrubland and major modified watercourses) and their associated fauna. However, the majority of recorded species within the assessment area were generalist species habituated to various levels of disturbance and that utilise a wide range of habitats. Moreover, the construction phase would be divided into four stages to further minimise disturbance impacts (refer to **Chapter 2.9**). Therefore, the disturbance impact is unlikely to be significant.

9.5.33 The construction of the four bridges across TSW Main Channel and its tributary would generate potential water quality impact to the channel and could affect the wetland-dependent species utilising this area including some recorded fauna species of conservation importance. Six avifauna species of conservation importance (Black-faced Spoonbill, Northern Shoveler, Chinese Pond Heron, Grey Heron, Eastern Great Egret and Little Egret) were recorded foraging and/or commuting near the bridge location. Disturbance impacts (e.g. noise/vibration and visual) during the bridge construction could also reduce the use of the channel by avifauna. This is particularly pertinent for the proposed bridge adjacent to Site 1-4 given the higher use of the TSW Main Channel at the downstream section. As the scale of the works would only affect a small area of the channel and the impact would be temporary, without mitigation measures, the impact to the wetland-dependent species of conservation importance would be low.

9.5.34 TSW Main Channel is important for breeding birds (e.g. Little Egret) as they are typically restricted in their foraging locations and have to forage more efficiently, or for longer, to collect sufficient food for themselves and their young. The site formation and construction works of the four bridges could result in disturbance impacts. Although the channel provides a foraging resource for ardeids from the San Sang San Tsuen egretry, the construction works of bridge piers would be temporary, small in scale and localised, the construction impact is anticipated to be low.

9.5.35 A total of 128 individuals of 12 avifauna species, either migrants or overwintering species, were recorded at the TSW Main Channel during the overwintering period (October to March). TSW Main Channel is structurally linked to the Mai Po Inner Deep Bay Ramsar Site and may contribute as a potential foraging ground for migrants and

overwintering waterbirds. However, the utilisation of TSW Main Channel by the migratory birds was relatively low, when compared to the Mai Po Inner Deep Bay Ramsar Site (over 100,000 migratory waterbirds recorded each year) (AFCD, 2015d). Moreover, the construction works of bridge piers would be temporary, small in scale and localised. Thus, the construction impact is anticipated to be minor.

9.5.36 Given the intertidal habitats, including mangrove, mudflat and coastal water body, were located more than 200 m away from the Project area, construction disturbance (i.e. construction dust, increase of human activities) is unlikely to cause significant impact to the intertidal habitats and its associated fauna.

9.5.37 Dust generated during the construction phase (e.g. construction machinery, improper storage or dumping of construction materials) could degrade the habitats adjacent to works areas. Construction dust could cover the leaves of adjacent habitats and may affect photosynthesis, respiration, transpiration and allow the penetration of phytotoxic gaseous pollutants, thus changing the vegetation structure and / or reducing the quality of habitat (Farmer, 1993). Details of the air quality assessment are separately presented in **Chapter 3**. The predicted dust impact to natural habitats (e.g. "CA" at west of San Sang San Tsuen, Ngau Hom Shek knoll) would be acceptable during the construction phase, with proper implementation measures implemented (e.g. recommended dust suppression measures and good site practices).

9.5.38 Artificial lighting (glare) has been shown to affect some wildlife and can result in a reduction in the density of a faunal population in an area through disorientation from, and attraction to artificial light, and effects on the light-sensitive cycles of a species (e.g. bats, birds). This can affect migration, foraging/predation and breeding success of species. However, since the majority of the Project area was already developed (i.e. developed area or village/orchard), the surrounding habitats (e.g. woodland, shrubland, plantation) were already disturbed by the artificial lighting from existing residential building, roads and village housings under the current condition. Given most recorded nocturnal species were common and generalist and habituated to various levels of disturbance, the impact is minor.

9.5.39 The mitigation ponds would be subjected to construction disturbance (i.e. noise and increase of human activity, during the construction of slip roads between Road P1 and the KSWH under DP5), which could discourage their use, particularly by avifauna. However, as this habitat was already highly disturbed from the existing roads and open storage area, the expected disturbance during the construction would be similar with the current condition. As the majority of recorded species were species habituated to various levels of disturbance and utilise a wide range of habitats, the impact is anticipated as minor.

Water Quality Impacts

9.5.40 Site formation and construction works which could affect the water quality of the watercourses and ponds include.

Accidental Spillage

9.5.41 Accidental spills of oil, fuel and other chemicals could affect aquatic communities at retained watercourses as well as downstream intertidal flora and fauna at Deep Bay area. This could result in lethal/sublethal impacts (abnormal structures and reproductive retardation) on aquatic organisms. Larger particles could cause physical injury to aquatic organisms, while small particles could clog the respiratory and feeding systems of fish and invertebrates. TSW Main Channel and Deep Bay area are potential foraging resources for waterbirds (including species of conservation importance such as Little Egret) therefore water quality impacts could indirectly impact these species.

Removal and Realignment of Modified Watercourses

9.5.42 During the construction phase, a small section of Tin Sam Channel (approximately 500 m) will be realigned under this Project (refer to **Section 5.4**). A section of existing watercourse would be diverted and replaced by new open water channel (refer to **Figure 5.2.9**). Downstream water quality impacts could potentially affect foraging resources for avifauna species.

9.5.43 A small section of Watercourse 3 will be modified as part of the covered drainage network within the new development area (refer to **Section 5.4**). The proposed works could impact the water quality as well the fauna (e.g. Predaceous Chub) at downstream areas. Without mitigation, the impact is anticipated to be minor. To avoid potential water quality impacts downstream, it is recommended that construction works are undertaken during the dry season.

Construction Works near Watercourses and Ponds

9.5.44 Construction works near retained watercourses and mitigation ponds may pollute the waters due to potential release of construction waste. Construction waste are generally characterised by high concentration of suspended solids and elevated pH, which may result in lethal/sublethal impacts (e.g. clogging respiratory and feeding systems of aquatic organisms, extreme acidic/alkaline aquatic environment) to aquatic communities within watercourses, mitigation ponds and downstream intertidal area at Deep Bay.

9.5.45 As stated in **Chapter 5**, the water quality impacts can be controlled by implementing the recommended mitigation measures. Inland water quality monitoring and regular site inspections should be undertaken to ensure that any unacceptable water quality impacts due to the construction activities could be readily detected and timely action be taken to rectify the situation. With the mitigation measures suggested in **Section 5.14**, no significant indirect ecological impact from deterioration of water quality is anticipated during the construction phase.

Operational Phase - Direct Impacts

Impacts to Fauna Species / Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.5.46 The majority of the nesting ardeids at the San Sang San egretry were recorded to have a northeastward flight path towards the major watercourses such as TSW Main Channel. The provision of a wide eco-corridor comprising “Local Open Space” (“LO”) (Site 3-37 and 3-36) (approximately 100 m wide) and “GB” (Site 3-35) to the northeast of the egretry would accommodate the majority of the flight paths observed (approximately 70% of the total flight paths) (**Appendix 9.13** refers). As the proposed “LO” would only be used for passive recreational uses (e.g. local park, plant nursery) and integrated with landscape features which are in favour of the egrets (e.g. planting of trees and bamboo suggested in **Section 9.8.2**), it would provide a suitable buffer from adjacent buildings in “Enterprise and Technology Park” zoning which comprised of buildings up to 10 storeys in height. Any recreational facilities/uses that would attract human activities in close proximity of the egretry would be avoided during the detailed design stage of the “LO”. Thus, significant impact is therefore not anticipated.

Harm to Wildlife

9.5.47 Increased risk of bird collision with man-made structures, which causes direct harm to or even the mortality of birds, is anticipated during the operational phase of the Project. Bird collision can be caused by transparent noise barriers on the road network, excessive use of transparent/reflective building façade and excessive artificial lighting during daytime and nighttime. Although most recorded species were common and generalist that habituated to various levels of disturbance, due to the extent of development throughout the Project area, the impact is anticipated to be low to moderate. Therefore, precautionary measures, such as wildlife friendly lighting installation, use of tinted materials and superimposing dark patterns on noise barrier as stated in **Sections 9.7.16 - 9.7.17,** are recommended to minimise the disturbance.

Operational Phase - Indirect and Secondary Ecological Impacts

Impacts to Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.5.48 San Sang San Tsuen egretry is situated within the "GB" zone within the Revised RODP and will not be directly impacted by the development. The nearest distance between the egretry and the proposed "Enterprise and Technology Park" site boundary is about 25 m. During the operational phase, this habitat may be subjected to human, noise and visual disturbance generated from the surrounding facilities (e.g. daily operation of the "Enterprise and Technology Park" and human activities in the "LO"). However, given that the egretry is currently surrounded by open storage area with frequent human disturbance (refer to **Appendix 9.12**), it is expected that inclusion of the egretry within "GB" of the Revised RODP buffered with "LO" zones compared to its existing zoning of "R(D)" would be an improvement upon the current condition. The operation of the proposed "Enterprise and Technology Park" and human activities in the "LO" located to the southeast could have minor disturbance impacts to the egretry.

9.5.49 The "CA" at west of San Sang San Tsuen is located adjacent to the new "Industry" ("I") zone. In addition, part of Road P1 and a slip road would be constructed at the developed area within this "CA" (refer to **Section 9.5.5**). Indirect impacts including noise, air quality deterioration, visual impact generated from the above facilities/infrastructure are anticipated. However, this "CA" was already subjected to high levels of disturbance from existing open storage and roads under the current condition. Moreover, the plantation area of low ecological value at the slope toe acts as a buffer zone to minimise the noise, air and visual impacts. Therefore, the impact is considered as minor.

Habitat Fragmentation

9.5.50 The majority of retained habitats, including village/orchard, agricultural area (dry and wet), ponds and woodland were already highly fragmented within the Project area under the current condition. Most were surrounded by developed areas, which comprised approximately 75% of the Project area, and had limited ecological linkage with other natural habitats such woodland and shrubland. Habitat loss during the construction phase is therefore unlikely to significantly increase habitat fragmentation in the assessment area.

Impacts to Flora and Fauna Species of Conservation Importance

9.5.51 During the operational phase, human activities would increase at west of Fung Kong Tsuen (i.e. the planned "Other Specified Use" ("OU") and "Government" ("G") land zoning). This may cause indirect impacts from noise, glare and visual impacts to Crested Serpent Eagle as its territory is located nearby. Given the Crested Serpent Eagle was already subjected to high levels of disturbance from the existing open storages, roads and residential buildings under the current condition. As the current situation was already disturbed, minor impact is anticipated.

9.5.52 TSW Main Channel provides foraging resources for waterbirds including some of the avifauna species of conservation importance. During the operational phase, disturbance impacts from the newly developed areas (e.g. pedestrian walkway along the riverside and the four bridges) may occur. However, as TSW Main Channel was already disturbed by the existing features including inflatable dam, bridges and pedestrian walkways, it is expected the level of the disturbance during the operational phase would be similar to the current condition. As during the 12-month monitoring period, waterbirds (including species of conservation importance) were commonly recorded near the existing bridge and inflatable dam, this may indicate these species were already adapted to the existing disturbance. In addition, the construction of new bridges across the channel will reduce the distance to approximately 250 - 350 m between the existing and proposed bridges over the channel. Given the new separation and the height of the proposed bridges would be similar to those existing which were also commonly utilised by waterbirds for foraging, including four wetland dependent bird species of conservation importance (e.g. section at south of San Uk Tsuen). The new separation of bridges are not likely to significantly affect the avifauna flight path and affect the channel utilisation, including overwintering/migratory waterbirds and nesting ardeids. Therefore, a minor impact is anticipated.

9.5.53 One seagrass, twenty-two avifauna and two Horseshoe Crab species of conservation importance, were recorded at mangrove, mudflat and coastal water body habitats (**Sections 9.4.126** and **9.4.127** refers). During operational phase, increase in population arising from the Project may result in disturbance such as air quality deterioration, increase of traffic and artificial lighting to the intertidal habitats and its associated flora and fauna. However, Deep Bay Road is already frequently used by vehicles, and the disturbance level caused by Project generated traffic during the operational phase is anticipated to be similar to the current condition. Moreover, given the intertidal habitats were located more than 200 m away from the Project area and buffered by developed area (e.g. Sha Kong Tsuen and Hang Hau Tsuen) and Ngau Hom Shek knoll, the impact of air quality deterioration and artificial lighting is anticipated to be acceptable.

Disturbance Impacts to Habitat, Fauna and Vegetation

9.5.54 During the operational phase, the increased traffic flow and human activities from the newly developed residential and commercial areas could result in disturbance to the habitat, fauna and flora within and in the vicinity of the Project area. The associated noise pollution, glare, deterioration in air quality can affect the quality of the habitats and the fauna. For instance, light pollution (glare) is expected to increase during the operational phase from residential, commercial, street and traffic lighting. Artificial lighting can disorient wildlife or cause disturbance to their diurnal cycle (e.g. bat, mammal, bird species). However, since most of the habitats within the Project area were already subjected to human disturbance and artificial lighting under the current condition, the disturbance impact during the operational is minor.

9.5.55 During the operational phase, given the intertidal habitats (i.e. Deep Bay area) and the associated fauna were located more than 200 m away from the Project area and buffered by developed area (e.g. Sha Kong Tsuen and Hang Hau Tsuen) and Ngau Hom Shek knoll, any indirect impact including air/dust, noise and artificial lighting is anticipated to be minor.

9.5.56 All four mitigation ponds near San Sang San Tsuen would be retained under the development, currently these ponds are bisected by the KSWH and the ecological value is considered low to moderate. The nearest distance between the proposed new Road P1 and the existing vegetation at pond bund was approximately 4 m. The presence of the new Road P1, the slip road structure (linking the existing KSWH to the at-grade Road D3) and associated traffic may reduce the use of the ponds by avifauna and cause shadowing effect to the vegetation and pond. However, these mitigation ponds were already highly disturbed by the existing roads, and the recorded species were common and disturbance-tolerant. The impact is anticipated to be minor.

9.5.57 The majority of existing brownfield operations would be planned to relocate at the northwest and southwest part of the Project area (i.e. the "Port Back-up Storage and Workshop Uses" ("PBU & SWU") and "Industry" ("I")). Given both the northwest and southwest habitats were already subjected to a high level of disturbance (i.e. noise human disturbance, air/dust) from nearby construction works and open storage area under the current condition, the disturbance level is anticipated to be similar to the current condition during the operational phase.

Water Quality Impacts

9.5.58 As the existing brownfield operations within the Project area would be relocated into the "PBU & SWU" and "I" zonings with proper drainage system, direct discharge of contaminated water or sewage to watercourses could be minimised. Any contaminated water or sewage would be either properly treated or properly recharged into the ground under various blue-green infrastructure and best storm water management practices (**Section 5.14** refers). Therefore, no significant adverse water quality is anticipated for the relocation of brownfield operations.

9.5.59 No adverse water quality impacts are expected to arise to North Western WCZ or the TSW Main Channel and its downstream mudflat and mangrove habitats, as well as Deep Bay at the Deep Bay WCZ. As mentioned in the Water Quality Impact Assessment (**Section 5**), discharge into the North Western and Deep Bay WCZ would be kept within the discharge limit of the previously approved San Wai Sewage Treatment Works EIA report. As such, it is assumed that the pollution load that would be discharged into these WCZs would be in compliance with water quality standards and thus would not significantly deteriorate the water quality of these marine waters.

9.5.60 As stipulated in **Section 5.12**, sewage generated from the Project during the operational phase would be treated in the new HSK STW, and discharged at Urmston Road Outfall. No sewage would be discharged to the TSW Main Channel and therefore no adverse sewage impacts would be induced. Meanwhile, as discussed in **Section 5.11.46**, this Project would not increase in the paved surface and rainwater run-off in the Project area. Thus, no adverse impact on the drainage regime in TSW Main Channel and the overall flow regime in Inner Deep Bay would result from the Project. The use of reclaimed water from the new HSK STW is proposed for toilet flushing and irrigation. With the mitigation measures suggested in **Section 5.14** and **Section 9.7.18 - 9.7.20**, such as adequate storm water drainage system, blue-green infrastructure and standard storm water management practices, no significant adverse indirect ecological impact from deterioration of water quality and hydrological condition (e.g. drainage flow, velocity, water depth, salinity) in TSW Main Channel and its downstream Deep Bay is anticipated during the operational phase.

Prediction and Evaluation of Ecological Impacts

9.5.61 Potential ecological impacts (unmitigated) to habitats in the assessment area resulting from the Project have been evaluated according to Table 1 of Annex 8 of the EIAO-TM, and are summarised in **Table 9.29** to **Table 9.43** below. The severity of direct and indirect impacts on each habitat type during construction and operational phase is represented in **Table 9.44**. A summary of potential impacts to sites and species of conservation importance is presented in **Table 9.45**.

Table 9.29 Potential Ecological Impacts to Developed Area/Wasteland within the Assessment Area

Criteria	Impacts to Developed Area/Wasteland
Habitat Quality	Moderate - for area supporting San Sang San Tsuen egretry was, and this area would be retained Low - for all other developed area/wasteland
Species	Moderate floral and faunal diversity, but most of the recorded flora species were planted Current Surveys No plant species of conservation importance were recorded Ten avifauna (Chinese Pond Heron, Little Egret, Black Kite, Crested Serpent Eagle, Besra, Eastern Buzzard, Greater Coucal, Collared Crow, Red-billed Starling and White-cheeked Starling), two mammal (Japanese Pipistrelle and *Myotis* spp.) and one butterfly (Grass Demon) species of conservation importance were recorded Literature Review Six avifauna (Little Grebe, Chinese Pond Heron, Black Kite, Greater Coucal, Collared Crow and Red-billed Starling), two mammal (Greater Bandicoot Rat and Japanese Pipistrelle) and one odonate (Giant Hooktail) species of conservation importance were recorded
Size / Abundance	Direct impact to approximately 338.94 ha of this habitat would be permanent
Duration	Direct impact of habitat loss within footprint of proposed development would be permanent Indirect impact (noise, human and construction dust) during the construction phase would be temporary Indirect impact (noise and human disturbance) during the operational phase would be permanent
Reversibility	Direct impact within footprint of proposed development would be irreversible Construction phase disturbance to San Sang San Tsuen egretry and flight paths would be temporary and reversible Operational phase disturbance from “Enterprise and Technology Park” and “LO” to San Sang San Tsuen egretry would be permanent and irreversible
Magnitude	Minor to Moderate - for San Sang San Tsuen egretry during construction Minor - for San Sang San Tsuen egretry during operation Negligible - for the remaining habitat
Overall Impact Evaluation before mitigation	**Minor to Moderate** - for San Sang San Tsuen egretry due to small and disturbed nature of egretry during construction **Minor** - for San Sang San Tsuen egretry due to small and disturbed nature of egretry during operation **Negligible** - for the remaining developed area/wasteland

Table 9.30 Potential Ecological Impacts to Village/Orchard within the Assessment Area

Criteria	Impacts to Village/Orchard
Habitat Quality	Low
Species	Low to moderate floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Six avifauna (Chinese Pond Heron, Little Egret, Crested Goshawk, Greater Coucal, Red-billed Starling and White-cheeked Starling), three mammal (Short-nosed Fruit Bat, Japanese Pipistrelle and Chinese Noctule) and three butterfly (Cornelian, Dark Brown Ace and Grass Demon) species of conservation importance were recorded <u>Literature Review</u> Seven avifauna species of conservation importance (Chinese Pond Heron, Little Egret, Black Kite, Eastern Buzzard, Greater Coucal, Collared Crow and Red-billed Starling) were recorded
Size / Abundance	Approximately 61.60 ha would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Temporary indirect impact (noise and vibration, construction dust) to the remaining village/orchard during the construction phase would be anticipated Permanent indirect impact (noise and human disturbance) to the remaining village/orchard during the operational phase would be anticipated
Reversibility	Direct impact within the proposed development would be irreversible Indirect impact during the construction phase would be temporary and reversible Indirect impact during the operational phase would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.31 Potential Ecological Impacts to Agricultural Area (Dry) within the Assessment Area

Criteria	Impacts to Agricultural Area (Dry)
Habitat Quality	Low
Species	Low floral and faunal diversity Current Surveys No plant species of conservation importance were recorded Five avifauna (Chinese Pond Heron, Little Egret, Greater Coucal, Red-billed Starling and White-cheeked Starling) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded Literature Review Three avifauna (Black Kite, Eastern Buzzard and Peregrine Falcon) and one mammal species of conservation importance (Small Asian Mongoose) was recorded
Size / Abundance	Approximately 6.26 ha would be lost permanently
Duration	Direct impact within the footprint of the proposed development would be permanent Construction disturbance (e.g. noise and vibration, air quality/dust) to the remaining dry agricultural area at Tin Sam Tsuen would be temporary Operational phase impact (noise and human disturbance) to the remaining dry agricultural area at Tin Sam Tsuen would be permanent
Reversibility	Direct impact within the proposed development would be irreversible Construction disturbance to the remaining dry agricultural area at Tin Sam Tsuen would be temporary and reversible Operational phase impact during to the remaining dry agricultural area at Tin Sam Tsuen would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.32 Potential Ecological Impacts to Agricultural Area (Wet) within the Assessment Area

Criteria	Impacts to Agricultural Area (Wet)
Habitat Quality	Low
Species	Low floral and faunal diversity Current Surveys No plant species of conservation importance were recorded Three avifauna (Greater Coucal, Red-billed Starling and White-cheeked Starling) and one butterfly (Danaid Eggfly) species of conservation importance were recorded Literature Review Four avifauna species of conservation importance (Chinese Pond Heron, Grey Heron, Eastern Great Egret and Little Egret) were recorded
Size / Abundance	Approximately 1.33 ha would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Short-term disturbance (noise, construction dust, fragmentation) to wet agriculture area at Sik Kong Wai during the construction phase Permanent disturbance impact (noise, human disturbance, fragmentation) to wet agriculture area at Sik Kong Wai during the operational phase
Reversibility	Direct impact within the proposed development would be irreversible. Construction disturbance to wet agriculture area at Sik Kong Wai during would be temporary and reversible Indirect impact during the operational phase to wet agricultural area at Sik Kong Wai would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.33 Potential Ecological Impacts to Plantation within the Assessment Area

Criteria	Impacts to Plantation
Habitat Quality	Low to moderate - for hillside plantation at Ngau Hom Shek and west of San Sang San Tsuen Low - for urban plantation
Species	Low to moderate floral diversity Low faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Five avifauna (Chinese Pond Heron, Grey Heron, Little Egret, Crested Serpent Eagle and Greater Coucal) and two mammal (Japanese Pipistrelle and Chinese Noctule) species of conservation importance were recorded <u>Literature Review</u> Four mammal species of conservation importance (Chinese Pangolin, Small-toothed Ferret Badger, Japanese Pipistrelle and Short-nosed Fruit Bat) were recorded The plantation west of Fung Kong Tsuen was a part of potential territory of Crested Serpent Eagle as repeated sightings of the species hovering over the habitat were recorded
Size / Abundance	Approximately 2.37 ha of hillside plantation at Ngau Hom Shek and west of San Sang San Tsuen; and 12.35 ha of urban plantation would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Construction disturbance (noise and vibration, air quality/dust and human disturbance) to the retained plantation at west of Fung Kong Tsuen, Hung Uk Tsuen and the “CA” near San Sang San Tsuen would be temporary Operational phase disturbance (noise and human disturbance) to the retained plantation at Ngau Hom Shek, west of Hung Uk Tsuen and the “CA” near San Sang San Tsuen would be permanent
Reversibility	Direct impact within the proposed development would be irreversible Indirect impact during the construction phase to the retained plantation at Ngau Hom Shek, Hung Uk Tsuen and the “CA” near San Sang San Tsuen would be temporary and reversible Indirect impact to the retained plantation at Ngau Hom Shek, Hung Uk Tsuen and the “CA” near San Sang San Tsuen during the operational phase would be minor
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.34 Potential Ecological Impacts to Woodland within the Assessment Area

Criteria	Impacts to Woodland
Habitat Quality	Moderate - for woodland at hillside area at Ngau Hom Shek Low to moderate - for woodland at Tung Tau Tsuen and west of Fung Kong Tsuen
Species	Typically supported moderate floral diversity and higher structural complexity <u>Current Surveys</u> Two plant species of conservation importance (Incense Tree and Luofushan Joint fir) was recorded. The individual of Incense Tree at Tung Tau Tsuen would be preserved together with the woodland Four avifauna (Crested Serpent Eagle, Common Emerald Dove, Greater Coucal and Rufous-capped Babbler) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded <u>Literature Review</u> Two plant species of conservation importance (Incense Tree, Black Olive) were recorded Three avifauna (Black Kite, Crested Serpent Eagle and Crested Goshawk), one mammal (Short-nosed Fruit Bat) and two herpetofauna (Indo-Chinese Rat Snake and Common Rat Snake) species of conservation importance were recorded The woodland west of Fung Kong Tsuen was a part of the potential territory of Crested Serpent Eagle as repeated sightings of the species hovering over the habitat were recorded
Size / Abundance	0.1 ha of woodland at Ngau Hom Shek; 0.11 ha of woodland at west of Fung Kong Tsuen would be lost permanently
Duration	Direct impact (habitat loss) within the footprint of the proposed development would be permanent Indirect impact (noise and vibration, air/dust) to retained woodlands at Ngau Hom Shek, Tung Tau Tsuen and west of Fung Kong Tsuen during the construction phase would be temporary Indirect impact (fragmentation, noise and human disturbance) to retained woodlands at Ngau Hom Shek, Tung Tau Tsuen and west of Fung Kong Tsuen during the operational phase would be permanent
Reversibility	Direct impact within the proposed development would be irreversible Indirect impact to retained woodlands at Ngau Hom Shek, Tung Tau Tsuen and west of Fung Kong Tsuen during the construction phase would be temporary and reversible Indirect impact to retained woodlands at Ngau Hom Shek, Tung Tau Tsuen and west of Fung Kong Tsuen during the operational phase would be permanent and irreversible
Magnitude	Minor to moderate
Overall Impact Evaluation before mitigation	**Minor** – for woodlands at Ngau Hom Shek, Tung Tau Tsuen and west of Fung Kong Tsuen

Table 9.35 Potential Ecological Impacts to Shrubland within the Assessment Area

Criteria	Impacts to Shrubland
Habitat Quality	Low to moderate - for shrubland at hillside areas at Ngau Hom Shek and southeast of Yuen Long Highway as well as outside the Project area Low - for other shrubland within the Project area near developed area
Species	Low to moderate floral and faunal diversity <u>Current Surveys</u> Four plant species of conservation importance (Lamb of Tartary, Incense Tree, Luofushan Joint-fir and Hong Kong Pavetta) were recorded, all of which were located outside the Project area One avifauna (Greater Coucal) and one mammal (Japanese Pipistrelle) species of conservation importance was recorded <u>Literature Review</u> Five avifauna (Chinese Pond Heron, Black Kite, Crested Serpent Eagle, Crested Goshawk and Peregrine Falcon) and one mammal (Japanese Pipistrelle) species of conservation importance were recorded
Size / Abundance	Approximately 0.25 ha of shrubland at Ngau Hom Shek, 0.5 ha at southeast of Yuen Long Highway and 2.09 ha shrubland at urban area would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Temporary construction disturbance (noise and vibration, air/dust) to retained shrubland at Ngau Hom Shek, "CA" near San Sang San Tsuen and southeast of Yuen Long Highway Permanent operational phase disturbance (fragmentation, noise and human disturbance) at Ngau Hom Shek, the "CA" near San Sang San Tsuen and southeast of Yuen Long Highway
Reversibility	Direct impact within the proposed development would be irreversible Construction disturbance to retained shrublands at Ngau Hom Shek, the "CA" near San Sang San Tsuen and southeast of Yuen Long Highway would be temporary and reversible Operational phase disturbance to retained shrublands Ngau Hom Shek, the "CA" near San Sang San Tsuen and southeast of Yuen Long Highway would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.36 Potential Ecological Impacts to Grassland within the Assessment Area

Criteria	Impacts to Grassland
Habitat Quality	Low
Species	Low floral and faunal diversity <u>Current Surveys</u> One plant (Incense Tree) species of conservation importance was recorded Six avifauna (Chinese Pond Heron, Little Egret, Besra, Greater Coucal, Red-billed Starling and White-cheeked Starling), two mammal (Japanese Pipistrelle and Small Indian Civet) and one butterfly (Common Cerulean) species of conservation importance were recorded <u>Literature Review</u> One plant species of conservation importance (Cycad-fern) was recorded
Size / Abundance	Approximately 14.19 ha would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Construction disturbance (noise and vibration, air quality/dust) to the retained grassland at Sik Kong Wai and west of Fung Kong Tsuen would be temporary Operational phase disturbance (noise and human disturbance) to the retained grassland at Sik Kong Wai and west of Fung Kong Tsuen would be permanent
Reversibility	Direct impact within the proposed development would be irreversible Construction disturbance to the retained grassland at Sik Kong Wai and west of Fung Kong Tsuen would be temporary and reversible Operational phase disturbance to the retained grassland at Sik Kong Wai and west of Fung Kong Tsuen would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.37 Potential Ecological Impacts to Marsh within the Assessment Area

Criteria	Impacts to Marsh
Habitat Quality	Low
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded One avifauna (Chinese Pond Heron) and one mammal (Chinese Noctule) species of conservation importance were recorded <u>Literature Review</u> One mammal (Japanese Pipistrelle) and one herpetofauna (Two-striped Grass Frog) species of conservation importance were recorded The marsh at Fung Kong Tsuen was recorded as part of the potential territory of Crested Serpent Eagle
Size / Abundance	Only the larger marsh of 0.56 ha at west of Fung Kong Tsuen would be lost permanently
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent Construction disturbance (noise and vibration, air quality/dust) to the remaining marshes at Sik Kong Wai and Fung Kong Tsuen and Hang Hau Tsuen would be temporary Operational phase disturbance (noise and human disturbance) to the remaining marshes at Sik Kong Wai and Fung Kong Tsuen and Hang Hau Tsuen would be permanent
Reversibility	Direct impact within the proposed development would be irreversible
Magnitude	Minor - for larger Fung Kong Tsuen marsh Negligible - for other mashes
Overall Impact Evaluation before mitigation	**Minor**

Table 9.38 Potential Ecological Impacts to Fishponds and Mitigation Ponds within the Assessment Area

Criteria	Impacts to Fishponds and Mitigation Ponds
Habitat Quality	Moderate - for fishponds at Deep Bay Low to moderate – for mitigation ponds Low – fishponds near San Sang San Tsuen and San Wai
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded. Seven avifauna (Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Besra, Greater Coucal and Red-billed Starling), two mammal (Japanese Pipistrelle and Chinese Noctule) and one odonate (Scarlet Basker) species of conservation importance were recorded <u>Literature Review</u> Eleven avifauna (Little Grebe, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Black Kite, Greater Coucal, Red-billed Starling, White-cheeked Starling, Grey-chinned Minivet and Zitting Cisticola), one mammal (Japanese Pipistrelle), two herpetofauna (Burmese Python, Chinese Bullfrog) and two odonate (Blue-spotted Dusk-hawker and Scarlet Basker) species of conservation importance were recorded
Size / Abundance	Approximately 0.32 ha of fishpond near San Sang San Tsuen and 0.34 ha of fishponds near San Wai would be lost permanently
Duration	Direct impact of habitat loss to fishponds within the footprint of the proposed development would be permanent. Indirect impact (noise, air quality/dust, run-off and water quality impact) to retained mitigation ponds other fishponds outside Project area during the construction phase would be temporary Indirect impact (fragmentation, noise and human disturbance) to retained mitigation ponds other fishponds outside Project area during the operational phase would be permanent
Reversibility	Direct impact within the proposed development would be irreversible Indirect impact to retained mitigation ponds and other fishponds outside Project area during the construction phase would be temporary and reversible Indirect impact to retained mitigation ponds and other fishponds outside Project area during the operational phase would be permanent and irreversible
Magnitude	Minor – for all fishponds within Project area as they would be filled for development Minor – for mitigation ponds Negligible – for other fishponds outside Project area
Overall Impact Evaluation before mitigation	**Minor** – for the fishponds near San Sang San Tsuen and San Wai **Minor** – for mitigation ponds **Negligible** – for other fishponds outside Project area

Table 9.39 Potential Ecological Impacts to Storm Water Drain/Recreational Pond within the Assessment Area

Criteria	Impacts to Storm Water Drain/Recreational Pond
Habitat Quality	Low
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Three avifauna (Chinese Pond Heron, Grey Heron and Little Egret) species of conservation importance were recorded <u>Literature Review</u> No flora and fauna species of conservation importance were recorded
Size / Abundance	No habitat loss
Duration	No direct impact is anticipated Temporary indirect impact (noise and vibration, air/dust) to the ponds at Hung Uk Tsuen and Hang Mei Tsuen during the construction phase Permanent indirect impact (noise and human disturbance) to the ponds at Hung Uk Tsuen and Hang Mei Tsuen during the operational phase
Reversibility	Indirect impact to the ponds at Hung Uk Tsuen and Hang Mei Tsuen during the construction phase would be temporary and reversible Indirect impact to the ponds at Hung Uk Tsuen and Hang Mei Tsuen during the operational phase would be permanent and irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.40 Potential Ecological Impacts to Natural Watercourse within the Assessment Area

Criteria	Impacts to Natural Watercourse
Habitat Quality	Low to moderate - for the Watercourse 1, Watercourse 7, undisturbed upper sections of Watercourse 5 and Watercourse 6 Low - for the disturbed lower sections of Watercourse 5 and Watercourse 6
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded One freshwater fish (Predaceous Chub) species of conservation importance was recorded in downstream of Watercourse 1 out of the site boundary <u>Literature Review</u> One mammal (Japanese Pipistrelle) and one odonate (Indochinese Copperwing) species of conservation importance were recorded
Size / Abundance	Lower sections of Watercourse 5 of approximately 8 m in total would be lost permanently
Duration	Direct impact of habitat loss of Watercourse 5 within the footprint of the proposed development would be permanent Temporary construction disturbance (run-off and water quality impact) to Watercourse 1, Watercourse 7, Watercourse 6 and the unaffected upper sections of Watercourse 5 during the construction phase Indirect impact to Watercourse 1, Watercourse 7, Watercourse 6 and the unaffected upper sections of Watercourse 5 (disturbance, water quality impact, hydrological disruption) during the operational phase would be negligible
Reversibility	Direct impact within the proposed development area would be irreversible Indirect impact during the construction phase would be reversible Indirect impact during the operational phase would be irreversible
Magnitude	Minor
Overall Impact Evaluation before mitigation	**Minor**

Table 9.41 Potential Ecological Impacts to Modified Watercourse within the Assessment Area

Criteria	Impacts to Modified Watercourse
Habitat Quality	Moderate - for TSW Main Channel Low - for other modified watercourses
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Nine avifauna (Northern Shoveler, Black-faced Spoonbill, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Wood Sandpiper, Collared Crow and Red-billed Starling), one mammal (Japanese Pipistrelle) and one freshwater fish (Predaceous Chub) species of conservation importance were recorded. However, Predaceous Chub is a common species in Hong Kong and considered as "Least Concern" in China by the latest Red List of China's Vertebrates <u>Literature Review</u> No flora and fauna species of conservation importance were recorded
Size / Abundance	Approximately 8.52 km of upper and middle section of Watercourse 3, Watercourse 4 and other minor modified watercourses would be lost permanently (1.36 ha in total) Approximately 0.50 km of Tin Sam Channel would be lost temporarily Approximately 0.2 ha of TSW Main Channel would be lost temporarily during the bridge pier construction
Duration	Direct impact of habitat loss within the footprint of the proposed development would be permanent, except it is temporary for the loss at Tin Sam Channel and TSW Main Channel Indirect impact (water quality, site run-off and dust) to other modified watercourses during the construction phase would be temporary Indirect impact (fragmentation, water quality deterioration, hydrological disruption and human disturbance) to other modified watercourses during the operational phase would be permanent
Reversibility	Direct impact within the proposed development would be irreversible for the affected modified watercourse, except it is reversible for the temporary loss at Tin Sam Channel Indirect impact during the construction phase to other modified watercourses would be reversible Indirect impact during the operational phase to other modified watercourses would be irreversible
Magnitude	Minor - for TSW Main Channel Minor - for other modified watercourses
Overall Impact Evaluation before mitigation	**Minor** - for TSW Main Channel **Minor** - for other modified watercourses

Table 9.42 Potential Ecological Impacts to Mangrove within the Assessment Area

Criteria	Impacts to Mangrove
Habitat Quality	Moderate to High
Species	Low floral and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Six avifauna (Chinese Pond Heron, Eastern Great Egret, Little Egret, Pied Avocet, Common Greenshank and Red-billed Starling) species of conservation importance were recorded <u>Literature Review</u> No flora and fauna species of conservation importance were recorded
Size / Abundance	No habitat loss
Duration	Direct and indirect impacts are unlikely due to the distance between the proposed development and mangrove
Reversibility	N/A
Magnitude	N/A
Overall Impact Evaluation before mitigation	**Negligible**

Table 9.43 Potential Ecological Impacts to Mudflat/Coastal Water Body within the Assessment Area

Criteria	Impacts to Mudflat/Coastal Water Body
Habitat Quality	High
Species	Low flora and faunal diversity <u>Current Surveys</u> No plant species of conservation importance were recorded Nine avifauna (Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Pied Avocet, Eurasian Curlew, Marsh Sandpiper, Collared Crow and Collared Crow) and one intertidal (*Carcinoscorpius rotundicauda*) species of conservation importance were recorded <u>Literature Review</u> One plant species of conservation importance (*Halophila beccarii*) was recorded Eighteen avifauna species of conservation importance (Great Crested Grebe, Black-faced Spoonbill, Chinese Pond Heron, Grey Heron, Eastern Great Egret, Little Egret, Great Cormorant, Western Osprey, Crested Goshawk, Bar-tailed Godwit, Eurasian Curlew, Common Redshank, Wood Sandpiper, Great Knot, Greater Coucal, Collared Crow, Red-billed Starling and White-cheeked Starling) and two intertidal (*Tachypleus tridentatus* and *Carcinoscorpius rotundicauda*) species of conservation importance were recorded
Size / Abundance	No habitat loss
Duration	Direct and indirect impacts are unlikely due to the distance between the proposed development and the mudflat/coastal waterbody
Reversibility	N/A
Magnitude	N/A
Overall Impact Evaluation before mitigation	**Negligible**

Table 9.44 Severity of Direct and Indirect Impacts on Each Habitat Type during Construction and Operational Phase

Habitats	Construction Phase			Operational Phase Impact		
	Direct Impact Severity	Indirect Impact Severity	Mitigation Measure	Direct Impact Severity	Indirect Impact Severity	Mitigation Measure
Developed Area/Wasteland	Permanent loss of 338.94 ha (**Minor severity**)	Disturbance impact, including air/dust, increase traffic and human activities and glare (**Minor to moderate severity** for San Sang San Tsuen egretry; **Negligible severity** for the remaining developed area/wasteland habitat)	**Yes** for San Sang San Tsuen egretry	No impact	Disturbance impact, including air/dust, increase traffic and human activities and glare: (**Minor severity** for San Sang San Tsuen egretry and "CA"; **Negligible severity** for other developed area/wasteland)	No. Enhancement measure is recommend to improve the habitat quality of San Sang San Tsuen egretry
Village/Orchard	Permanent loss of 61.60 ha (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Disturbance Impact (**Minor severity**)	No
Agricultural Area (Dry)	Permanent loss of 6.26 ha (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Disturbance Impact (**Minor severity**)	No
Agricultural Area (Wet)	Permanent loss of 1.33 ha (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Fragmentation, disturbance Impact (**Minor severity**)	No
Plantation	Permanent loss of 14.72 ha (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Fragmentation, disturbance Impact (**Minor severity**)	No
Woodland	Permanent loss of woodland at Ngau Hom Shek (0.10 ha) and at west of Fung Kong Tsuen (0.11 ha) (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Fragmentation, disturbance Impact (**Minor severity**)	No. Enhancement planting area with native species would be considered at the suitable area of the proposed Fung Kong Tsuen service reservoir site
Shrubland	No	No impact	Fragmentation, disturbance Impact (**Minor severity**)			No

Habitats	Construction Phase			Operational Phase Impact		
	Direct Impact Severity	**Indirect Impact Severity**	**Mitigation Measure**	**Direct Impact Severity**	**Indirect Impact Severity**	**Mitigation Measure**
Grassland	Permanent loss of 14.19 ha (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Disturbance Impact (**Minor severity**)	No
Marsh	Permanent loss of larger marsh of 0.56 ha at Fung Kong Tsuen (**Minor severity**)	Disturbance impact (**Minor severity**)	No	No impact	Disturbance Impact (**Minor severity**)	No
Fishponds and Mitigation Ponds	Permanent loss of fishponds near San Wai (0.11 ha and 0.23 ha) and San Sang San Tsuen (0.32 ha) (**Minor severity**)	Disturbance impact (**Minor severity**) Water quality impact (**Minor severity** for mitigation ponds and unaffected ponds)	No	No impact	Fragmentation, disturbance, water quality deterioration (**Minor severity**)	No. Precautionary measure is recommended to provide additional screen planting to mitigation ponds
Storm Water Drain/Recreational Pond	No direct impact	Disturbance impact (**Minor severity**) Water quality impact (**Minor severity**)	No	No impact	Disturbance, water quality deterioration (**Minor severity**)	No
Natural Watercourse	Permanent loss of 8 m of Watercourse 5 (**Minor severity**)	Disturbance, water quality impact (**Minor severity**)	No	No impact	Disturbance Impact, water quality impact and hydrological disruption (**Minor severity**)	No
Modified Watercourse	Permanent loss of 8.52 km of modified watercourse (**Minor severity** for TSW Main Channel and other modified watercourses) Temporary loss of 500 m of Tin Sam Channel and TSW Main Channel at the proposed new bridges piers (**Minor severity**)	Disturbance impact (**Minor severity**) Water quality impact (**Minor severity** for TSW Main Channel and its associated watercourses)	No	Revitalisation of 500 m of Tin Sam Channel (**Minor positive severity**)	Fragmentation, disturbance, water quality deterioration and hydrological disruption (**Minor severity** for TSW Main Channel and its associated modified watercourse)	No. Enhancement measure would be considered to be incorporated into the channel design during design stage

Habitats	Construction Phase			Operational Phase Impact		
	Direct Impact Severity	**Indirect Impact Severity**	**Mitigation Measure**	**Direct Impact Severity**	**Indirect Impact Severity**	**Mitigation Measure**
Mangrove	No direct impact	Disturbance impact (**Negligible severity**) Water quality impact and hydrological disruption (**Minor severity**)	No	No impact	Disturbance Impact (**Minor severity**)	No
Mudflat/Coastal Water Body	No direct impact	Disturbance impact (**Negligible severity**) Water quality impact (**Minor severity** for Deep Bay)	No	No impact	Disturbance, water quality impact and hydrological disruption (**Minor severity** for Deep Bay)	No

Table 9.45 Summary of Potential Impacts to Sites and Species of Conservation Importance

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
	Recognised Sites of Conservation Importance and Ecologically Sensitive Areas					
9.5.5	"CA"	A small area of "CA" at northwest of San Sang San Tsuen, which comprised of approximately 0.1 ha of developed area, would be directly impacted under the Project.	Negligible	No habitat loss during the operational phase.	No impact	No
9.5.31		"CA" in the vicinity of the proposed works areas (e.g. west of San Sang San Tsuen and southeast of the Project area) may be subjected to indirect disturbance impacts (e.g. arising from construction vehicles, noise and vibration, air/dust).	Minor	The "CA" at west of San Sang San Tsuen would be subjected to indirect impacts including noise, air quality deterioration, visual impact generated from the new "I" zone and slip road between Road P1 and the KSWH.	Minor	No mitigation measure is required. However, precautionary approach is recommended during construction and operational phases
9.5.30; 9.5.48 - 9.5.49	San Sang San Tsuen egretry	Although San Sang San Tsuen egretry would be retained under the development, construction activities (e.g. noise, dust and visual impacts) would likely disturb any nesting ardeids including the use of flight paths to and from foraging grounds.	Minor to Moderate	San Sang San Tsuen egretry may be subjected to noise and visual disturbance generated from the surrounding facilities during the operational phase.	Minor	Mitigation measure is recommended during the design stage and construction phase
	Species of Conservation Importance					
9.5.22	Plant Species of Conservation Importance – Incense Tree	One mature individual of Incense Tree was recorded at the woodland at Tung Tau Tsuen and would be affected under the proposed development.	Minor to Moderate	Assuming the plant would be retained *in-situ*, it is expected the level of the disturbance during the operational phase would be similar to the current condition.	Negligible	Avoidance measure is recommended during the design stage

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
9.5.23; 9.5.51	Avifauna Species of Conservation Importance - Crested Serpent Eagle	The mosaic of plantation, woodland, marsh, developed area/wasteland habitat at the west of Fung Kong Tsuen is likely to form a part of the Crested Serpent Eagle's territory. During the construction phase, a small area of the habitats would be permanently lost.	Minor	During the operational phase, human activities would increase at west of Fung Kong Tsuen (i.e. the planned "OU" and "G" land zoning). Indirect impacts from noise, glare and visual impacts to Crested Serpent Eagle would occur as its territory is located nearby.	Minor	Proper mitigation measures for control of dust, noise, glare and water quality are recommended during construction and operational phases
9.5.25; 9.5.52	Avifauna Species of Conservation Importance - foraging ardeids including Chinese Pond Heron, Little Egret	Loss of small fishponds, some sections of minor modified watercourses, marsh and wet agricultural area which contributed foraging habitat for ardeids. Disturbance impacts (e.g. noise/vibration and visual) during the bridge construction over the TSW Main Channel could reduce the use of the channel by avifauna.	Minor Low	During the operational phase, disturbance impacts from the newly developed areas (e.g. pedestrian pathway along riverside channel and the bridges) may occur. The construction of new bridges across the channel will increase the number of, and reduce the distance between physical structures over the channel. However, since the height of proposed bridges would be similar to those existing, the bridges are not likely to affect the avifauna flight path and utilisation, including nesting ardeids.	Minor Minor	Proper mitigation measures for control of dust, noise, glare and water quality are recommended during construction and operational phases
9.5.35; 9.5.52	Avifauna Species of Conservation Importance - overwintering/migratory waterbirds	Construction works of bridge piers may cause disturbance to overwintering/migratory waterbirds at TSW Main Channel. However, the utilisation of TSW Main Channel by the migratory waterbirds was relatively low and construction	Minor	The newly built bridges across the channel will reduce the distance between physical structures over the channel, which may create disturbance and affect the avifauna utilisation during the overwintering period.	Minor	Proper mitigation measures for control of dust, noise, glare and water quality are recommended during construction and operational phases

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
		works would be temporary, small in scale and localised. Thus, the construction impact is anticipated to be low.				
9.5.25	Mammal Species of Conservation Importance within the Project area - Short-nosed Fruit Bat, Japanese Pipistrelle and Chinese Noctule	Demolition of buildings or removal of trees during the construction phase may impact roosting bats, if present.	Minor	It is expected the level of the disturbance during the operational phase would be similar to the current condition.	Negligible	No mitigation measure is required. However, precautionary approach is recommended during construction and operational phase
9.5.27	Fish Species of Conservation Importance within the Project area - Predaceous Chub	Predaceous Chub recorded in the upper section of Watercourse 3 and lower section of Watercourse 4 would be impacted directly. Demolition of the upper section of watercourses could result in water quality deterioration and impact Predaceous Chub in the lower section of Watercourse 3.	Minor Minor	It is expected that the level of the disturbance during the operational phase would be similar to the current condition.	Minor	Mitigation measure for control of water quality is recommended during the construction and operational phases
	General Impacts					
9.5.8	Habitats Loss - Woodland	Only two small areas of woodlands located at Ngau Hom Shek (0.1 ha) and west of Fung Kong Tsuen (0.1 ha) would be lost.	Minor	No habitat loss during the operational phase.	No impact	No mitigation measure is required. But enhancement planting area with native species would be considered at the suitable area of the proposed Fung Kong

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
						Tsuen service reservoir site
9.5.12	Habitats Loss – Natural Watercourse	Loss of approximately 8 m of the lower sections of Watercourse 5 (low ecological value).	Minor	No habitat loss during the operational phase.	No impact	No
9.5.14	Habitats Loss – Modified Watercourse	Permanent loss of minor modified watercourses of approximately 7 km	Minor	Minor	Minor	No
		A section of Tin Sam Channel (approximately 500 m) which is located to the west of San Lee Uk Tsuen would be realigned due to the development.	Minor	Environmentally-friendly design of the realigned channel could improve the ecological value compared to the baseline condition.	Minor positive	No mitigation measure is required. The enhancement measure is recommended during the design stage
		Temporary loss of approximately 0.2 ha of TSW Main Channel at the proposed bridge pier	Minor	Minor	Minor	No
9.5.50	Habitat Fragmentation	N/A	N/A	Most habitats were surrounded by developed area, which comprised approximately 75% of the Project area, and had limited ecological linkage with other natural habitats such as woodland and shrubland.	Minor	No
9.5.29; 9.5.47	Injury/Mortality to Wildlife	Construction phase activities (e.g. site clearance and formation) have the potential to cause direct injury/mortality to common wildlife.	Minor	The installation of roadside noise barrier, transparent/reflective windows and excess artificial lightings during daytime and nighttime could result in harm to avifauna species due to chance of bird collision may occur.	Low to moderate	Precautionary measures are recommended during construction and operational phase

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
9.5.32; 9.5.54	Disturbance Impacts to habitats/fauna	Construction activities would increase human activity, noise and vibration disturbance from traffic and construction machinery, and result in indirect and secondary impacts.	Minor	The increased traffic flow and human activities from the newly developed residential and commercial areas could result in disturbance (e.g. noise pollution, glare, deterioration in air quality). The presence of the new Road P1, the slip road structure (linking the existing KSWH to the at-grade Road D3) and associated traffic could further reduce the use of the mitigation ponds by avifauna.	Minor Minor	Mitigation measure for control of dust, noise and glare is recommended to further minimise the disturbance during the construction and operational phase
9.5.32 - 9.5.35; 9.5.52		Disturbance impacts (e.g. noise/vibration and visual) during the bridge construction at TSW Main Channel could reduce the use of the channel by avifauna.	Minor	During the operational phase, disturbance impacts from the newly developed areas (e.g. pedestrian pathway along the riverside and the four bridges) may occur.	Minor	No
9.5.40 - 9.5.42; 9.5.58	Water Quality Impacts	Site formation and construction works near retained watercourses and mitigation ponds may pollute the waters due to potential release of construction waste. A small section of Tin Sam Channel will be diverted and replaced by new open water channel, which could affect the downstream water quality temporarily. A small section of Watercourse 3 will	Minor Minor	It is expected the level of the disturbance during the operational phase would be similar to the current condition.	Minor	Mitigation measure for control of water quality is recommended to further minimise the disturbance during the construction and operational phase

Reference to Relevant Section	Sensitive Receiver / Impact	Construction Phase Impacts		Operational Phase Impacts		Mitigation Measures
		Description	Evaluation	Description	Evaluation	
		be modified as part of the covered drainage network and may impact the water quality as well the fauna (e.g. Predaceous Chub) at downstream areas temporarily.	Minor			
9.5.42 - 9.5.45		Accidental spills of oil, fuel and other chemicals could result in water quality impacts.	Minor	During operational phase, sewage generated from the commercial, domestic sections may cause deterioration of water quality.	Minor	Mitigation measure for control of water quality is recommended to further minimise the disturbance during the construction and operational phase

9.6 Cumulative Impacts

Concurrent Projects

9.6.1 This section assesses the cumulative ecological impacts that may arise during the construction and operation of the proposed Project, due to interaction with other developments in the area. The major site formation and construction works is targeted to commence in Year 2019 and be completed by Year 2037/2038 for full population intake. The assessment area covers a large portion of the North West New Territories where several major development projects would have commenced or would be at various stages of construction /operation during the development of the Project.

9.6.2 According to the latest information, the proposed project "Planning and Engineering Study for Housing Sites in Yuen Long South – Investigation" is located near the Project and is under investigation stage. Most of the ecologically sensitive resources, including natural streams, woodland and agricultural lands are proposed to be planned in "Green Zone" and preserved under the Yuen Kong South development. Yet confirmation is required if this project would overlap with the construction period of the Project, the cumulative impact on the terrestrial environment would be needed to be further assessed when the project programme and final design of the Yuen Long South development are available.

9.6.3 Given all concurrent projects listed in **Table 2.18** in **Chapter 2** would be land-based, with the implementation of proper mitigation measures as recommended in the EIA Report, no significant cumulative water quality impacts would be anticipated during construction phase. As stated in **Section 5.12**, during the operational phase, the San Wai Sewage Treatment Works – Phase 1 which will tentatively start operation in 2020 and would discharge the treated sewage to the Urmston Road Outfall, which may result in potential cumulative water quality impacts upon the marine water of North Western WCZ. The potential cumulative water quality impacts have been preliminarily evaluated under the water quality impact assessment in **Chapter 5**. Detailed water quality modelling and detailed analysis will be conducted under the separate Schedule 2 EIA for HSK STW to fully assess any cumulative water quality impact due to discharge from concurrent projects in the assessment area.

9.6.4 Based on the latest available information, other projects with programmes likely to overlap with the Project construction include, "Proposed Development under the Study on the Enhancement of the Lau Fau Shan Rural Township and Surrounding Areas" (LFSRTSA); "Preliminary Land Use Study for Lam Tei Quarry and the Adjoining Areas – Feasibility Study"; and "Tuen Mun Western Bypass" (TMWB). However, the details of these projects are currently unavailable and therefore the cumulative impact assessment for these projects is not possible at this time.

9.7 Mitigation Measures

9.7.1 According to the EIAO-TM Annex 16 and EIAO Guidance Note. 3/2010, ecological impacts on important habitats and the associated wildlife caused by the proposed Project should be mitigated, in order of priority, avoidance, minimisation, and compensation approaches to the maximum practical extent.

Avoidance Measures for Recognised Sites of Conservation Importance / Ecologically Sensitive Areas / Habitats with High Ecological Value

9.7.2 In order to avoid environmentally sensitive areas, detailed evaluation of site selection and layout of the Project was undertaken during the design phase to arrive at the optimum layouts for the Project. Most of the sites of conservation importance as well as habitats with high ecological value were excluded from the Project area in order to avoid any direct impacts. Under the Revised RODP, "GB" would be retained during the

development to avoid the loss of natural habitats and recognised sites of conservation importance (i.e. San Sang San Tsuen egretry) (refer to **Chapter 2.10**). Moreover, the alignment of the Road P1 and a slip roads were adjusted to avoid direct loss of natural or semi-natural habitats in the "CA" and indirect loss of the mitigation ponds (refer to **Chapter 2.10**).

Mitigation Measures for Recognised Sites of Conservation Importance and Ecologically Sensitive Areas

9.7.3 San Sang San Tsuen egretry falls within the Project area but would be retained within the "GB" zoning (Site 3-38) under the proposed development to avoid direct impacts to the egretry (refer to **Section 2.10.13 – 2.10.18**). To minimise potential impacts (e.g. noise/vibration, visual disturbance) on the egretry during the construction phase, it is recommended that site formation and construction works at Sites 3-32, 3-33, 3-37, 3-39 and 3-40 are scheduled outside the breeding season of the ardeids (i.e. between March and August). In addition to the wide "LO" corridor provided to maintain the egretry flight paths, buffer planting at the boundaries of Sites 3-32, 3-33, 3-37, 3-39 and 3-40 with the "LO" will provide screening and further reduce disturbance impacts to the egretry and associated flight paths.

9.7.4 The "CA" at northwest of San Sang San Tsuen is located adjacent to the new "I" zone. Whilst there would be no direct loss of natural/semi-natural habitats, the provision of screening (e.g. hoarding) during the construction phase is recommended to minimise disturbance impacts (e.g. noise/vibration, visual) to adjacent habitats within the "CA". In addition, the provision of buffer planting at the "I" zone and slip road/"CA" interface would minimise disturbance impact during the operational phase.

Mitigation Measures for Plant Species of Conservation Importance

9.7.5 Only one mature individual of Incense Tree, which is a plant species of conservation importance, recorded from the woodland at Tung Tau Tsuen would be potentially impacted by the proposed development. It is recommended to preserve the woodland at Tung Tau Tsuen as well as the Incense Tree *in situ* (see also **Chapter 11.15.14**).

Mitigation Measures for Fauna Species of Conservation Importance

Crested Serpent Eagle

9.7.6 To minimise habitat loss and disturbance impacts (e.g. visual, noise, human activity) to Crested Serpent Eagle habitats (i.e. in "GB" zone, Site 3-2) during the construction phase, provision of screening (e.g. hoarding) at the boundary of the adjoining "OU" zones (i.e. Site 3-1, 3-5, 3-6, 3-7 and 3-8) should be provided. Practices to minimise noise and dust impacts (see **Sections 9.7.14** and **9.7.15**) should also be employed during the construction phase. In addition, the provision of screening planting at the boundary of the "OU" sites adjacent to the Site 3-2 will help to minimise disturbance impacts during the operation phase.

Other Avifauna Species

9.7.7 The revitalisation of the river channels should seek to find a balance between design for human requirements (e.g. aesthetics, recreation/leisure and access) and provision of ecological enhancements. The design should not result in disturbance impacts to the habitats/fauna utilising the channels (e.g. foraging avifauna). In addition, to minimise water quality impacts during construction, the construction method and sequence of the proposed pier in the watercourses should be carefully designed so that all the site formation and construction works including any pilling and excavation would be undertaken within a dry zone and physically separated from the watercourse downstream.

Bat Species

9.7.8 The majority of the bats foraging and commuting habitats (e.g. watercourses, "GB", "V") would be retained or not be changed under the development. Under the operational phase, buffer planting would be provided to minimise the disturbance (e.g. glare) from the adjacent land use zonings and enhance the ecological value of the existing habitats. All lighting along the river channel (e.g. TSW Main Channel and its associated watercourses) or near vegetated areas in "CA" or "GB" zones (e.g. Ngau Hom Shek knoll, Tung Tau Tsuen woodland, San Sang San Tsuen egretry, mitigation ponds) should be minimised or incorporate wildlife-friendly lighting to avoid light spill. Long wavelength lights such as amber lamps are recommended, as they are visible to human but invisible to most animals. The lights should be directed away from the "CA"/"GB" zones and installed with a shield and at a level as close to the ground as possible to prevent extensive light from cast up into the sky. Moreover, lights with motion sensors can further minimise the disturbance to the surrounding habitats as lights that are not in use would be automatically turn off. Bat species are protected under the Wild Animals Protection Ordinance (Cap. 170), hence any removal of roosting bats would require a special permit. To avoid potential injury to bat species, if any evidence of roosting bats is found in any buildings or trees that would be cleared during the construction phase, an ecologist with relevant experience should be consulted and involved to develop appropriate mitigation strategies.

Predaceous Chub

9.7.9 A small part of Watercourse 3, where Predaceous Chub inhabits, will be modified as part of the covered drainage network under the development (refer to **Chapter 5**). During the demolition of the existing watercourses, potential impacts of downstream water quality can be avoided by undertaking the construction and demolition works in dry conditions and dry season. The proposed permanent drainage for carrying the diverted flow should be constructed and completed before demolition and dewatering of the existing watercourse. Construction of all the proposed permanent and temporary drainage would be conducted in a dry zone. The flow diversion works should be conducted in dry season, where possible, when the flow in the watercourse is low. Normal operation of the existing watercourse would be maintained during construction of the new or temporary drainage required for the flow diversion works. With adoption of the proposed construction design and sequence, there should not be significant sediment and contaminant release to the downstream water.

Mitigation Measures for Habitat, Vegetation and Fauna

General Habitat Loss

9.7.10 Within the Project area, most of the habitats that would be lost are of low or low to moderate ecological value such as village/orchard, agricultural area (dry and wet), modified watercourse, plantation, grassland and shrubland. To minimise the impacts and enhance the overall habitat quality in the future, any vegetation loss will be replaced by new or restored back to native shrub and woodland plantings in areas of open space.

Mitigation Ponds

9.7.11 Mitigation ponds located within the Project area would be retained in the "GB" zone to avoid direct habitat loss (**Chapter 2** refers). However, the mitigation ponds would be subjected to construction disturbance, i.e. noise and potential site run-off. Provision of hoarding for proper delineation of works boundary is recommended before the commencement of the works. With implementation of proper mitigation measures stated in **Section 9.7.13 – 9.7.15** below, the construction disturbance would be minimised.

9.7.12 During the operational phase, the presence of the new Road P1, the slip roads between the KSWH and the at-grade Road D3 near the two eastern mitigation ponds could result in a further decrease of avifauna usage of these ponds. The mitigation ponds are already subject to some disturbance due to the existing, adjacent container storage, road and highway. To minimise disturbance impacts the existing tree belt (including some tall trees) on the eastern side of the larger pond would be retained (**Figure 9.5** refers). The proposed amenity strip and additional tree planting along the new Road P1 would provide screening for the existing ponds. Other screening measures (e.g. vertical greening walls, green roof, noise barriers) are recommended to be incorporated at the buildings and roads in the vicinity of the mitigation ponds to minimise the disturbance impact (e.g. noise, visual impact).

Air Quality, Noise and Glare

9.7.13 In general, the disturbance impacts to the surrounding habitats and associated wildlife arising from the construction activities could be minimised by adopting the following mitigation measures.

9.7.14 Construction dust should be suppressed to avoid and minimise the dust covering leaves of plants that would affect their photosynthesis, and thus their health and growth (refer to **Section 3.8**):

- Regular spraying of haul roads;
- Proper storage of construction materials; and
- Covering trucks or transporting waste in enclosed containers to minimise windblown litter and dust during transportation of waste.

9.7.15 Noise impact during construction phase should be avoided and minimised to reduce the disturbance to the habitats adjacent to the works areas (refer to **Chapter 4**):

- Machines and plant (e.g. trucks) that may be in intermittent use should be shut down between work periods or should be throttled down to a minimum;
- Machines and plant known to emit strong directional noise should, wherever possible, be orientated so that the noise is directed away from the nearby habitats;
- Material stockpiles and other structures should be effectively utilised, wherever practicable, in screening noise from on-site construction activities;
- Using Quiet Mechanical Plant (QMP) to limit noise emissions at source; and
- QMP and other machines and plants (e.g. air compressors, concrete pumps) should be covered by noise enclosure to further reduce noise impact.

9.7.16 Through night-time lighting control during construction phase, glare disturbance to wildlife would be avoided. During operational phase, wildlife friendly lighting is recommended to be installed as the precaution measure to further minimise adverse impact resulted from artificial lightings (**Section 9.7.8** refers).

9.7.17 During the operational phase, the road networks and associated noise barriers may result in bird collision and mortality. Mitigation measures such as use of tinted materials and superimposing dark patterns or strips on the barrier, as per EPD/Highways Department requirements stated in “Guidelines on Design of Design of Noise Barriers” (2003), would be employed to minimise bird mortality from collision.

Measures to Minimise Water Quality Impacts

9.7.18 As stated in the Water Quality section, good site practices during the construction phase should be adopted to avoid any pollution entering any nearby watercourses. Practices to minimise surface run-off and to reduce suspended solid levels should be undertaken during construction (refer to **Section 5.13**):

- Surface run-off from construction sites should be discharged into storm drains via adequately designed sand/silt removal facilities such as sand traps, silt traps and sedimentation basins;
- Open stockpiles of construction materials (e.g. aggregates, sand and fill material) on sites should be covered with tarpaulin or similar fabric during rainstorms;
- General refuse and construction waste should be collected and disposed of in a timely and appropriate manner;
- Drainage arrangements should include sediment traps to collect and control construction run-off;
- Silt removal facilities, channels and manholes should be maintained and the deposited silt and grit should be removed regularly, at the onset of and after each rainstorm to prevent local flooding;
- All works and storage areas should be restricted to the site boundary;
- All vehicles and plant should be cleaned before they leave a construction site to minimise the deposition of earth, mud, debris on roads; and
- Regular check of the construction boundary to avoid unmitigated impacts imposed on nearby watercourse.

9.7.19 The construction method and sequence of the proposed bridge piers within the TSW Main Channel should be carefully designed, to ensure all excavation and pilling operations would be undertaken within a dry zone and physically separated from the watercourse downstream. All the construction activities in watercourses should be fully enclosed by impermeable sheet pile cofferdam walls. Silt removal facilities should be used before discharging the effluent generated from the dewatering operation to the watercourses. Silt curtains should also be deployed around the construction works area inside the watercourses, where practicable, to minimise the water quality impacts. It is recommended that the construction works in watercourses should be undertaken in dry seasons, where practicable, to minimise the potential water quality impact.

9.7.20 As stated in the Water Quality section (**Section 5.14**), it is anticipated that the water quality impacts associated with the operational phase would be minimal and acceptable, provided that the recommended precautionary measures such as adequate storm drainage system and sewage pumping stations, blue-green infrastructure, and best storm water management practices and storm water pollution control plan are properly implemented. Ecological impacts associated with the change in water quality from this Project are considered negligible to minor.

9.7.21 Although proper site practice and good site management could prevent site run-off, in view of the sensitive nature of the Deep Bay WCZ, a Contingency Plan should be developed to minimise the potential impact sewage discharges under failure of treatment facilities (e.g. new HSK STW) during emergency situations or inclement weather at construction and operational phases (**Section 5.14** refers). To minimise the chance of emergency sewage discharge, standby pump would be provided to cater for emergency breakdown or maintenance of the duty pump. Backup power supply in the form of dual / ring circuit power supply or generator would be provided to secure electrical power supply. With incorporation of the recommended preventive measures, the chance of emergency sewage bypass would be minimised.

9.8 Potential Enhancement Recommended for Further Study

9.8.1 Enhancement could be achieved by incorporating habitat creation through the landscaping plan (e.g. at "LO" and "District Open Space" ("DO") zoning), particularly at where areas connected to "GB" (i.e. natural habitats). Introduction of native plant species as buffer planting could be adopted within the developed areas, to provide connectivity between any isolated "GB" zones within the Project area. This can help to improve the current habitat quality of the "GB" zones as the majority of them were of low ecological value plantation.

9.8.2 Landscape planting in the "LO" zone adjacent to the San Sang San Tsuen egretry should include planting of trees and bamboo, such as Chinese Hackberry and Da Ngan Bamboo (*Bambusa eutuldoides*) that are potentially utilised by ardeids for nesting, to enhance nesting resources for breeding ardeids. To maximise the chance of use, this should be located in an area with limited human disturbance.

9.8.3 Consideration could also be given to the inclusion of suitable planting within the flood retention lake at Site 1-27 to provide foraging and roosting habitats for wetland dependent species (e.g. ardeids). This could include appropriate planting of suitable trees and bamboo on the proposed island to potentially provide additional nesting resources for breeding ardeids. The island embankments (e.g. slope surface, vegetation planting, etc.) should be appropriately designed to provide suitable foraging opportunities for ardeids. For example, shallow water with gentle sloping banks covered with marshy species may provide foraging ground for breeding ardeids.

9.8.4 During detailed design for the realignment of the Tin Sam Channel at Tin Sam Tsuen, opportunities for ecological enhancement (e.g. bioengineering, creating meanders) would be considered to improve to the ecological value. Provision of natural substrate that would encourage colonisation of flora and freshwater fauna in the bottom and banks of the revitalised watercourses would be considered subject to detailed design of the proposed revitalisation measure.

9.8.5 Although the impact was considered as minor for the small piece of woodland lost at Ngau Hom Shek, to enhance the overall habitat in the future, an enhancement planting area with native species would be considered at a suitable area of the proposed Fung Kong Tsuen service reservoir site.

9.9 Environmental Acceptability of the Schedule 2 Designated Projects

9.9.1 Under the Schedule 2 Designated Projects identified in **Chapter 1** (**Table 1.1**, **Figure 1.3**), the summary of potential habitats that would be affected under Schedule 2 Designated Projects in the Project area was presented in the **Table 9.46**.

Table 9.46 Summary of Potentially Affected Habitats under Schedule 2 Designated Projects Subject to EP under this Study

Designated Project Reference No.	Work Component / Reference in Revised RODP	Potentially Affected Habitats
DP1	Construction of new primary distributor road (Road P1)	Directly impacted (e.g. habitat loss) • Developed area/wasteland (Low ecological value) (10.0 ha) • Village/orchard (Low ecological value) (0.2 ha) • Agricultural land (dry) (Low ecological value) (<0.1 ha) • Plantation (Low ecological value) (0.3 ha) • Shrubland (Low ecological value) (<0.1 ha) • Grassland (Low ecological value) (0.3 ha) • Natural watercourse (Low ecological value) (< 0.1 ha) • Modified Watercourse (Low ecological value) (<0.1 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Village/orchard (Low ecological value) • Agricultural land (dry) (Low ecological value) • Modified Watercourse (Low ecological value) • Plantation (Low ecological value) • Shrubland (Low ecological value) • Grassland (Low ecological value) • Mitigation ponds (Low to moderate ecological value)
DP2	Construction of eight new distributor roads (Road D1–D8)	Directly impacted (habitat loss) • Developed area/wasteland (Low ecological value) (25.7 ha) • Village/orchard (Low ecological value) (3.5 ha) • Agricultural land (dry) (Low ecological value) (0.2 ha) • Modified watercourse (Low ecological value) (< 0.1 ha) • Shrubland (Low ecological value) (0.3 ha) • Grassland (Low ecological value) (1.0 ha) • Fishponds (Low ecological value) (0.1 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Village/orchard (Low ecological value) • Agricultural land (dry) (Low ecological value) • Modified watercourse (Low ecological value) • Plantation (Low ecological value) • Grassland (Low ecological value) • Fishponds (Low ecological value) • Marsh (Low ecological value)

Designated Project Reference No.	Work Component / Reference in Revised RODP	Potentially Affected Habitats
DP5	Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and KSWH connection to Road D3	Directly impacted (habitat loss) • Developed area/wasteland (Low ecological value) (0.63 ha) • Village/orchard (Low ecological value) (0.1 ha) • Shrubland (Low ecological value) (0.1 ha) • Grassland (Low ecological value) (<0.1 ha) • Modified watercourse (Low ecological value) (< 0.1 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Village/orchard (Low ecological value) • Shrubland (Low ecological value) • Grassland (Low ecological value) • Modified watercourse (Low ecological value) • Mitigation ponds (Low to moderate ecological value)
DP6	Construction of partly depressed and partly decked-over road located at Road D2, Road D4, and Road D6	Directly impacted (habitat loss) • Developed area/wasteland (Low ecological value) (0.5 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Village/orchard (Low ecological value)
DP9	Construction of four new sewage pumping stations	Directly impacted (habitat loss) • Developed area/wasteland (Low ecological value) (0.5 ha) • Village/orchard (Low ecological value) (0.2 ha) • Plantation (Low ecological value) (<0.1 ha) • Modified watercourse (Low ecological value) (< 0.1 ha) • Fishponds (Low ecological value) (<0.1 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Village/orchard (Low ecological value) • Modified watercourse (Low ecological value) • Fishponds (Low ecological value)
DP12	"Construction of Road P1 and Slip-Road from Kong Sham Western Highway to Road D3 partly located within the "Conservation Area" of Yuen Tau Shan	Directly impacted (habitat loss) • Developed area/wasteland (Low ecological value) (0.4 ha) Indirectly impacted (e.g. human disturbance, air/dust, noise and traffic, and water quality) • Developed area/wasteland (Low ecological value) • Plantation (Low ecological value)

9.9.2 A small section of modified watercourse (i.e. Watercourse 3) that would be lost under DP2 and DP5 was of low ecological value. Although Predaceous Chub was recorded within this habitat, it is common species in Hong Kong and thus the impact is considered minor.

9.9.3 A small section of natural watercourse, i.e. Watercourse 5, would be unavoidably affected under DP1. However, since this affected area was small in size (<0.1 ha), immediately downstream of a culvert/modified section of the watercourse, and subjected to a high level of disturbance from nearby village housing and developed area, the direct impact of habitat loss is anticipated to be minor.

9.9.4 The eastern, larger mitigation pond near KSWH would be indirectly affected under the construction and operation of slip road under DP5 and new Road P1 under DP1. However, this mitigation pond was already subject to some disturbance due to the existing, adjacent container storage, road and highway. Mitigation measures as presented in **Sections 9.7.13 - 9.7.20** (e.g. retaining and planting of tall tree belt along the new Road P1 and slip road to provide screening, vertical greening wall, and noise barrier) were proposed to minimise the impact.

9.9.5 A small area of 0.1 ha at the fringe of the "CA" at northwest of San Sang San Tsuen would be affected by the construction of slip road under DP12. However, as this habitat was mainly comprised of developed area and no species of conservation importance recorded, the direct impact is expected to be negligible. Mitigation/precautionary measures were recommended in **Sections 9.7.4**, **9.7.8**, **9.7.15 - 9.7.20** to minimise potential indirect impacts to adjacent habitats within the "CA" and associated wildlife.

9.9.6 The remaining majority of the habitat that would be lost directly under the DP1, DP2, DP5, DP6, DP9 and DP12 were developed area/wasteland, village/orchard, dry agricultural land, plantation, shrubland, grassland, modified watercourses and fishponds of low ecological value (**Table 9.46** refers). These habitats were highly disturbed by human activities (e.g. farming activities, roads, noise) and supported low species diversity. Thus, no significant adverse impact is anticipated for these affected habitats.

9.9.7 Indirect impacts including human disturbance, air/dust, noise and traffic, and water quality may result at the habitats in the vicinity of DP works sites. The habitats included developed area/wasteland, village/orchard, dry agricultural land, plantation, shrubland grassland, modified watercourse, marsh, mitigation ponds and fishponds of low to moderate or low ecological value (**Table 9.46** refers). Therefore, with the implementation of proper mitigation measure as suggested in **Section 9.7**, the potential indirect impact is anticipated to be minor.

9.10 Evaluation of Residual Ecological Impacts

9.10.1 This section addresses the requirements of Section 3.4.10.4(xii) and (xiii) of the Study Brief. Mitigation measures and predicted residual impacts are summarised below in accordance with Section 5.4.5(b) and (c) of Annex 16 of the EIAO-TM, residual impacts with on-site mitigation measures are defined, quantified, where possible, and evaluated.

9.10.2 Direct impact on habitats to be permanently lost by the project are identified and quantified in **Table 9.28**. All are assessed as minor severity in **Section 9.5.4 - 9.5.21** and **Table 9.44**. While mitigation measures are considered not required for the direct impact arising from habitat loss, enhancement measures are suggested in **Section 9.8**. The residual direct impact of habitat loss due to the project is considered to be minor and acceptable.

9.10.3 With proper implementation of the mitigation measures suggested in **Section 9.7**, potential indirect impact (i.e. human disturbance, noise, air/dust, water quality deterioration), arising from the construction and operation of the Project, to the nearby habitats and its associated fauna would be minimised and the residual indirect impact is considered to be acceptable.

9.10.4 After the revitalisation of the Tin Sam Channel to incorporated opportunities for ecological enhancement, the overall habitat quality is anticipated to be improved. Therefore, a minor positive residual impact is expected.

9.11 EM&A Requirements

9.11.1 Implementation of the recommended mitigation measures described in **Section 9.7** would be audited monthly throughout the construction phase. Site audit should be also carried out monthly throughout the construction phase. In case of non-compliance, the contractor should be informed to strengthen the proposed mitigation measures accordingly. Details of environmental monitoring and audit (EM&A) requirements are discussed in the separate EM&A Manual. The summary of the EM&A requirements are presented below.

9.11.2 A monitoring programme covering construction phase for monitoring the condition and integrity of the San Sang San Tsuen egretry and use of the site by ardeids and any other species of conservation significance should be specified during the detailed design stage of the development. No site formation and construction works would be undertaken in Sites 3-32, 3-33, 3-37, 3-39 and 3-40 and during the breeding season of ardeids. During the construction phase, the egretry would be monitored monthly in the ardeid breeding season i.e. from March to August by qualified ecologists with at least 10 years relevant local experience to confirm if it is active during the breeding season and to ensure the egretry is not significantly disturbed by nearby construction activities other than the aforementioned sites.

9.12 Conclusion

9.12.1 The Project area and boundary have evolved and adjusted, during the course of the Study, to avoid recognised sites of conservation importance or ecologically sensitive areas. Four sites of conservation importance were identified within the assessment area, including “CPA”, Ngau Hom Shek egretry and Shenzhen Bay Bridge egretry which are located outside the Project area, and San Sang San Tsuen egretry which is located within the Project area in “GB” zoning that would be retained. Although Road P1 and a slip road (under DP12) would be constructed in a small area of the “CA”, northwest of San Sang San Tsuen, the affected area was small (0.1 ha) and developed (i.e. roads and reinforced concrete flood storage pond) without any species of conservation importance recorded, therefore the impact is negligible.

9.12.2 Fifteen habitats were identified within the assessment area including developed area/wasteland, village/orchard, agricultural area (dry), agricultural area (wet), plantation, woodland, shrubland, grassland, marsh, fishponds and mitigation ponds, storm water drain/recreational pond, natural watercourse, modified watercourse, mangrove, and mudflat/coastal water body. Among them, approximately 90% of the Project area (595 ha out of 714 ha Project area) were comprised of developed area/wasteland and village/orchard which is of low ecological value.

9.12.3 Habitats with high, moderate to high, or moderate ecological value (i.e. hillside woodlands, fishponds at the Deep Bay shore, mangrove and mudflat/coastal water body) are either located outside the Project area or would be retained within the Revised RODP. The majority of habitats, including woodland, shrubland, plantation, natural and modified watercourse, marsh and village/orchard would be retained based on the Revised RODP, which would also minimise impacts to fauna species (e.g. by protecting their foraging/roosting grounds).

9.12.4 The Project will involve proposed developments adjacent to the San Sang San Tsuen egretry and the existing mitigation ponds adjacent to KSWH; however, the implementation of proper mitigation measures would be required to minimise the impacts to an acceptable level. The San Sang San Tsuen egretry would be retained in "GB" and an eco-corridor comprising "LO" zone would be designated to maintain a development-free zone for the ardeid flight path. This would allow the ardeids to freely access their favorable foraging habitats such as TSW Main Channel. Moreover, the construction works in close proximity to the egretry should be scheduled outside the ardeid breeding months from March to August to minimise any disturbance impacts.

9.12.5 The four existing mitigation ponds which were developed under the DBL project would be retained in the "GB" zone. While the Project would not have direct impact on these ponds, the proposed slip roads between Road P1 and KSWH would be located adjacent to the two eastern ponds which could result in a potential decrease of usage of these ponds by avifauna species (due to disturbance and potential barrier effect). To minimise the impacts an amenity strip is proposed adjacent to the eastern side of these ponds and additional buffer tree planting along the new Road P1 would provide screening minimise disturbance impacts from the surrounding roads.

9.12.6 To avoid direct impacts (loss) of one mature individual of Incense Tree recorded from the woodland at Tung Tau Tsuen, the woodland would be preserved *in situ* and the tree would be protected.

9.12.7 The proposed revitalisation of the riverside channels should seek to find a balance between design for human requirements (e.g. aesthetics, recreation/leisure, access) and provision of ecological enhancements. The design should not result in disturbance impacts to the habitats/fauna utilising the channels (e.g. foraging avifauna).

9.12.8 Indirect and secondary impacts during the construction phase would comprise human disturbance, construction noise and vibration, construction dust, glare and construction site run-off. With proper implementation of good site practices, no significant adverse ecological impact is anticipated. The nature of disturbance during the operational phase would be comparable to the existing condition. No significant adverse impact is therefore expected.

9.12.9 The Project area covers a total area of about 714 ha. Excluding the retained areas (e.g. existing roads, major watercourses, hillside areas, villages and the committed developments), the development area is about 441 ha. Of which, about 200 ha (45% of the development area) is currently occupied by brownfield operations. The existing brownfield operation had created various environmental impacts, including ecology, air and noise, visual and water quality impacts. Therefore, the consolidation of the existing brownfield operations to modern facilities will help to reduce the current impacts and improve general quality of environment.

9.12.10 Ecological enhancement could be achieved by incorporating habitat creation through the landscaping plan (e.g. at "LO" and "DO" zoning), particularly at areas connected to "GB" (i.e. natural habitats). Incorporation of ecological enhancement into channel design of the realigned Tin Sam Channel could improve the ecological value of the habitats during the operational phase. Consideration could also be given to the inclusion of suitable planting within the flood retention lake at Site 1-27 to provide foraging and/or roosting/nesting habitats for wetland dependent species (e.g. ardeids). For example, shallow water with gentle sloping banks covered with marshy species may provide foraging ground for breeding ardeids.

9.12.11 The Revised RODP has generally avoided impacts to habitats and species through its layout and considered the retention of higher ecological value habitats (e.g. the egretry, woodland). With the implementation of the recommended mitigation measures (e.g. measures to avoid/minimise impacts to San Sang San Tsuen egretry, measures to reduce disturbance from construction activities, etc.), no unacceptable residual impacts including both direct and indirect residual impacts during construction and operational phases would be expected.

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